Do Mainstream Party Positions Cause Far-Right Support?*

Laurenz Guenther† Salvatore Nunnari‡

July 17, 2026

Abstract

Far-right parties have gained substantial electoral support across Western democracies. One explanation emphasizes political supply: when mainstream parties are perceived as diverging from voters on salient issues, they create electoral space for far-right competitors. We identify this channel in an information-provision experiment conducted during the 2025 German federal election. We randomly assign 5,040 German citizens truthful signals about the immigration position of the center-right CDU and use these signals as instruments for voters' beliefs. A signal portraying the CDU as one unit more liberal on a common 0–10 scale increases AfD vote intention by 1.1 percentage points. Standard two-stage least squares implies an effect of 2.1 percentage points per unit of perceived CDU positioning. Because standard IV overweights strong belief updaters, who respond least to their beliefs, we apply a Local Least Squares estimator that recovers an average population effect of 4.7 percentage points—nearly a quarter of the AfD's vote share in the 2025 election. The estimates also imply significant responses in donation decisions, consequential campaign-ad allocations, and newsletter sign-ups; effects on vote intention and party feelings persist in an obfuscated follow-up. Responses are concentrated among conservative voters and are asymmetric across parties, as predicted by the location of voters and parties in the policy space.

Keywords: Far-Right Parties, Elections, Information Provision Experiment, Immigration.
JEL Codes: D72, C93, K37.

*We are grateful to Felix Chopra, Mathilde Emeriau, Luca Henkel, Timothy Neal, Giacomo Ponzetto, Chris Roth, Johannes Wohlfart and audiences at the 2025 Bocconi-Columbia Political Economy Conference, 2025 CSEF-IGIER Symposium on Economics and Institutions, 2026 Resilient Democracy Workshop, 2026 CEPR Political Economy Symposium, Bergamo, UNSW, Melbourne, and Monash for useful comments and discussions. This study was approved by the Bocconi Research Ethics Committee (Application RA000845). The pre-registration is available at https://osf.io/5kjmt. We acknowledge financial support from the European Research Council (SN; POPULIZATION Grant No. 852526) and the French Agence Nationale de la Recherche (LG; Investissement d'Avenir Programme, ANR-17-EURE-0010). A previous version of this paper circulated under the title "Do Political Representation Gaps Cause Populism? Evidence from the 2025 German Election."

†Toulouse School of Economics; laurenz.guenther@tse-fr.eu.

‡Bocconi University, Department of Economics and IGIER; CEPR; CESifo; salvatore.nunnari@unibocconi.it.

1 Introduction

Across Western democracies, far-right parties have surged in electoral support. In Europe, their vote shares have more than doubled since 2000, reshaping political competition and carrying consequential economic effects (Funke et al., 2023; Guriev and Papaioannou, 2023). One influential explanation emphasizes political supply: on cultural issues, most notably immigration, mainstream parties in many European countries are perceived as systematically more liberal than the median voter (Guenther, 2026). Standard spatial voting models predict that such gaps create electoral opportunities for parties positioned to represent otherwise underserved voters (Downs, 1957; Persson and Tabellini, 2002). Far-right parties appear to have exploited precisely this opportunity. But the resulting correlation does not establish that mainstream-party positioning causes far-right support. Both may instead reflect economic grievances, cultural anxiety, dissatisfaction with political elites, or other changes in voter demand (Guriev and Papaioannou, 2023). Does perceiving a mainstream party as more liberal on immigration cause voters to shift toward the far right?

Germany provides a particularly informative setting. For much of the postwar period, the CDU was the rightmost major party in a stable national party system. The AfD, founded in 2013, entered the Bundestag in 2017, becoming the first party to the right of the CDU to gain national parliamentary representation since the consolidation of the postwar party system. In the 2025 federal election, it became the second-largest party, with 20.8% of the vote. Figure 1 illustrates the representational gap that accompanied this rise. Using data from the German Longitudinal Election Study (GLES, 2023), the figure reports the average difference between citizens' preferred immigration policy and their perceived immigration stance of each major party. From 2009 to 2023, voters consistently perceived every mainstream party—including the CDU—as more liberal on immigration than themselves on average. When the AfD emerged, it became the only major party perceived as lying to the right of the average voter, occupying the space left open by the mainstream parties. The gap widened during the 2015–16 refugee crisis (shaded in the figure), when the CDU was perceived as shifting toward a more liberal immigration stance while AfD support roughly doubled, from about 6% in early 2015 to 12% by late 2016. Similar gaps characterize other European countries, where far-right parties systematically occupy

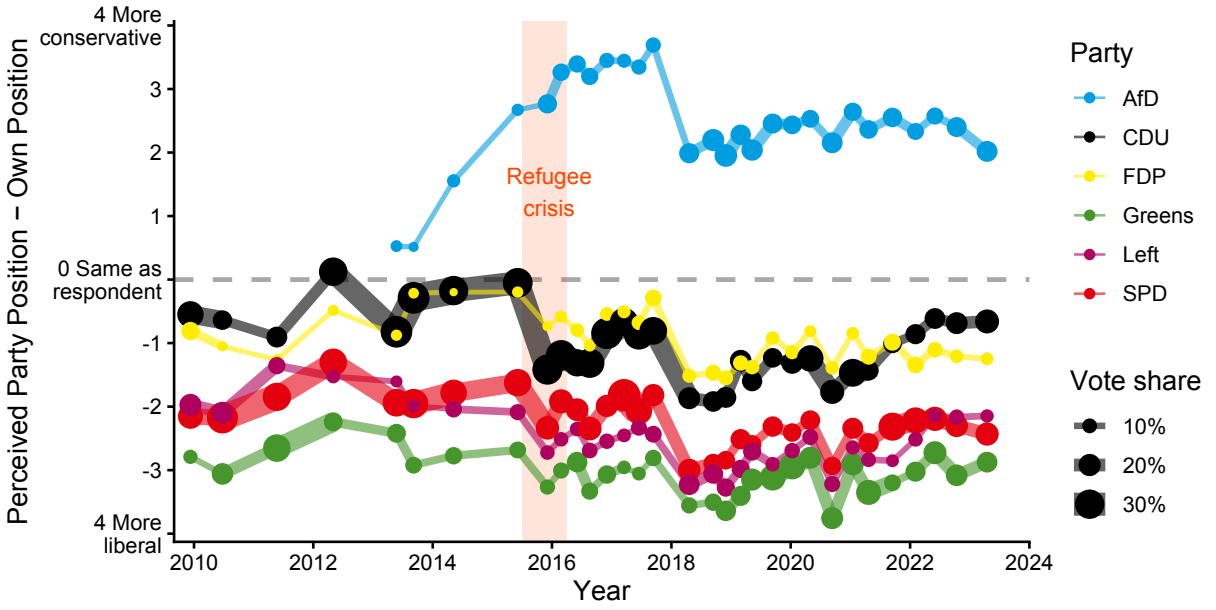


Figure 1: Perceived Immigration Stance of German Parties Relative to Voters
Note: Each point plots the weighted average of (perceived party position − respondent's own position) on a 0–10 immigration scale across all respondents in a given GLES wave. Positive values indicate that voters perceive the party as more conservative than themselves; negative values indicate more liberal. Point size is proportional to the party's vote intention share. The shaded region marks the 2015–16 refugee crisis. Source: GLES (2023).

the space between mainstream-party positions and voter preferences on cultural issues (Guenther, 2026). These patterns motivate our question but cannot answer it because party positions and voter demand are jointly determined.

We provide causal evidence from an information-provision experiment conducted during the 2025 German federal election. Two features of the setting are especially useful for our design. First, the CDU is Germany's main center-right party and the AfD's closest mainstream competitor on immigration, while an informal cordon sanitaire separates the AfD from the mainstream parties. Second, conflicting signals from the CDU's leadership generated substantial uncertainty about the party's immigration stance. We exploit this uncertainty to create exogenous variation in voters' beliefs about where the CDU stands. Using an anonymous survey of parliamentary candidates, we construct four truthful signals reporting the median immigration position of different sets of CDU candidates. Participants are randomly assigned to signals placing the CDU at 4, 5, 6, or 7 on a common 0–10 immigration scale, while information about every other party is held constant. Moving the signal by one unit shifts posterior beliefs by approximately half a unit in the same direction, with a first-stage F-statistic of 375.

The reduced-form results show that portraying the CDU as one unit more liberal increases the probability of intending to vote for the AfD by 1.1 percentage points and raises feelings toward the AfD by 0.025 standard deviations. Instrumenting posterior beliefs with the randomized signal, standard two-stage least squares (2SLS) implies that perceiving the CDU as one unit more liberal increases AfD vote intention by 2.1 percentage points. These estimates establish that beliefs about a mainstream party's position can causally affect support for the far right.

Standard 2SLS does not, however, recover the average effect of beliefs in this setting. It places greater weight on respondents whose beliefs respond most strongly to the signal. In our experiment, these strong updaters are also the respondents whose political behavior responds least to their beliefs, a pattern documented in other information-provision experiments by Balla-Elliott (2026).[1] We therefore also apply the Local Least Squares (LLS) estimator of Balla-Elliott (2026), which, under their identifying assumptions, recovers the average population effect rather than the updater-weighted 2SLS estimand.

LLS implies that perceiving the CDU as one unit more liberal increases AfD vote intention by 4.7 percentage points, an effect equivalent to nearly one quarter of the AfD's 20.8% vote share in the 2025 election. The LLS estimates are statistically significant for all ten party-specific outcomes and both summary indexes. They include two behavioral measures with direct stakes—donation decisions and consequential campaign-ad allocations—as well as newsletter sign-ups. The LLS effects on vote intentions and party feelings also persist in an obfuscated follow-up survey conducted approximately one week later. Thus, random assignment establishes the existence of a causal effect on AfD vote intention and party feelings without relying on LLS; the LLS analysis shows that standard IV understates its average magnitude and provides evidence that the effect extends to consequential political behavior.

The effects follow the distributional predictions of a spatial voting model. Immigration preferences in the German electorate are concentrated on the conservative side of the policy space, and most participants place themselves to the right of the CDU. For these voters, perceiving the CDU as more liberal increases its distance from their ideal point and makes the AfD

[1] Conditional Average Partial Effect (CAPE) curves make this heterogeneity visible: the marginal effects of CDU beliefs decline sharply with updating intensity. Formal slope tests with bootstrap inference reject flatness for all ten party-specific outcomes and both summary indexes; for the two PCA indexes, $p < 0.001$.

relatively more attractive. The CDU simultaneously becomes more attractive to some voters with liberal immigration preferences, partially offsetting its losses on the conservative side. The aggregate response is therefore asymmetric: a more liberal perceived CDU position increases AfD support much more than it decreases CDU support. Consistent with this mechanism, the effects concentrate among voters with conservative immigration attitudes, are amplified among voters who consider immigration important, and are largest among those who initially perceived the CDU as relatively liberal.

Our paper contributes to three literatures. First, we identify a political-supply channel behind support for the far right. A large literature attributes the rise of far-right parties to changes in voter demand generated by trade exposure (Autor et al., 2020; Colantone and Stanig, 2018a,b), economic insecurity (Algan et al., 2017; Fetzer, 2019; Funke et al., 2016), and new communication technologies (Campante et al., 2018; Guriev et al., 2021; Liberini et al., 2025; Zhuravskaya et al., 2020); see Guriev and Papaioannou (2023) for a review. Existing evidence also documents widespread gaps between mainstream-party positions and voter preferences, particularly on immigration and other cultural issues, and relates these gaps to far-right support (Guenther, 2026). We show that this relationship is causal: holding voters' own preferences fixed, experimentally induced beliefs about the position of an actual mainstream party affect support for its far-right competitor during a national election. The results connect the forces that generate political discontent to a strategic feature of political supply. Party positioning is directly adjustable by political actors, and its electoral consequences operate through where voters believe parties stand.

Second, our paper contributes to the growing literature that uses randomized information provision to identify how beliefs affect behavior. This approach has been applied to beliefs about immigration (Alesina et al., 2023; Grigorieff et al., 2023; Guenther, 2025), climate change (Dechezleprêtre et al., 2025; Nyhan et al., 2022), labor markets (Jäger et al., 2024), income distributions (Bottan and Perez-Truglia, 2022), and macroeconomic outcomes (Link et al., 2023; Roth and Wohlfart, 2020; Roth et al., 2022).[2] Within this literature, we provide a direct

[2]Other applications include health behavior (Akesson et al., 2022), protest mobilization (Hager et al., 2022), and prosocial behavior toward shareholders (Henkel and Pugnaghi-Zimpelmann, 2026). See Haaland et al. (2023) for a review and discussion of experimental design.

distributional test of spatial voting. A large empirical literature examines how the distance between voters and parties shapes electoral behavior (Adams, 2012; Gallego and Schofield, 2016; Lee et al., 2004; Schofield, 2005; Schofield and Zakharov, 2010; Di Tella et al., 2025), and recent information-provision, vignette, and conjoint experiments provide causal evidence that party positions affect political support (Chou et al., 2021; Grewenig et al., 2020; Hjorth and Larsen, 2022). Hjorth and Larsen (2022) vary whether respondents expect the Danish Social Democrats to maintain or relax a restrictive immigration policy, identifying how voters respond to the direction of accommodation. In a German conjoint experiment, Chou et al. (2021) vary the refugee-policy positions of hypothetical mainstream-party candidates and show how restrictive positions can attract far-right voters while alienating mainstream-party supporters. Our design shifts posterior beliefs about the actual CDU during a live national election and measures voters' ideal points, party beliefs, and experimental signals on the same scale. This allows us to quantify a unit change in perceived positioning, locate that change relative to each voter and the competing parties, and test which voters respond and why the aggregate effects are asymmetric.

This distributional focus also distinguishes our estimand from the one studied by Castanho Silva and Wratil (2023). They examine whether learning that some party represents a respondent's position on European integration changes populist attitudes. They find no average effect, although respondents initially low in populism become more populist, primarily through increased anti-elite sentiment. We study a different margin: how beliefs about the position of an identified mainstream party affect electoral substitution between that party and a specific far-right competitor. Our results show that representational gaps can have substantial consequences for choices among available parties, even when related information has limited average effects on broader political attitudes. This interpretation is consistent with experimental evidence that the electoral appeal of far-right candidates depends importantly on their policy platforms rather than only on populist rhetoric (Baskaran et al., 2025; Chou et al., 2021; Dai and Kustov, 2024).

Third, our findings inform the debate over whether mainstream parties should accommodate or demarcate against the radical right (Abou-Chadi and Krause, 2020; Hjorth and Larsen, 2022). Recent quasi-experimental studies estimate the electoral consequences of salient real-world changes in mainstream-party strategy. Becker and Hager (2025) study the CDU's decision to pass

a parliamentary motion with AfD support in January 2025 and find that the CDU gained support while the AfD lost it; Heyna et al. (2026) document similar responses to cooperation at the local level. In the United Kingdom, Turnbull-Dugarte et al. (2026) study Labour leader Keir Starmer's "Island of Strangers" speech and find electoral costs for Labour without corresponding gains for Reform UK. These studies identify the overall electoral response to politically consequential events as voters experienced them. Our experiment identifies a different estimand: the effect of perceived policy distance, holding information about the other parties and other dimensions of party strategy fixed. The distinction helps explain why accommodation need not have uniform effects. As our model and heterogeneity results show, the response depends on the accommodating party's initial position, the distribution of voters, and the location of the far-right competitor. A shift by a center-right party can draw conservative voters away from the far right, whereas an otherwise similar shift by a party beginning elsewhere in the political space can produce a different pattern of electoral substitution.

The remainder of the paper proceeds as follows. Section 2 develops the spatial-voting framework and derives our hypotheses. Section 3 and Section 4 present the experiment and empirical strategy. Section 5 reports the results, and Section 6 examines their robustness. Section 7 concludes.

2 Conceptual Framework and Empirical Predictions

Before presenting our experiment, we outline the theoretical framework guiding our analysis and derive predictions for the effect of our manipulation. We first derive individual-level comparative statics from a spatial model of voting. Because aggregate effects also depend on where voters sit relative to the parties, we then use the measured distribution of immigration attitudes to state an aggregate hypothesis. Finally, we contrast these predictions with alternative channels linking mainstream-party positions to far-right support.

2.1 A Spatial Voting Model

We consider a single salient policy dimension: immigration. Voter i holds an ideal policy position $x_i \in [0, 10]$, where higher values indicate more restrictive preferences. Each party p is perceived by voter i to hold position π_{ip}. Voter i's utility from party p is

$$U_i(p) = -(x_i - \pi_{ip})^2 + v_p + \varepsilon_{ip}, \tag{1}$$

where v_p is a party-specific valence term (capturing, e.g., perceived competence, incumbency advantages, or social acceptability) and ε_{ip} is an idiosyncratic preference shock. The voter selects the party that maximizes utility:

$$p_i^* = \underset{p \in \mathcal{P}}{\text{argmax}} \; U_i(p). \tag{2}$$

Consider the choice between the CDU and the AfD. The voter prefers the AfD over the CDU if and only if

$$(x_i - \pi_{i,\text{CDU}})^2 - (x_i - \pi_{i,\text{AfD}})^2 > v_{\text{CDU}} - v_{\text{AfD}} + \varepsilon_{i,\text{CDU}} - \varepsilon_{i,\text{AfD}}. \tag{3}$$

Because the AfD is a newcomer party operating under a cordon sanitaire, we expect $v_{\text{CDU}} > v_{\text{AfD}}$: voters require the AfD to be substantially closer to their ideal point to overcome the CDU's valence advantage. This assumption plays no role in our formal results, which concern only the left-hand side of (3); it matters for interpretation, reconciling the model with the fact that the AfD's vote share falls well short of its spatial potential (Figure 4) and implying that the voters marginal between the CDU and the AfD sit to the CDU's right. The left-hand side of (3) equals $(2x_i - \pi_{i,\text{CDU}} - \pi_{i,\text{AfD}})(\pi_{i,\text{AfD}} - \pi_{i,\text{CDU}})$, which is increasing in x_i when $\pi_{i,\text{AfD}} > \pi_{i,\text{CDU}}$: more conservative voters are more likely to prefer the AfD.

Our experiment shifts $\pi_{i,\text{CDU}}$—the perceived immigration stance of the CDU—while holding all other perceived party positions and the voter's own attitude fixed. The following proposition characterizes how this shift affects the relative attractiveness of the AfD at the individual level.

Proposition 2.1 (Effect of CDU positioning on party preference)**.** *Assume $\pi_{i,AfD} > \pi_{i,CDU}$ (the AfD is perceived as more conservative than the CDU). Define $\Delta_i \equiv (x_i - \pi_{i,CDU})^2 - (x_i - \pi_{i,AfD})^2$ as the relative policy cost of the CDU versus the AfD. Then $\frac{\partial \Delta_i}{\partial \pi_{i,CDU}} = 2(\pi_{i,CDU} - x_i)$, so that:*

(a) a marginal leftward shift in $\pi_{i,CDU}$ increases Δ_i—making the AfD relatively more attractive—for all voters with $x_i > \pi_{i,CDU}$, and decreases Δ_i for voters with $x_i < \pi_{i,CDU}$;

(b) the effect of a marginal leftward shift, $-\frac{\partial \Delta_i}{\partial \pi_{i,CDU}} = 2(x_i - \pi_{i,CDU})$, is increasing in the voter's conservatism: $\frac{\partial}{\partial x_i}\left[-\frac{\partial \Delta_i}{\partial \pi_{i,CDU}}\right] = 2 > 0$;

(c) for voters with $x_i > \pi_{i,CDU}$, the magnitude of the effect, $2(x_i - \pi_{i,CDU})$, is strictly decreasing in $\pi_{i,CDU}$: a voter who initially perceives the CDU as more liberal experiences a larger effect.

The intuition is straightforward. Quadratic loss implies that moving the CDU one unit to the left hurts a voter in proportion to how far that voter already is to the right of the CDU. A voter close to the CDU's position barely notices the shift; a voter far to its right sees the CDU pull away sharply. Because the AfD stays in place, the relative cost of sticking with the CDU rises most for conservative voters—this is parts (a) and (b). Part (c) reflects the curvature of the same loss function in the party's position: a voter who already perceives the CDU as liberal operates on a steep region of the loss function, where a further leftward shift produces a large additional loss, while a voter who perceives the CDU as conservative sits near the minimum, where the same shift barely registers. Parts (a) and (b) yield our first, individual-level hypothesis.

Hypothesis 1 (Spatial voting). *Perceiving the CDU as more liberal on immigration increases AfD support among voters to the CDU's right, with larger effects for voters with more conservative immigration attitudes.*

These predictions vary in how much of the model they use. Part (a) requires only that utility decline in the distance between voter and party. Parts (b) and (c) additionally use the convexity of the loss function: they hold for any increasing convex loss, not only the quadratic, but fail, for example, under linear loss.[3] The aggregate hypothesis we derive below relies only on part (a)

[3]At the level of observed vote choice, the prediction that conservative voters respond more does not rest on curvature alone: switching parties requires being near the CDU–AfD margin, and—given the measured distribution of attitudes—those voters sit on the conservative side of the space.

and the measured distribution of ideal points, and is therefore robust to the shape of the loss function. Part (c), by contrast, is a sharper implication that effectively tests its curvature: effects should be larger for voters who initially perceive the CDU as more liberal on immigration, whose priors place the party on a steeper region of the loss function. Because this prediction leans on functional form rather than on the general logic of spatial competition, we treat it as secondary and examine it in Subsection 5.4.

Proposition 2.1 characterizes individual-level effects; it does not by itself sign aggregate effects on party support. A leftward CDU shift moves voters at different electoral margins in opposite directions: the CDU becomes less attractive relative to the AfD for voters to its right, but more attractive relative to the SPD, the Greens, and the Left for voters to its left. Aggregate AfD support therefore rises when voters marginal between the CDU and the AfD lie predominantly to the CDU's right. The aggregate effect on the CDU is ambiguous because the party can lose conservative voters to the AfD while gaining liberal voters from parties to its left.

Our design measures the policy distribution relevant to these electoral margins. Most respondents place themselves to the right of the CDU (Figure 4), and the CDU is the closest mainstream party for much of the conservative side of the electorate. Policy positions alone do not reveal which voters are electorally marginal, because vote choice also depends on valence and idiosyncratic preferences. Nevertheless, the measured configuration motivates the following aggregate prediction, which we take to the data.

Hypothesis 2 (Asymmetry). *Given the measured distribution of voters and perceived party positions, perceiving the CDU as more liberal increases aggregate AfD support by more than it decreases aggregate CDU support.*

Appendix B applies this spatial perspective descriptively to the German policy space, using fifteen years of survey data to show that the AfD occupies a distinct, immigration-defined region of the space and that the CDU is its closest mainstream competitor.

2.2 Alternative Predictions

A prominent alternative explanation holds that far-right support is driven by generalized frustration or alienation toward mainstream parties, rather than by policy preferences (Mudde

and Rovira Kaltwasser, 2017). Under this *protest voting* view, voters choose far-right parties primarily to express dissatisfaction with political elites. Changes in the perceived positions of mainstream parties should have limited or no effect on far-right support, because dissatisfaction with the political establishment persists regardless of specific policy distances. In its starkest form, this view thus predicts no effect of our manipulation.

A second alternative is a *legitimation* channel: when a mainstream party is perceived as holding more conservative positions, voters may interpret this as validating positions previously considered controversial or extreme. This can operate through two related mechanisms. First, voters who are uncertain about the appropriate immigration policy may update positively about conservative positions when they learn that a mainstream party embraces them. Second, positions perceived as extreme or socially undesirable may become more acceptable when they are associated with a mainstream party. Under either mechanism, a more conservative CDU signal could increase AfD support alongside CDU support—the opposite of the spatial-voting prediction for the AfD.

Our design allows us to test these competing predictions about voter responses to perceived positional change. It is not designed to evaluate the protest or legitimation perspectives wholesale: both may operate through channels that our one-shot informational manipulation holds fixed, such as rhetoric, coalition behavior, or repeated elite cues over longer horizons. Our results should accordingly be read as evidence on the positional channel, not as a rejection of these mechanisms in all settings.

3 Experimental Design

Our design follows the principles for information-provision experiments discussed by Haaland et al. (2023). This section is organized as follows: Subsection 3.1 gives a general overview of the survey structure. Subsection 3.2 and Subsection 3.3 provide details on the key independent and dependent variables we use in the analysis, respectively. Subsection 3.4 explains our efforts to maximize high-quality responses and Subsection 3.5 provides information on the survey implementation. Finally, Subsection 3.6 describes the design of the follow-up study.

3.1 Structure

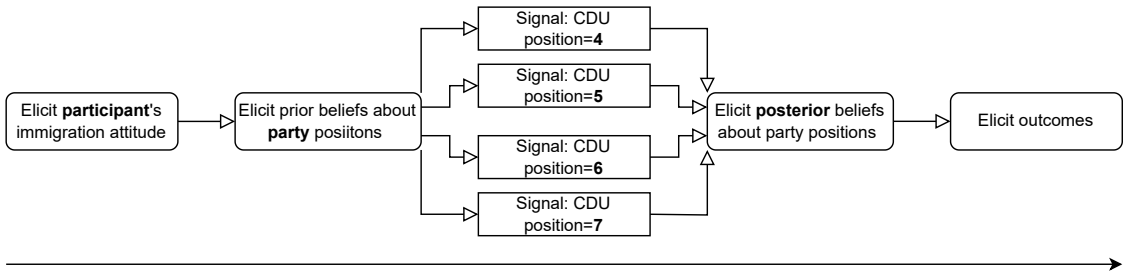


Figure 2: Survey Structure

We conduct an information provision experiment with an active control group (Haaland et al., 2023). In this design, every participant receives information about the parties' immigration stances; the four treatment arms differ only in the value reported for the CDU. Any effect of merely receiving immigration information, such as raising the salience of the issue, is therefore common to all arms and cancels out of treatment comparisons, which isolate the positional content of the signal. As visualized in Figure 2, the experiment follows a five-step procedure:

1. After initial screening questions (18+, citizenship, attention check), we elicit participants' own immigration attitudes using their response on a scale from 0 to 10 to the following question: *"What is your personal opinion on the topic of immigration opportunities? Please use the following scale from 0 to 10. A value of 0 means that, in your opinion, immigration opportunities should be strongly facilitated. A value of 10 means that, in your opinion, immigration opportunities should be strongly restricted. You can use the values between 0 and 10 to grade your opinion."*
2. We elicit participants' prior beliefs about the immigration stance of the main German parties on the same scale.
3. We randomly assign participants to one of four treatment conditions, each providing a distinct signal about the parties' immigration stances. These signals vary solely in the reported position of the CDU (4, 5, 6, or 7) holding all other party positions constant.
4. We elicit posterior beliefs about the parties' immigration stances, capturing how participants update their perceptions in response to the treatment.
5. We measure outcome variables, variables used to test for heterogeneous treatment effects, and variables used for validation exercises.

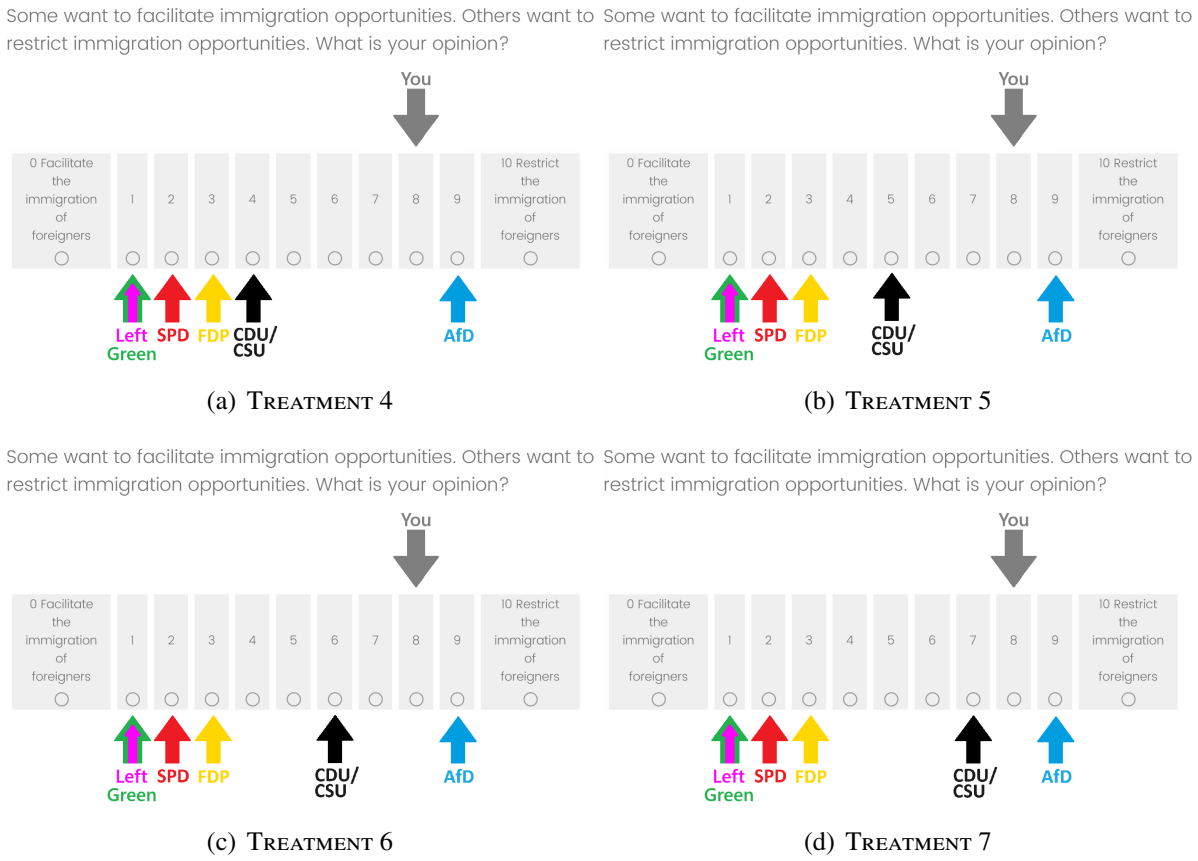
These steps share an important feature: participants' own immigration attitudes, their prior and posterior beliefs about each party, and the signal they receive are all measured on the same 0–10 scale. This places voters and parties in the same space, which is what makes the spatial voting model testable. It underlies much of what follows—locating voters relative to the CDU and the AfD, measuring how the signal moves the CDU through the voter distribution, and testing the model's sharper predictions: that conservatives respond more than liberals (Hypothesis 1), that the aggregate effect is asymmetric given the concentration of voters on the conservative side of the scale (Hypothesis 2), and that voters with more liberal priors about the CDU respond most (Proposition 2.1, part c).

Our central objective is to estimate the causal effect of belief changes on outcomes of interest—such as attitudes toward parties and behavioral responses—by inducing exogenous variation in participants' information sets. We achieve this by randomized allocation into estimates of parties' immigration stances. To generate these estimates, we employ survey data on candidates to the previous German federal election who responded to the same question on attitudes toward immigration as our participants. Following studies in political science and economics (Costello et al., 2020; Dalton, 2017; Guenther, 2026; Vasilopoulou and Gattermann, 2013; Walczak and Van der Brug, 2013), we estimate the position of a party as the median position of its candidates. To achieve exogenous variation in the estimate for the CDU while keeping estimates for other parties' constant, we draw a subsample of 11 CDU candidates (out of the 141 we have information for) and calculate their median stance but use the whole sample of available candidates for other parties. Participants are informed about this procedure—including the exact number of candidates sampled per party—in language designed to convey scientific credibility (the full text shown to participants is reproduced in Appendix K and summarized in the footnote below).[4] A potential concern is that participants find it implausible that only 11 CDU candidates are sampled while other parties have 58–131. Self-reported trust in the provided information does

[4]This is the language we used in the survey (translated from German): "Scientists have long been researching how to measure the positions of political parties. As recent studies have shown, one valid method is to evaluate the responses of parties' candidates in an anonymous survey. We used this method to measure the immigration stance of German parties. For this purpose, we analyzed the answers of some Bundestag candidates to the same question we just asked you. The currently available data is for candidates to the 2021 Bundestag election, so we have no data for Sarah Wagenknecht's Alliance yet. We will provide you with information on the responses of 503 candidates. These are broken down by party as follows CDU/CSU: 11, AfD: 58, SPD: 65, Greens: 131, FDP: 117, Left: 121."

not differ across treatment conditions (Table A13), suggesting that the small CDU subsample does not undermine perceived credibility. The uniform random sampling of 11 CDU candidates would result in 4 different median stances with high likelihoods: 4, 5, 6, and 7. To maximize statistical power, we assign participants to each of these four treatments with equal probability (rather than with the exact probabilities from uniform random sampling).

3.2 Independent Variable: The Signal



(a) Treatment 4

(b) Treatment 5

(c) Treatment 6

(d) Treatment 7

Figure 3: Information Provision Screen by Treatment

Figure 3 shows the four possible signals on the CDU immigration stance (that is, the four *treatments*), as seen by our participants on their screens. Notice that, in this same screen, we also remind them of their own answer to the same question and give them a short description of each party's position in words (for instance: the CDU/CSU wants to restrict immigration opportunities). See Subsection K.3 for the complete screenshots. Our main endogenous variable is the *CDU posterior belief*: the participant's post-treatment belief about the CDU's immigration

stance, measured on the same 0–10 scale. The signal shifts this belief, and we use the random assignment as an instrument.

3.3 Dependent Variables

As pre-registered, we examine 11 main dependent variables, which we categorize into three groups: voting intentions and attitudes, engagement with party materials, and party support.

3.3.1 Voting Intentions and Attitudes

To measure voting intention, we use two standard items. First, participants indicate via an 11-point scale how likely it is that they are turning out at the upcoming federal election. Thereafter, independent of their previous answer, we ask them which party they would vote for if they voted. Based on this item, we calculate indicators for each major party that equal one if the participant states to vote for this party and zero otherwise. We also include a more continuous measure of party attitudes: a standard feeling thermometer item that asks participants how positive or negative they feel toward each party on an 11-point scale.

3.3.2 Engagement with Party Materials

To measure engagement with party materials, we give participants links to pages where they can sign up for each party's newsletter. We observe whether participants click each sign-up link, but not whether they subsequently complete the subscription. We refer to this revealed choice as a newsletter sign-up for brevity, while interpreting it as an intention to subscribe rather than a verified subscription. As pre-registered, we construct binary indicators for newsletter sign-ups for the CDU and AfD, respectively.

3.3.3 Party Support

We use two incentivized measures for direct party support. First, we follow earlier work and adopt a donation measure (Braghieri and Eichmeyer, 2024; Haaland et al., 2023; Stantcheva, 2023). In particular, we elicit participants' willingness to accept to allow us make a donation of €5 to (i) the CDU and (ii) the AfD on their behalf. This is the minimum amount we need to pay

them (in addition to the reward for completing the survey) for them to agree with the donation. We construct two such measures, one for the donation to the CDU and one for the donation to the AfD.[5] A greater amount indicates a lower support for or greater aversion to the party.

Second, we employ a measure that has, to our best knowledge, not been used so far: we (truthfully) tell participants that, just before an upcoming election, we will conduct another study in which we will show official campaign ads to 1,000 voting-age German citizens. A random participant in this study decides what parties' campaign ads we will show to participants in the future study. To make the measure more continuous, we let participants in the present study allocate campaign ads freely to participants in the future study. For instance, a participant could assign 400 future participants to the CDU ad, 300 to the AfD ad, and 300 to the SPD ad. We interpret higher allocations to a party's ads as reflecting greater support for that party. One potential concern is that participants may allocate strategically—for example, assigning ads to a party they oppose in the hope of "inoculating" future participants against it. We view this as unlikely given the complexity of such reasoning in a survey context.

3.3.4 Indexes

The preregistration listed the 11 individual outcomes as the main outcomes and described the construction of party-support indexes as an exploratory analysis. To summarize the party-specific outcomes concisely, we calculate separate indexes of support for the AfD and CDU. To this end, we perform a Principal Component Analysis (PCA). For both indexes, we include all variables discussed above with the exception of likelihood of turnout. We scale all variables to have a standard deviation of one and such that higher values indicate greater support for the party. We then apply PCA on the standardized variables and extract the first principal component—which captures the maximum variance across these variables—and use it as a single index of party support. This index is then also standardized to have a mean of 0 and a standard deviation of 1.

[5] We focus on these two parties because this measurement is time consuming and, theoretically, our treatment should mostly affect attitudes toward these two parties.

We follow Falk et al. (2018) and use a "staircase method" for elicitation of this willingness to pay. In particular, participants answer a series of 4 or 5 binary questions where the amount we offer them to agree with the donation adapts to their previous answers. This allows us to create a fine-grained measure with a small number of questions and without asking multiple questions in the same screen (as it would be the case with a Multiple Price List.) See Appendix K for details.

3.4 Ensuring High-Quality Responses

We added several items to increase the internal and external validity of our findings. As pre-registered, participants who either failed to solve a captcha or an attention check at the beginning of the study were routed out of the survey and contributed no data. Moreover, to obtain a broadly representative sample, we elicited socio-demographic characteristics at the beginning and at the end of the survey. Table A12 compares the socio-demographic characteristics of our sample with the adult German population and Table A13 shows balance tests across treatments. To increase the likelihood of true reporting, we followed Alesina et al. (2023) and warned respondents that their responses could be flagged as low-quality to the recruiting platform if they did not read the instructions carefully and did not answer to the best of their knowledge.

In addition, we included a second attention check at the end of the survey, which around 90% of participants passed. To examine the possibility of experimenter demand effects, we asked participants whether they found the survey politically biased in either direction. As Figure A37 in the Appendix shows, nearly 80% perceive no survey bias, and the distribution of perceived political bias is symmetric. To check whether participants trusted the information, we elicited posterior beliefs and asked them directly to what extent they trusted the information at the end of the survey. Participants trust the provided information to a similar degree across all treatment conditions (Table A13). We use these three items for pre-registered robustness checks below.

Finally, following standard practices by the market research company we collaborated with, participants who completed the survey very quickly (that is, in less than one-third of the median completion time in the soft-launch phase) were classified as speeders, replaced with new participants during data collection, and excluded from the main analysis sample. This exclusion, which was not part of the pre-registered exclusion criteria, removes 87 participants (1.7% of all participants who completed the survey). We include these participants in an additional robustness check below.

3.5 Procedure

We designed the survey using Qualtrics and recruited participants via the established market research company Cint. Our study includes two surveys, a main study and an obfuscated

follow-up study (described in the next section), for which we invited only those who completed the main surveys.

The survey was fielded from December 16, 2024 to February 22, 2025. The 2025 German federal election took place on February 23, 2025.

The timing was deliberate: in the weeks before the election attention to political content was high, and the CDU's ideological direction was genuinely uncertain—Friedrich Merz had a conservative history, but recent CDU leaders had oscillated between conservative and liberal positions. This ambiguity, reflected in the high variance of pre-treatment beliefs, made all four signal values ex ante plausible descriptions of the CDU's stance.

As pre-registered, we aimed for a sample of 5,000 observations which is broadly representative of the German adult population in gender, age, and education. We planned to over-sample the Eastern states, aiming for 3,000 participants from Eastern Germany and 2,000 from Western Germany because the AfD has traditionally had a stronger support in Eastern Germany (as shown in Figure A36). Since multiple participants were allowed to begin the survey simultaneously (with some attrition during the survey, e.g., because of the eligibility screeners and the initial attention check) and socio-demographic quotas were soft, we ended up with a slightly larger than planned sample: 5,040 participants (2,003 from West Germany and 3,037 from East Germany). The median completion time was 10.4 minutes.

Ethics. The study was approved by the Bocconi University Research Ethics Committee (Application RA000845) and pre-registered before data collection. All information provided to participants was computed from actual candidate responses, and every statement shown to participants was accurate: for each treatment, the displayed position is the median immigration stance of a set of 11 CDU candidates in the candidate survey—exactly as the survey text stated—and the positions communicated for the other parties are the medians of all their available candidates. As described above, we assigned participants to the four signal values with equal probability, rather than with the probabilities implied by uniform random sampling of candidate subsamples, to maximize statistical power. Providing participants with truthful information from purposively selected sources or subsamples is standard practice in information-provision experiments in economics, where researchers routinely choose which truthful piece of information

(for example, which expert's forecast or which subsample's statistic) each participant receives, in order to generate exogenous variation in beliefs (Haaland et al., 2023).

3.6 Follow-Up Study

To mitigate potential experimenter demand effects and assess the persistence of treatment impacts, we conducted an obfuscated follow-up study approximately one week after the main experiment. Data collection for the follow-up study, like the main study, concluded on February 22, 2025—the day before the federal election. The follow-up was distributed to the same participants via the same survey firm but designed to appear unrelated to the initial study. We altered the visual layout, tone, and wording to obscure any connection, making it difficult for respondents to link the two surveys. The follow-up was intentionally brief, including only two outcome measures: a feeling thermometer rating for parties and vote intentions. These were selected as standard survey questions commonly encountered in political polls, reducing the likelihood of suspicion. The thermometer question was rephrased differently from the main survey to further minimize associations.[6]

Even though participants who completed the main study were invited repeatedly and the hourly wage for completing the follow-up survey was higher than in the main study (as the median completion time was 2.4 minutes), the recontact rate was low. Out of 5,040 participants to the main study, only 2,182 ($\approx 43\%$) contributed at least one outcome measure in the follow-up study. This 57% attrition rate is a significant limitation of the follow-up design. Table A12 compares the two samples and it shows that attrition was non-random: the follow-up sample is notably younger than the main sample and has a greater proportion of Western participants. While the consistency of point estimates between the main and follow-up studies is reassuring, the non-random attrition means the follow-up results should be interpreted with caution, as they may not generalize to the full sample. Nonetheless, we present the results from the follow-up study as pre-registered. In Section 6, we find no evidence that attrition is related to treatment assignment, and the follow-up estimates are robust to inverse-probability weighting on observables.

[6]While both versions asked participants how they feel about each party (with wording taken from GLES 2023), participants in the follow-up study had to enter a number between 0 and 100 while they were confronted with 11 horizontally arranged answer options in the main study (as in Braghieri and Eichmeyer 2024).

4 Empirical Strategy and Identification

This section describes how we identify the causal effects. Our analysis leverages the randomized assignment of information about the CDU's immigration stance to estimate both the reduced-form effect of the signal and the causal effect of perceived CDU positioning.

4.1 Reduced Form

As pre-registered, we estimate the direct effect of information provision on a range of outcome variables using ordinary least squares (OLS). The treatment variable is a four-level variable corresponding to the randomly assigned signal about the CDU's immigration position. We treat it as a continuous variable and estimate:

$$y_i = \alpha + \beta \cdot \text{Signal}_i + \theta \cdot X_i + \varepsilon_i, \tag{4}$$

where y_i is an outcome of interest (e.g., intention to vote for the AfD), Signal_i denotes the treatment assignment (4, 5, 6, or 7), and X_i is a vector of covariates including age, gender, education, and region of residence.

The coefficient β captures the average causal effect of receiving a signal that portrays the CDU as one unit more conservative on immigration. This specification answers the question: *How does informing participants that CDU candidates are more conservative on immigration change their political attitudes and behaviors?*

4.2 Estimating the Effect of CDU Positioning

While the reduced form estimates the direct effect of the information, our primary interest is the causal effect of *perceived CDU positioning*. This variable is the primitive object in our conceptual framework: the experiment shifts $\pi_{i,\text{CDU}}$, the position at which voter i perceives the CDU, while holding the voter's own position and information about the other parties fixed. The CDU posterior also has a direct interpretation because a one-unit change has the same meaning as a one-unit movement on the policy scale used to locate voters and parties.

The first stage of our IV strategy is:

$$\text{CDU Posterior}_i = \gamma_0 + \gamma_1 \cdot \text{Signal}_i + \delta \cdot X_i + \eta_i, \tag{5}$$

where CDU Posterior_i is the participant's post-treatment belief about the CDU's immigration stance (0–10 scale). The instrument is the randomly assigned signal, which generates exogenous variation in beliefs. The second stage is:

$$y_i = \alpha + \beta \cdot \widehat{\text{CDU Posterior}}_i + \theta \cdot X_i + \varepsilon_i. \tag{6}$$

The exclusion restriction requires that the signal affects outcomes only through its effect on beliefs about the CDU's immigration stance—an assumption aligned with the experimental design that we assess using the pre-registered cross-learning checks in Subsection 5.2.

The preregistration instead specified the perceived representation gap—the distance between the voter's own attitude and the perceived position of the closest mainstream party—as the endogenous variable. We depart from that specification for substantive and measurement reasons. The CDU posterior is the belief directly manipulated in the experiment and the party position entering our conceptual framework. It is observed on the original policy scale and has a linear, readily interpretable unit. The representation gap is instead a constructed minimum-distance measure whose value and identity-defining party can change nonlinearly as beliefs move. Both specifications have strong first stages ($F = 375$ for the CDU posterior and $F = 97$ for the representation gap). Appendix H reports the complete results using the pre-registered endogenous variable and shows that the substantive conclusions are unchanged.

4.3 The LLS Estimator

It is well known that standard 2SLS estimators target weighted averages of individual causal effects (Imbens and Angrist, 1994). In information provision experiments, these weights are proportional to the first-stage effect of information on beliefs. This creates a potential problem: strong dependence between belief updating and treatment effects makes standard 2SLS estimates misrepresent average effects. To address this, we adopt the Local Least Squares (LLS) estimator

proposed by Balla-Elliott (2026), which recovers the average population effect. Our experiment is an *active control* design: participants are randomly assigned to different signals (4, 5, 6, or 7), so that identification requires a learning rate updating assumption but does not require elicitation of prior variance. The learning rate is directly identified from observed belief updates, making the LLS estimator straightforward to implement in our setting.

LLS is a control function estimator. It constructs local regressions that condition on each respondent's learning rate—defined as $\alpha_i = (\text{posterior}_i - \text{prior}_i)/(\text{signal}_i - \text{prior}_i)$, measuring how much each respondent updated their CDU belief toward the signal—thereby isolating the purely experimental variation in beliefs that comes from random assignment to different signals.[7] By aggregating these local regressions, LLS weights all individuals equally rather than overweighting strong updaters, and thus recovers the average partial effect (APE).[8]

Both estimators in our presentation were specified in the pre-registration, which committed to estimating the local average treatment effect (standard 2SLS) and the adjusted average effect based on Balla-Elliott (2026) for the perceived representation gap as endogenous variable (Appendix H). We apply the same pair of estimators to the CDU posterior specification.

5 Results

5.1 First Stage

We begin by describing where voters and parties sit in the policy space. Figure 4 plots participants' own immigration attitudes alongside their post-treatment beliefs about each party's position, on the same 0–10 scale. Participants' own attitudes are concentrated on the conservative side of the scale: the AfD's perceived stance aligns most closely with the sample mean, median, and modal attitude, while all other parties, including the CDU, are perceived as more liberal than the average voter. This is the configuration that gives a spatial voting model sharp predictions about

[7] For the 572 respondents whose prior equals the signal, the learning rate is undefined (0/0); following our main specification we set $\alpha_i = 0$, assigning them the lowest learning-rate rank, and retain them so that all estimates target the average effect over the full sample. Dropping these non-learners instead leaves the headline estimates nearly unchanged (e.g., the LLS effect on AfD voting moves from 4.7 to 4.6 percentage points).

[8] A companion R package implementing the estimator is available at `dballaelliott.github.io/lls/`; the estimates in this paper were produced using code kindly shared by Dylan Balla-Elliott before publication of his paper.

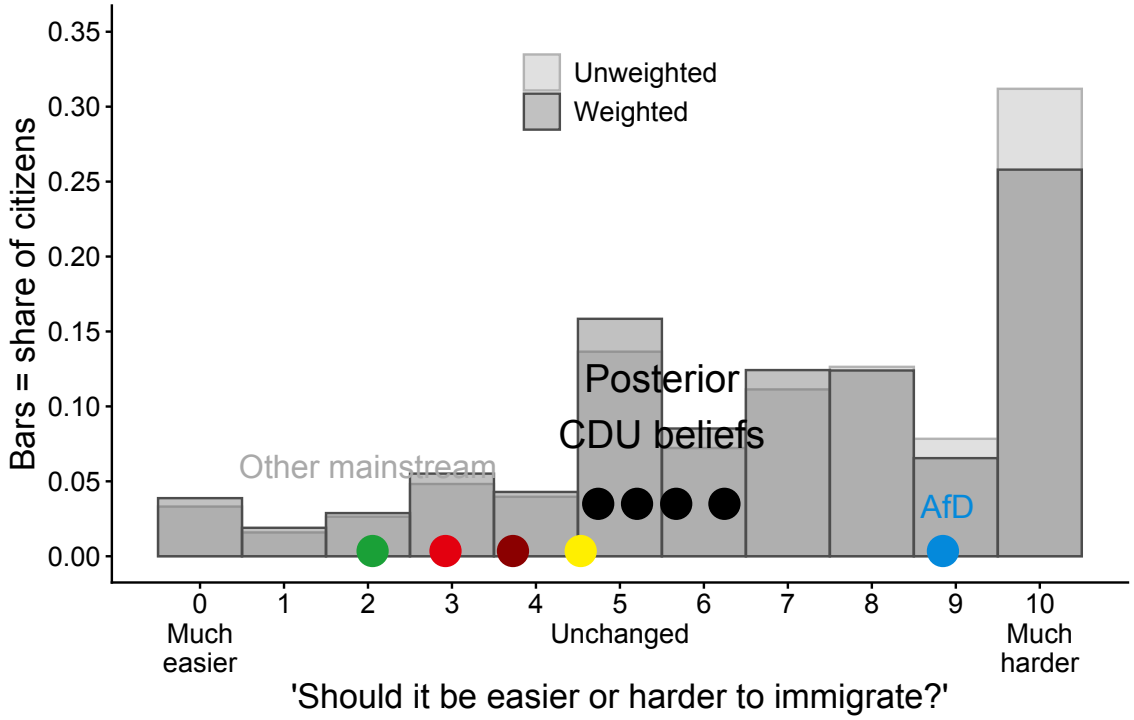


Figure 4: Post-Treatment Beliefs about CDU Immigration Stance by Treatment
Note: Bars show participants' own immigration attitudes. Unweighted bars include the original sample (East Germans over-represented); weighted bars correct for this. Black circles show average post-treatment beliefs about the CDU immigration stance for each treatment. Colored circles show average post-treatment beliefs about other parties, pooling treatments.

how shifting the CDU's perceived position reallocates votes—predictions our design is built to test.

The signal moves these posteriors as intended: participants receiving a signal of 4 update their CDU belief to approximately 4.7 (from a prior of 5.8), while a signal of 7 raises the posterior to 6.2. The first stage is correspondingly strong: a one-unit increase in the signal moves the CDU posterior by approximately 0.5 units, with a first-stage F-statistic of 375, well above conventional thresholds for strong instruments and far stronger than for the representation gap specification ($F \approx 97$; see Appendix H). Table A1 in the Appendix reports the full first-stage regression, and Figure A1 confirms that the signal affects only beliefs about the CDU's position, with no spillovers to other parties.

5.2 Exclusion Restriction

The exclusion restriction requires that the signal affects outcomes only through its effect on beliefs about the CDU's immigration stance. A key threat is *cross-learning*: respondents

may update beliefs other than the one the researcher intends to shift (Haaland et al., 2023). We implemented pre-registered cross-learning checks along four dimensions: (i) whether the signal changes participants' *own* immigration attitudes, (ii) whether it shifts perceptions of *other parties' positions* (especially the AfD), (iii) whether it alters beliefs about *post-election governing coalitions*, and (iv) whether it shifts beliefs about the *electoral support* the parties will receive in the upcoming election. Figure A2 in the Appendix shows no evidence of economically meaningful effects on any of these channels: own attitudes, beliefs about the AfD's position, coalition expectations, and beliefs about the AfD's and CDU's vote shares are all unaffected, and the handful of coefficients that reach conventional significance (perceived distances to three smaller parties and the perceived vote share of the Greens) are an order of magnitude smaller than the first-stage effect on CDU beliefs ($\leq$ 0.03 SD versus 0.08–0.11 SD), supporting the exclusion restriction.[9]

5.3 Main Results

We present three estimators in sequence—reduced form, standard 2SLS, and LLS.

Reduced Form. Figure 5 shows the reduced-form effects of the signal on all outcomes, estimated via equation (4). A signal portraying the CDU as one unit more liberal on immigration significantly increases the AfD thermometer rating and the likelihood of voting for the AfD. Effects on CDU outcomes have the predicted negative sign but are generally not statistically significant. The behavioral outcomes (donation decisions, campaign help, and newsletter sign-ups) have the expected signs but are individually imprecise. The PCA indexes—which aggregate all outcomes—are not statistically significant in the reduced form for either party. These reduced-form estimates are the cleanest causal effects, requiring no assumptions beyond random assignment.

Standard 2SLS. Figure 6 instruments the CDU posterior belief with the signal and estimates the effect of perceiving the CDU as one unit more liberal on immigration. Two outcomes are

[9]These short-run null effects do not imply that parties cannot shape voters' attitudes in the long run. Rather, they support the exclusion restriction in our setting, bolstering confidence that the instrument primarily operates through updating beliefs about where the CDU stands.

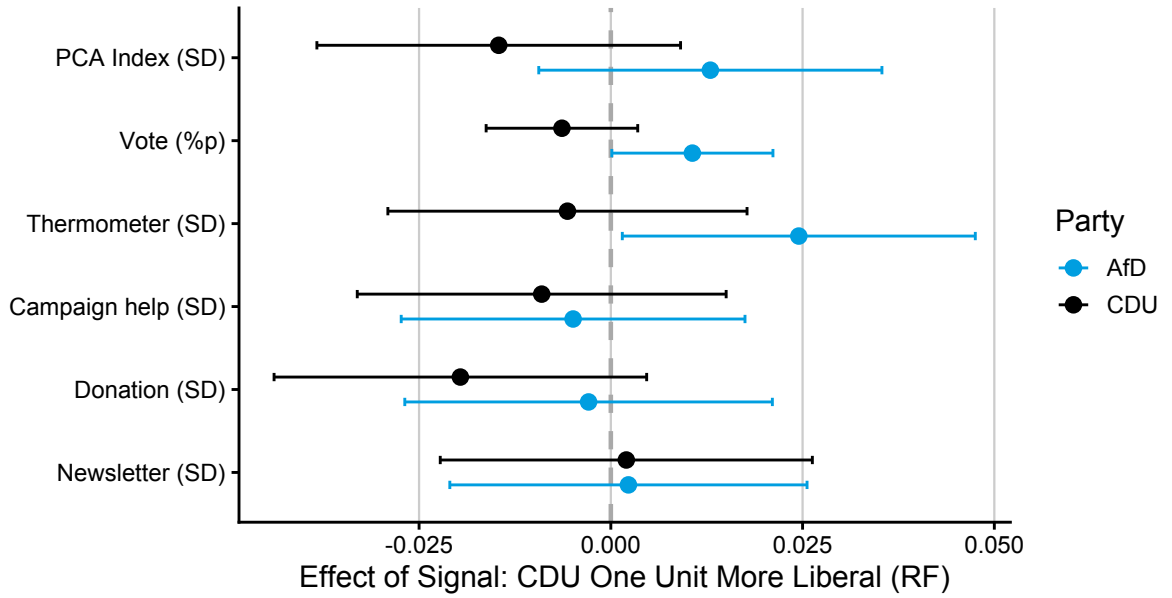


Figure 5: Reduced-Form Effects of the Signal on All Outcomes
Note: OLS estimates. Each coefficient shows the effect of the signal portraying the CDU as one unit more liberal on immigration (i.e., a one-unit decrease in the signal). Bars indicate 95% confidence intervals. Controls: age, gender, immigration background, East Germany indicator, education, and prior-belief fixed effects. N = 5,040.

statistically significant at the 5% level: AfD voting (2.1 percentage points per unit of perceived positioning) and the AfD feeling thermometer (0.049 SD). The remaining AfD outcomes—including the PCA index and the three behavioral measures—carry the predicted signs but are imprecisely estimated, and CDU outcomes are uniformly insignificant, consistent with the asymmetry predicted by Hypothesis 2.[10] Taken at face value, these estimates would suggest that perceived CDU positioning moves votes and feelings toward the AfD but little else. The next paragraphs show that this conclusion would be premature: standard 2SLS is attenuated in our setting.

CAPE Diagnostic. To assess whether 2SLS is attenuated, we examine the Conditional Average Partial Effect (CAPE) curves following Balla-Elliott (2026). The CAPE plots the marginal effect of the CDU posterior on outcomes as a function of each respondent's rank in the distribution of belief updating (the "learning rate"). If the CAPE is flat, all respondents share the same marginal effect and 2SLS is unbiased for the average effect; if it varies with the learning rate,

[10] In our just-identified setting (one instrument, one endogenous variable), the weak-instrument-robust Anderson–Rubin test is numerically equivalent to the reduced-form t-test (Keane and Neal, 2024). Statistical significance of the 2SLS estimates therefore mirrors the reduced form by construction; what 2SLS adds is the economic scaling of the effect per unit of belief change.

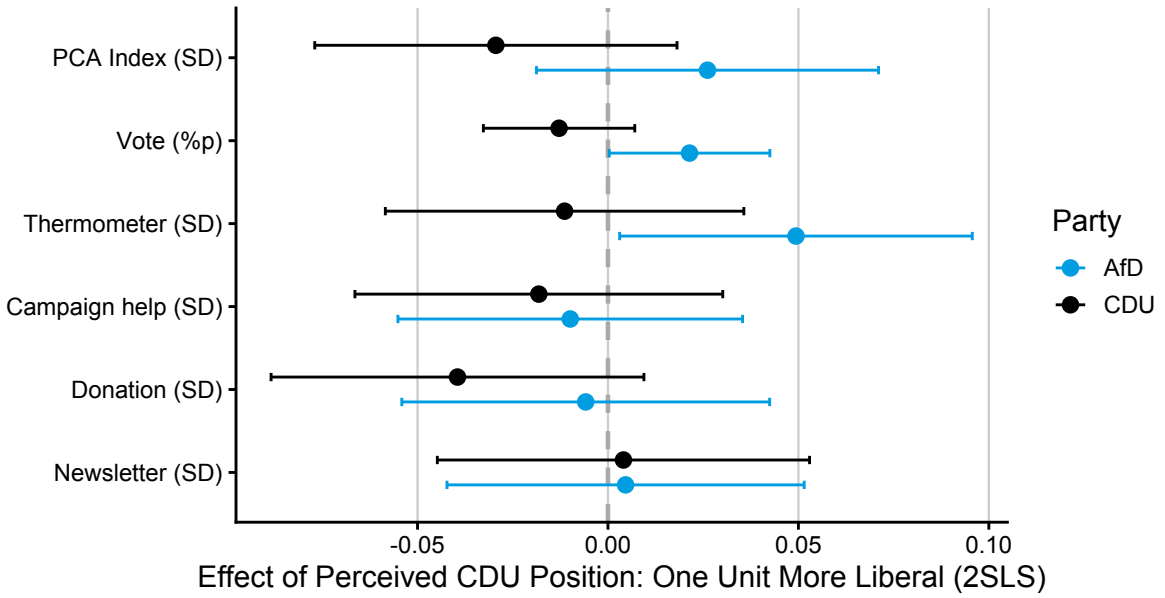


Figure 6: Standard 2SLS Estimates: Effect of Perceived CDU Positioning
Note: 2SLS estimates instrumenting the CDU posterior belief with the randomized signal. Each coefficient shows the effect of perceiving the CDU as one unit more liberal on immigration. Bars indicate 95% confidence intervals. Controls: age, gender, immigration background, East Germany indicator, education, and prior-belief fixed effects. N = 5,040. First-stage F = 375.

2SLS—which weights respondents in proportion to how strongly the instrument moves their beliefs—loads disproportionately on the strong updaters.

Figure 7 plots the CAPE for the AfD PCA index across four bandwidth choices. The curve is far from flat, and its most consequential feature is on the right: marginal effects collapse toward zero among the strongest updaters. Effects are moderate among the weakest updaters and largest in the middle of the learning-rate distribution. Because standard 2SLS places the greatest weight precisely on the strong updaters—the region where the CAPE is smallest—it understates the average effect. The figure overlays the 2SLS estimate and the LLS average: the former tracks the near-zero effects at the top of the learning-rate distribution, while the latter averages over the entire curve. This pattern is consistent with what Balla-Elliott (2026) documents across six other information-provision experiments: respondents whose beliefs most strongly affect their behavior tend to have invested in forming precise priors and consequently update little, whereas respondents who adopt the signal wholesale exhibit the weakest downstream responses.

Table 1 reports formal slope tests for all CAPE curves. Because the CAPE curve is itself estimated, inference must account for the estimation error in $\hat{\tau}(r)$: we re-estimate the CAPE on each of 500 Bayesian bootstrap draws, rerun the slope regression on every draw, and use the

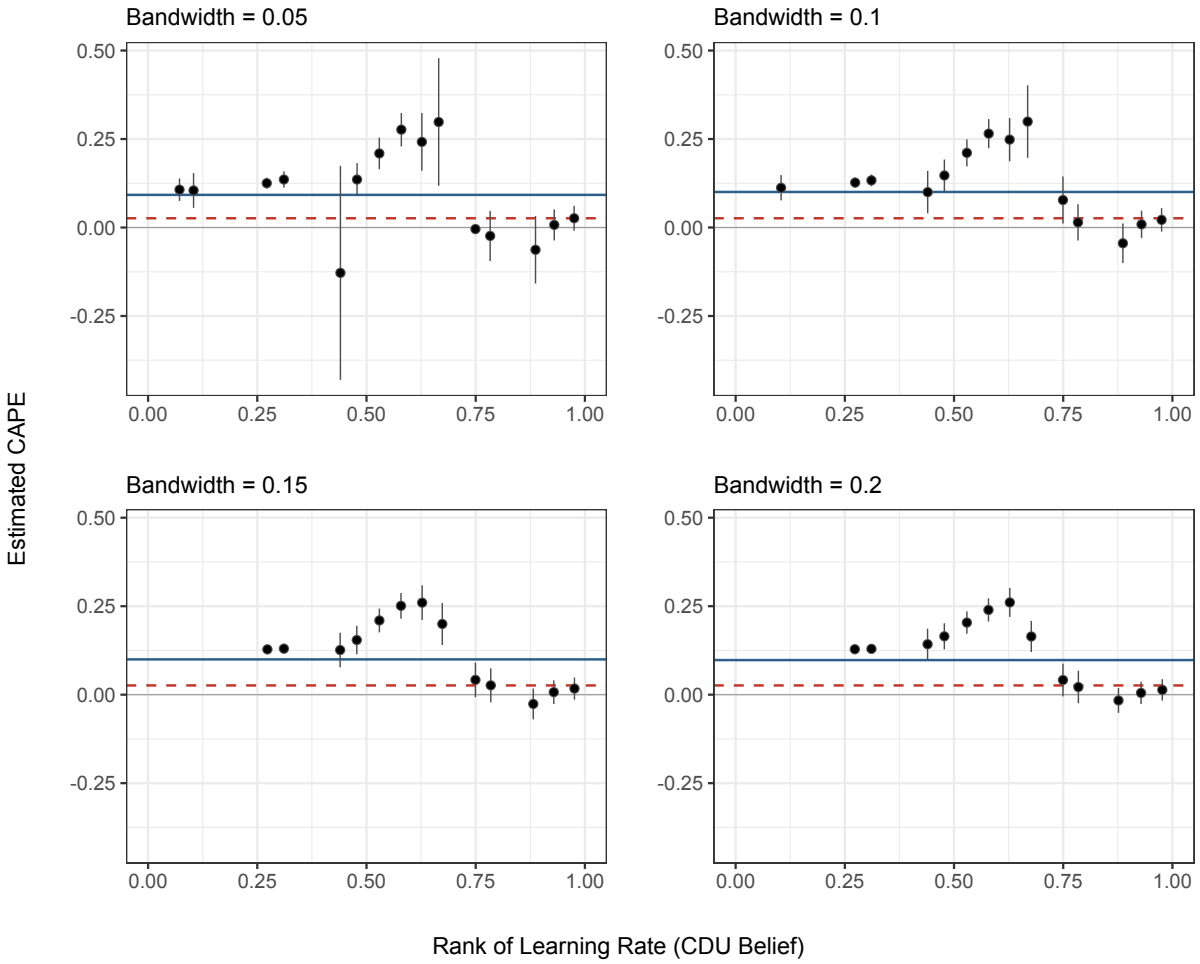


Figure 7: CAPE Estimates for AfD PCA Index

Note: Conditional Average Partial Effects of the CDU posterior belief on the AfD PCA index across four bandwidth choices. The x-axis is the rank of the CDU belief learning rate α_i = (posterior − prior)/(signal − prior), normalized to $[0, 1]$. Non-learners (prior = signal, $N = 572$) are assigned $\alpha_i = 0$, placing them at the bottom of the learning-rate distribution. Estimates are produced using the LLS framework of Balla-Elliott (2026). Points are kernel-weighted averages of the micro-level CAPE estimates within 20 equal-width bins of the learning-rate rank; vertical bars are pointwise 95% confidence intervals from a Bayesian bootstrap with 500 draws (the same draws as in Table 1). In each panel, the dashed red line marks the standard 2SLS estimate and the solid blue line the LLS average: 2SLS tracks the near-zero effects among the strongest updaters, while LLS averages over the entire curve. $N = 5{,}040$.

resulting bootstrap distribution of slope coefficients for standard errors.[11] All ten party-specific outcomes and both PCA indexes reject flatness at the 5% level, and the two indexes at the 1% level. Slopes are negative for AfD outcomes and positive for CDU outcomes; since the AfD effects are positive and the CDU effects negative, both signs indicate marginal effects that shrink toward zero among the strongest updaters. The rejection of flatness is not an artifact of the linear specification: the tercile panel of Table 1 shows that, in absolute value, the CAPE is closest

[11] We thank Dylan Balla-Elliott for suggesting this procedure. The same bootstrap draws underlie the confidence bands in Figure 7 and in the corresponding appendix figures.

Table 1: Formal Slope Tests for CAPE Curves

	Slope Test			CAPE by Learning-Rate Tercile			Middle − Top	
Outcome	Slope	SE	p-value	Bottom	Middle	Top	Diff.	p-value
PCA Indexes								
AfD PCA Index	-0.269***	(0.028)	<0.001	0.128	0.233	0.019	0.214***	<0.001
CDU PCA Index	0.183***	(0.027)	<0.001	-0.107	-0.134	-0.016	-0.117***	<0.001
Individual Outcomes								
AfD Vote	-0.123***	(0.013)	<0.001	0.055	0.111	0.010	0.101***	<0.001
AfD Thermometer	-0.232***	(0.028)	<0.001	0.109	0.203	0.019	0.185***	<0.001
AfD Donation	-0.150***	(0.029)	<0.001	0.067	0.150	0.011	0.138***	<0.001
AfD Campaign	-0.242***	(0.030)	<0.001	0.127	0.180	0.008	0.171***	<0.001
AfD Newsletter	-0.069**	(0.030)	0.022	0.035	0.065	0.014	0.051**	0.022
CDU Vote	0.044***	(0.011)	<0.001	-0.028	-0.046	-0.010	-0.037***	<0.001
CDU Thermometer	0.225***	(0.029)	<0.001	-0.108	-0.150	-0.002	-0.148***	<0.001
CDU Donation	0.080***	(0.029)	0.006	-0.047	-0.031	-0.008	-0.024	0.284
CDU Campaign	0.114***	(0.029)	<0.001	-0.087	-0.087	-0.019	-0.068***	<0.001
CDU Newsletter	0.065***	(0.025)	0.008	-0.035	-0.038	0.001	-0.038*	0.081

Note: The slope panel reports the slope from a weighted OLS regression of the micro-level CAPE estimate $\hat{\tau}(r)$ on the rank r of the learning rate, using LLS kernel weights. The null hypothesis of a zero slope is implied by a flat CAPE, under which 2SLS is unbiased for the average partial effect. The tercile panel reports the kernel-weighted average of $\hat{\tau}(r)$ within terciles of the learning-rate distribution and the difference between the middle and top terciles. Inference is from a Bayesian bootstrap with 500 draws: the CAPE curve is re-estimated on each draw and all statistics are recomputed, so that the estimation error in $\hat{\tau}(r)$ propagates to the tests. Standard errors are the trimmed standard deviations of the bootstrap distributions (trimming the 1% largest absolute deviations); p-values are from normal-approximation z-tests. The endogenous variable is the CDU posterior belief. Non-learners are included with the lowest learning-rate rank. Bandwidth = 0.1. *p< 0.1, **p< 0.05, ***p< 0.01.

to zero in the top tercile for every outcome—the region that 2SLS weights most heavily—and largest in the middle of the learning-rate distribution for eleven of the twelve outcomes; the middle-versus-top difference is significant for ten of the twelve.

The AfD and CDU CAPEs share the same qualitative shape in absolute value, which has a natural interpretation. Voters with strong anti-immigration preferences and entrenched political views update little (Appendix G documents the correlates of learning rates), consistent with holding more precise prior beliefs about the CDU—yet their behavior responds strongly to the beliefs they hold. At the opposite end of the distribution, respondents who adopt the signal wholesale appear to anchor their political behavior—whether toward the AfD or the CDU—only weakly on party positions. The practical implication is the same for all outcomes: 2SLS, which loads on the strong updaters, understates the magnitude of the average effect of perceived CDU positioning on support for both parties.

LLS Results. The non-flat CAPEs motivate the LLS estimator, which weights all respondents equally rather than overweighting high updaters. Figure 8 displays the LLS estimates, and Table 2 reports all three estimators side by side.

Under the LLS estimates, all ten party-specific outcomes and both exploratory summary indexes are statistically significant at the 5% level, and ten of these twelve estimates at the 1% level. Perceiving the CDU as one unit more liberal on immigration increases the AfD PCA index by 0.100 SD, increases AfD voting by 4.7 percentage points, and increases the AfD thermometer by 0.088 SD. For the CDU, the same leftward shift decreases the CDU PCA index by 0.066 SD and CDU voting by 2.3 percentage points. Behavioral outcomes also respond significantly: perceiving the CDU as more liberal increases AfD donations by 0.061 SD, AfD campaign-ad allocations by 0.078 SD, and AfD newsletter sign-ups by 0.033 SD. The LLS estimates are roughly twice as large as the corresponding 2SLS estimates across all outcomes, consistent with the CAPE diagnostic.

The existence of an effect on AfD voting and AfD feelings is established by the reduced form and standard 2SLS alone, and does not depend on the LLS. What the LLS adds is twofold: the average population effect is roughly twice the complier-weighted estimate, and it is precisely estimated also for the behavioral and CDU outcomes, for which the reduced form and 2SLS are too noisy to be informative.

To put these magnitudes in perspective: the AfD received 20.8% of the vote in the 2025 election, and the CDU received 28.5%. In our sample, the LLS estimate implies that a one-unit leftward shift in perceived CDU positioning changes AfD voting by an amount equal to nearly a quarter of that vote share; reweighting the sample to population East/West shares, the average effect of 3.6 percentage points corresponds to roughly a sixth (Subsection 5.4). For reference, by the height of the 2015–16 refugee crisis (mid-2016), the perceived position of the CDU had shifted approximately 1.4 units to the left on the same scale. Applying our population-reweighted estimate of 3.6 percentage points per unit—under the assumption that the per-unit effect is stable over time and persists beyond the one-week horizon we observe—this shift alone could have raised AfD voting by roughly 5 percentage points—of the same order as the AfD's actual rise over the crisis, from about 6% in early 2015 to 12% by late 2016. The relevance of our vote

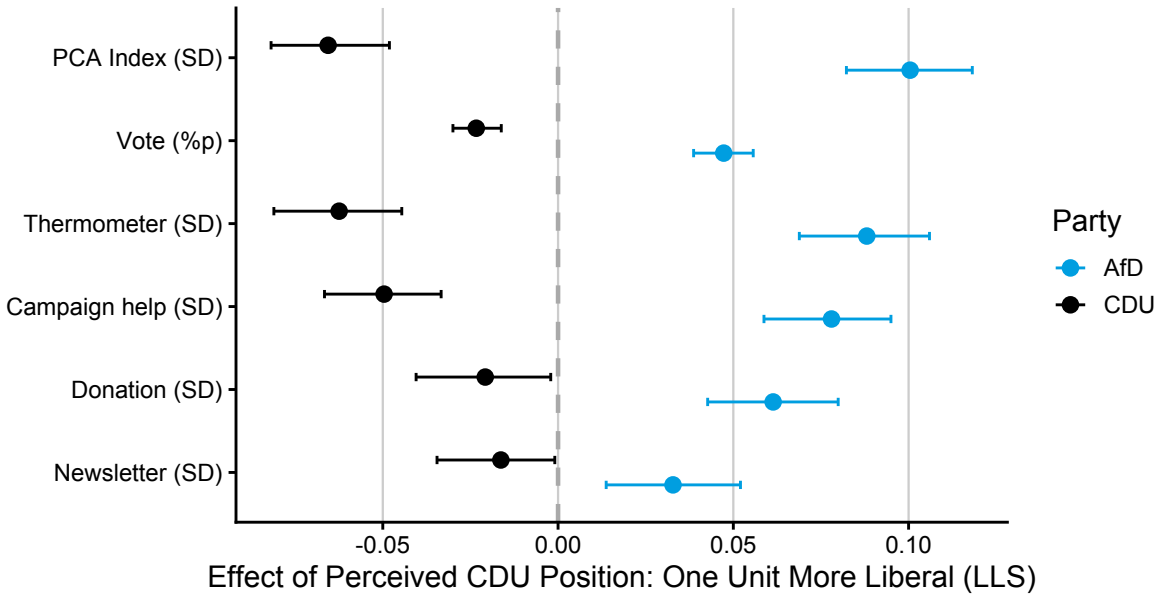


Figure 8: LLS Estimates: Effect of Perceived CDU Positioning
Note: LLS estimates of the effect of the CDU posterior belief on all outcomes. Each coefficient shows the effect of perceiving the CDU as one unit more liberal on immigration. Bars indicate 95% confidence intervals. Controls: age, gender, immigration background, East Germany indicator, education. N = 5,040. Bandwidth = 0.1.

intention measure is supported by the close correspondence between reweighted vote intentions from our sample and the official election results (Figure A39).

The results support the spatial voting hypotheses. Consistent with Hypothesis 1, perceiving the CDU as more liberal increases AfD support, with effects concentrated among voters with conservative immigration attitudes (Subsection 5.4). Consistent with Hypothesis 2, the aggregate response is asymmetric: AfD gains substantially exceed CDU losses. In our setting, the aggregate pattern is inconsistent with a legitimation channel (which would predict AfD gains alongside CDU gains from a rightward shift) and with the protest voting perspective (which predicts null effects). Table A15 in the Appendix confirms no effect on turnout, indicating that shifts in CDU positioning reallocate votes between parties rather than affecting participation.

5.4 Heterogeneous Effects

Figure 9 and Figure 10 examine how the effects differ across subgroups. We consider five dimensions of heterogeneity: four were pre-registered (the perceived importance of immigration, the representation ideal, issue voting, and East vs. West residence), and one—participants'

Table 2: Main Results: Effect of Perceived CDU Positioning on Political Outcomes

	CDU Outcomes			AfD Outcomes		
	RF	2SLS	LLS	RF	2SLS	LLS
PCA Index (SD)	-0.015	-0.029	-0.066***	0.013	0.026	0.100***
	(0.012)	(0.024)	[-0.082, -0.048]	(0.011)	(0.023)	[0.082, 0.118]
Vote (%p)	-0.006	-0.013	-0.023***	0.011**	0.021**	0.047***
	(0.005)	(0.010)	[-0.030, -0.016]	(0.005)	(0.011)	[0.039, 0.056]
Thermometer (SD)	-0.006	-0.011	-0.062***	0.025**	0.049**	0.088***
	(0.012)	(0.024)	[-0.081, -0.045]	(0.012)	(0.024)	[0.069, 0.106]
Campaign help (SD)	-0.009	-0.018	-0.050***	-0.005	-0.010	0.078***
	(0.012)	(0.025)	[-0.067, -0.033]	(0.011)	(0.023)	[0.059, 0.095]
Donation (SD)	-0.020	-0.040	-0.021**	-0.003	-0.006	0.061***
	(0.012)	(0.025)	[-0.040, -0.002]	(0.012)	(0.025)	[0.043, 0.080]
Newsletter (SD)	0.002	0.004	-0.016**	0.002	0.005	0.033***
	(0.012)	(0.025)	[-0.035, -0.001]	(0.012)	(0.024)	[0.014, 0.052]
Observations	5,040	5,040	5,040	5,040	5,040	5,040
First-stage F	—	375.3	—	—	375.3	—
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: RF: reduced-form OLS of outcome on signal. 2SLS: CDU posterior instrumented by signal. LLS: Local Least Squares estimator of Balla-Elliott (2026). All coefficients show the effect of perceiving the CDU as one unit more liberal on immigration. RF and 2SLS report heteroskedasticity-robust standard errors in parentheses; LLS reports 95% Bayesian-bootstrap percentile confidence intervals in brackets, with significance stars based on normal-approximation z-tests using trimmed bootstrap standard errors. Controls: age, gender, immigration background, East Germany indicator, education; RF and 2SLS additionally include prior-belief fixed effects. N = 5,040. *p< 0.1, **p< 0.05, ***p< 0.01.

own immigration attitudes—is motivated directly by the spatial voting model (Hypothesis 1). Regarding the perceived importance of immigration we distinguish between those who find immigration "Important" or "Very Important" (top 2 answer options on a 5-point scale; ≈ 72%) and all other subjects. Concerning participants' own immigration attitudes, we split the sample into three groups: Liberal (0–4), Moderate (5–6), and Conservative (7–10) on the 0–10 immigration scale. Moreover, we distinguish between participants who state that parliamentarians should vote per their own opinion (≈ 41%) vs. follow voters' opinion (≈ 59%). We also differentiate between issue voters (≈ 54%) and those who give any other main reason for their vote choice. Finally, we distinguish between West and East Germany.

Across all subgroups, the effects have the predicted signs: perceiving the CDU as more liberal decreases CDU support and increases AfD support. We perform formal interaction

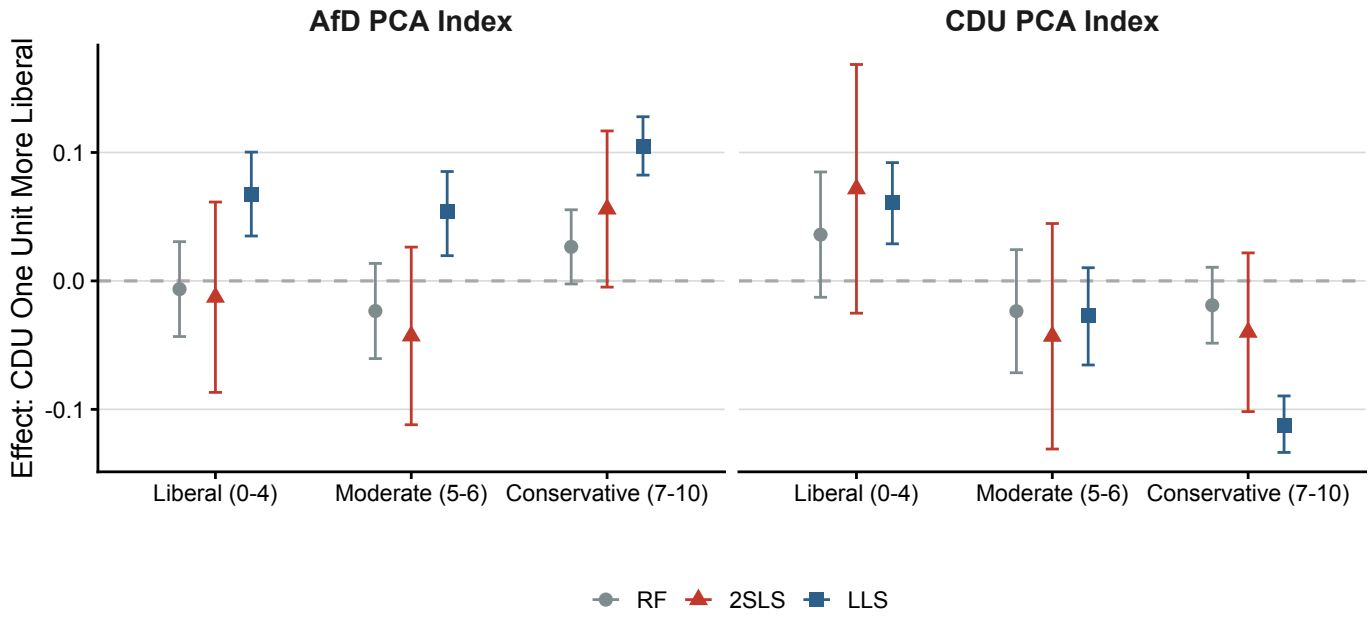


Figure 9: Effects on PCA Indexes by Immigration Attitude, Across Estimators
Note: RF, 2SLS, and LLS estimates of the effect of perceiving the CDU as one unit more liberal on immigration, separately for the three pre-treatment immigration-attitude groups (Liberal 0–4, Moderate 5–6, Conservative 7–10). RF and 2SLS: 95% robust confidence intervals; LLS: 95% bootstrap confidence intervals. Controls: age, gender, immigration background, East Germany indicator, education (RF and 2SLS additionally include prior-belief fixed effects).

tests (z-tests on coefficient differences across independent subsamples) for each dimension of heterogeneity and report the results below. Figure A9 in the Appendix reports the corresponding estimates for all individual outcomes.

Immigration attitudes. Figure 9 shows a clear gradient across immigration-attitude subgroups that holds up across estimators. For the CDU PCA index, the three estimators agree in sign within every group: a more liberal CDU gains support among liberal voters (LLS: 0.061 SD) and loses support among conservatives (LLS: −0.113 SD), with moderates in between (−0.027 SD). This is the signature of spatial voting, and it does not rely on any particular estimator: the Liberal–Conservative difference is marginally significant already in the reduced form and 2SLS ($p \approx 0.06$) and strongly significant under LLS (all three pairwise comparisons $p < 0.001$).

For the AfD PCA index, the estimators divide. Among conservatives—the group for which spatial voting predicts a response—all three agree: the reduced form and 2SLS are positive and marginally significant ($p = 0.07$), and the LLS average effect is 0.104 SD. Among liberals and moderates, by contrast, the reduced form and 2SLS are small, wrong-signed, and not significant

($p > 0.2$), exactly as a pure spatial voting model predicts for voters whose distance to the AfD margin is unaffected; only the LLS average effect is positive and significant for these groups, at the 1% level (0.067 and 0.054 SD). The Moderate–Conservative difference, by contrast, is significant under all three estimators (RF: $p = 0.037$; 2SLS: $p = 0.035$; LLS: $p = 0.011$), so the conclusion that conservatives respond most does not depend on the estimator.

The positive LLS estimate for liberals deserves scrutiny: a liberal voter who learns that the CDU is more liberal on immigration should, if anything, find the CDU closer to their ideal point and have less reason to turn to the far right. The pattern is confined to a single estimator: the reduced-form and 2SLS estimates for liberals are close to zero, and the arm-specific effects in the saturated reduced form show no dose-response pattern among liberals (Figure A35 in the Appendix), in contrast to the monotonic decline among conservatives that spatial voting predicts; we read them as descriptive corroboration of the interaction tests above rather than as a separate formal test. When interpreting the LLS coefficient, it is important to consider that far-right support is also rare in this subgroup: only 9.6% of liberal respondents (79 of 823) intend to vote for the AfD, compared with 43.5% of conservatives, so the liberal estimates rest on a small number of AfD-sympathetic respondents. This matters most for the LLS estimator, which averages belief–outcome slopes estimated within narrow windows of the learning-rate distribution. For liberals, each window contains approximately 74 respondents, of whom roughly seven express any support for the AfD, and in windows drawn from the weakest updaters, the slope is identified off little variation in beliefs, so a small number of respondents can move the local estimate substantially. Random imbalances in cells this small are unavoidable and generate large, erratically signed local slopes whose population-weighted average is poorly determined; standard 2SLS implicitly places little weight on weak updaters, while the LLS returns them to their population weight, which debiases the full-sample estimates but is fragile when the subgroup is small and the outcome rare. Neither condition binds in the full sample nor among conservatives or moderates, where each window contains many respondents and far-right support is common. However, both conditions bind in the case of liberals. Bootstrap inference is least reliable in exactly this configuration, because each resample redraws from the same few AfD-sympathetic respondents, so nominal significance can overstate precision in such

cells. If the estimate is nonetheless taken at face value, we can only think of one mechanism that could produce the pattern: learning that the CDU is more liberal than expected may signal to some voters that mainstream parties are converging on immigration, increasing openness to non-mainstream alternatives. Consistent with the aggregate results, such a channel, if present, would be confined to this small liberal segment.

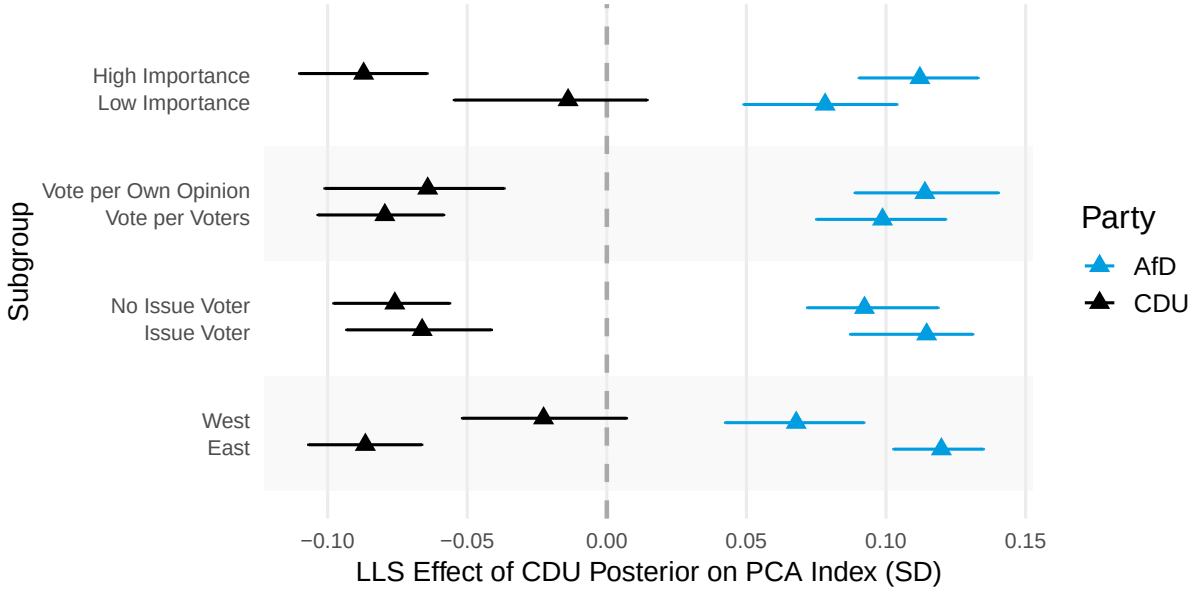


Figure 10: Heterogeneous Effects on PCA Indexes: Pre-Registered Subgroup Dimensions
Note: LLS estimates of the effect of perceiving the CDU as one unit more liberal on the AfD and CDU PCA indexes, for the four pre-registered heterogeneity dimensions (immigration importance, representation ideal, issue voting, East vs. West). Bars indicate 95% bootstrap confidence intervals. Heterogeneity by immigration attitude is shown in Figure 9; individual outcomes are in Figure A17 in the Appendix.

Immigration importance. Effects are stronger for participants who rate immigration as important. For the CDU index, the estimate is −0.087 SD for high-importance participants vs. −0.014 SD for low-importance participants ($p < 0.001$). For the AfD index, the corresponding estimates are 0.112 SD vs. 0.078 SD, a difference that is marginally significant ($p = 0.055$).

East vs. West Germany. Effects are significantly stronger in East Germany for both parties. The CDU index estimate is −0.086 SD in the East vs. −0.023 SD in the West ($p < 0.001$); the AfD index estimate is 0.120 SD in the East vs. 0.068 SD in the West ($p < 0.001$). Because our sample deliberately oversamples the East (60% of observations versus 19% of the German adult population), the pooled estimates weight this high-response region heavily. Reweighting the

sample to population East/West shares, the LLS average effect on AfD voting is 3.6 percentage points ($p < 0.001$; versus 4.7 unweighted), so the estimated effect remains substantial and precisely estimated after correcting for the study's East–West oversampling.

Representation ideal and issue voting. We find no significant differences between participants who think parliamentarians should vote per voters' opinion vs. their own opinion (CDU: $p = 0.443$; AfD: $p = 0.389$), nor between issue voters and non-issue voters (CDU: $p = 0.563$; AfD: $p = 0.173$). While the point estimates go in the expected direction for both dimensions, the differences are modest and imprecisely estimated.

Figure A7 and Figure A8 in the Appendix show that the effects are symmetric regarding the signal's direction and robust to dropping any single treatment arm.

5.4.1 Heterogeneity by Prior Beliefs

The spatial voting model makes a further, sharper prediction (Proposition 2.1, part c): effects should be strongest for voters who initially perceived the CDU as most liberal, for whom a further leftward signal reinforces an already large perceived distance from their conservative preferences. Unlike the model's sign and asymmetry predictions, this implication relies on the convexity of the loss function (Section 2); like the split by own immigration attitudes, it was not pre-registered. Splitting the sample into terciles of prior beliefs about the CDU's position yields estimates consistent with this gradient: the LLS estimate for the AfD PCA index is 0.093 SD in the "Low (liberal)" tercile, compared to 0.039 SD in the "High (conservative)" tercile, with the mirror-image pattern for the CDU PCA index (−0.077 vs. −0.023 SD). The Low–High differences are marginally significant (AfD: $p = 0.075$; CDU: $p = 0.085$), and, as with the behavioral and follow-up outcomes, the gradient is recovered only by the LLS average effect; within terciles the reduced form and 2SLS are too imprecise to detect it (Figure A10 in the Appendix reports all three estimators by tercile). We read this pattern as suggestive evidence on the model's curvature rather than as a confirmed prediction.

Appendix E complements these experimental results with descriptive binscatters of all outcomes against both the posterior belief about the CDU's position and respondents' own

immigration attitudes. The patterns are monotone and consistent with the spatial voting framework: AfD support falls and CDU support rises as the CDU is perceived as more conservative, and AfD support rises monotonically with respondents' own anti-immigration attitudes.

6 Robustness

6.1 Subsample Robustness Checks

To assess the robustness of our main findings, we re-estimate the LLS specification on four subsamples—the first three pre-registered, the fourth not: (a) participants who pass the second attention check (Figure 11(a)), (b) participants who find the signal credible (Figure 11(b)), (c) participants whose posterior beliefs update toward the signal (Figure 11(c)),[12] and (d) including participants classified as speeders (Figure 11(d)). Across all checks, results are highly consistent with the main findings. Figure A11 in the Appendix reports the corresponding reduced-form estimates.

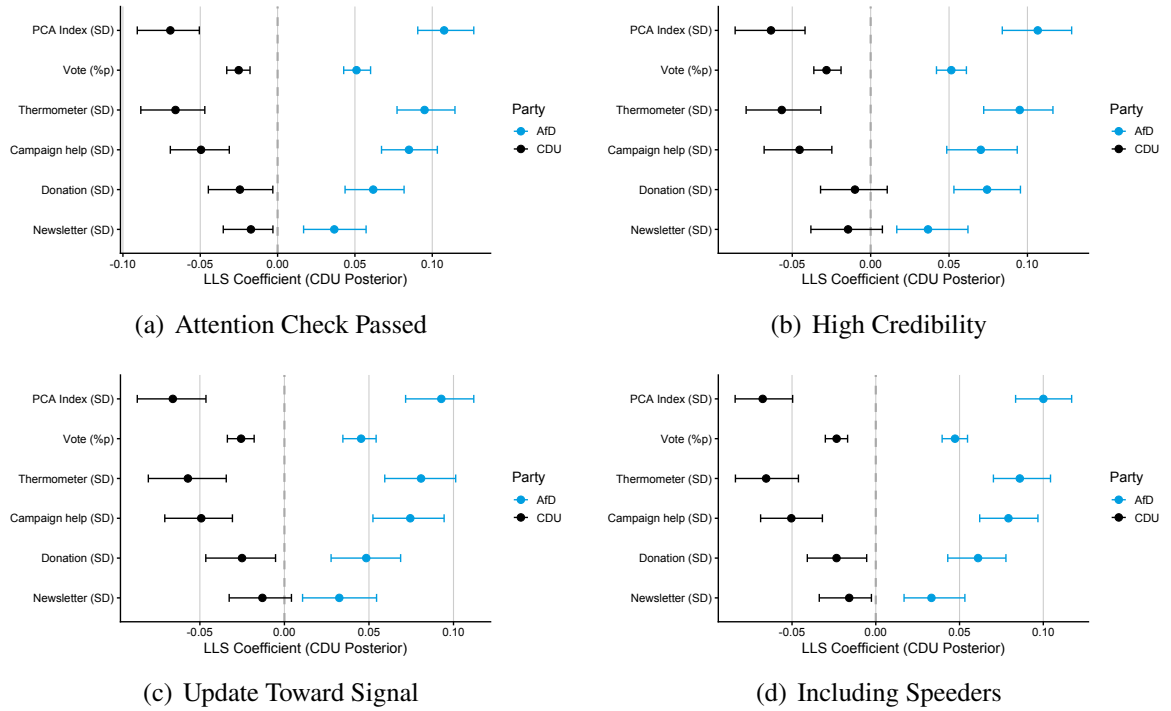
6.2 Bandwidth Sensitivity

The bandwidth parameter in the LLS estimation controls the degree of smoothing. Our main estimates use a bandwidth of 0.1, meaning approximately 10% of observations contribute to each local regression. Figure A25 in the Appendix reports estimates across four bandwidth choices (0.05, 0.1, 0.15, 0.2). The estimates are highly stable, confirming that our results are not driven by the choice of bandwidth.

6.3 Linearity

Our specifications treat the 0–10 position scale linearly. Because the signal takes only four values, we can probe this assumption directly: we estimate a fully saturated reduced form—each

[12] Table A14 in the Appendix compares consistent and inconsistent updaters. The two groups are demographically similar, but inconsistent updaters are substantially more likely to hold objectively incorrect beliefs about party positions—for example, that the AfD is not the most conservative party on immigration—and more likely to fail the second attention check, suggesting lower attentiveness.



(a) Attention Check Passed

(b) High Credibility

(c) Update Toward Signal

(d) Including Speeders

Figure 11: Robustness Checks: LLS Estimates
Note: Each panel shows LLS estimates for all AfD and CDU outcomes on a restricted sample. Bars indicate 95% bootstrap confidence intervals.

outcome on signal-arm indicators with the usual controls—and test whether the arm effects lie on a line. Figure A34 in the Appendix reports the results: for both PCA indexes and both voting outcomes, the arm effects are consistent with linearity (Wald tests; p ranges from 0.49 to 0.98 across the four outcomes, with each outcome's exact p-value reported in the figure). Two features of the design further limit the scope of the linearity assumption. First, the randomized signals span only the 4–7 segment of the scale, so the induced belief variation is concentrated in the interior of the scale, where linear scoring is least problematic; by the same token, our estimates are not informative about hypothetical shifts at the extremes of the scale. Second, the LLS estimator does not impose a marginal effect common to all respondents: it allows the effect of beliefs to vary freely across the learning-rate distribution, imposing a linear belief–outcome relationship only within each narrow learning-rate window.

6.4 Follow-Up Survey

Figure A13 and Table A6 in the Appendix present results from the obfuscated follow-up survey, conducted approximately one week after the main study and concluded the day before the 2025

federal election. Despite the smaller sample (N = 2,182) and the passage of time, the LLS estimates remain significant for all outcomes: perceiving the CDU as one unit more liberal increases AfD voting by 3.7 percentage points and the AfD thermometer by 0.062 SD. The reduced-form and 2SLS estimates are insignificant in the smaller follow-up sample, so the persistence results—like the behavioral outcomes in the main study—rest on the LLS average effects. The persistence of effects in an obfuscated follow-up—where participants were unlikely to connect the two surveys—strengthens our confidence that the results reflect genuine belief updating rather than experimenter demand effects.

A potential concern is the follow-up's 57% attrition rate. Two analyses address it. First, we find no evidence of differential attrition by treatment: a joint test of the signal-arm indicators in a regression of follow-up participation on treatment and controls does not reject the null (p = 0.68). This addresses the primary threat to the internal validity of the follow-up comparisons, though it cannot exclude that the composition of attriters differs across arms. Second, to address representativeness, we re-estimate all follow-up results weighting participants by the inverse of their estimated participation probability, from a logit of participation on age, gender, immigration background, East Germany residence, education, immigration attitudes, and prior beliefs. Table A11 in the Appendix shows that the weighted estimates are nearly identical to the unweighted ones—the LLS effect on AfD voting moves from 3.7 to 3.8 percentage points—indicating that selection on these observables does not drive the follow-up results; selection on unobservables cannot be ruled out.

7 Conclusion

This paper identifies how voters' beliefs about the position of an actual mainstream party affect support for its far-right competitor. In an information-provision experiment conducted during the 2025 German federal election, we randomly shift beliefs about the immigration stance of the CDU, Germany's main center-right party, while holding voters' preferences and information about the other parties fixed.

Perceiving the CDU as more liberal on immigration increases AfD support. Standard 2SLS

implies that a one-unit leftward shift in the perceived CDU position raises AfD vote intention by 2.1 percentage points. Because standard IV overweights the strongest belief updaters, who respond least to their beliefs, the LLS estimator of Balla-Elliott (2026) recovers a larger average population effect of 4.7 percentage points—nearly a quarter of the AfD's vote share in the 2025 election, or roughly a sixth after reweighting to national East–West population shares. The LLS estimates also reveal effects on donation decisions, consequential campaign-ad allocations, and newsletter sign-ups, while effects on vote intention and party feelings persist in an obfuscated follow-up. Responses are concentrated among conservative voters and are asymmetric across parties, as predicted by the location of voters and parties in the immigration-policy space.

The findings have three implications. First, political supply can be a first-order determinant of far-right support: conditional on voters' preferences, support for a far-right party depends on where voters believe its mainstream competitor stands. Second, the electoral consequences of repositioning depend on the distribution of voters and the party's initial location relative to its competitors. Measuring voters and parties on a common scale allows us to identify this distributional mechanism and explain both who responds and why aggregate effects are asymmetric. Third, information-provision experiments with heterogeneous updating require attention to the estimand: standard 2SLS recovers an updater-weighted effect, whereas the LLS estimator of Balla-Elliott (2026) recovers the average population effect of beliefs.

Several limitations point to avenues for future research. Our analysis does not address why so many voters hold restrictive views on immigration, nor why this issue is such a powerful driver of vote choice; our contribution is to show that, when mainstream parties and the electorate diverge on a salient issue, these views have first-order electoral consequences. Moreover, our results are identified on a single policy dimension—immigration—in a single country. Whether similar effects arise for other issues or in other national contexts remains an open question. We study *perceived* positions, but actual policy change involves credibility and commitment problems that our experiment does not capture. A CDU that merely signals conservatism without following through may face voter backlash, and the long-run electoral equilibrium could differ from the short-run effects we estimate. Finally, our analysis abstracts from normative considerations: whether mainstream parties *should* accommodate voter preferences on immigration—particularly

when doing so may conflict with liberal democratic values or international obligations—is a question our framework does not address, but one that policymakers must confront.

References

Abou-Chadi, Tarik and Werner Krause, "The Causal Effect of Radical Right Success on Mainstream Parties' Policy Positions: A Regression Discontinuity Approach," *British Journal of Political Science*, 2020, *50* (3), 829–847.

Adams, James, "Causes and Electoral Consequences of Party Policy Shifts in Multiparty Elections: Theoretical Results and Empirical Evidence," *Annual Review of Political Science*, 2012, *15* (1), 401–419.

Akesson, Jesper, Sam Ashworth-Hayes, Robert Hahn, Robert Metcalfe, and Itzhak Rasooly, "Fatalism, Beliefs, and Behaviors During the COVID-19 Pandemic," *Journal of Risk and Uncertainty*, 2022, *64* (2), 147–190.

Alesina, Alberto, Armando Miano, and Stefanie Stantcheva, "Immigration and Redistribution," *The Review of Economic Studies*, 2023, *90* (1), 1–39.

Algan, Yann, Sergei Guriev, Elias Papaioannou, and Evgenia Passari, "The European Trust Crisis and the Rise of Populism," *Brookings Papers on Economic Activity*, 2017, *2017* (2), 309–400.

Autor, David, David Dorn, Gordon Hanson, and Kaveh Majlesi, "Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure," *American Economic Review*, 2020, *110* (10), 3139–3183.

Bakker, Ryan, Seth Jolly, and Jonathan Polk, "Complexity in the European Party Space: Exploring Dimensionality with Experts," *European Union Politics*, 2012, *13* (2), 219–245.

Balla-Elliott, Dylan, "Identifying Causal Effects in Information Provision Experiments," *Review of Economics and Statistics*, 2026.

Baskaran, Thushyanthan, Zohal Hessami, and Alexander Sohl, "Populist Policymakers," *CESifo Working Paper No. 12199*, 2025.

Becker, Bastian and Anselm Hager, "Do Democratic Parties Lose Support by Cooperating with Far-Right Extremists?," *Unpublished Manuscript*, 2025.

Bonomi, Giampaolo, Nicola Gennaioli, and Guido Tabellini, "Identity, Beliefs, and Political Conflict," *The Quarterly Journal of Economics*, 2021, *136* (4), 2371–2411.

Bottan, Nicolas L and Ricardo Perez-Truglia, "Choosing Your Pond: Location Choices and Relative Income," *Review of Economics and Statistics*, 2022, *104* (5), 1010–1027.

Braghieri, Luca and Sarah Eichmeyer, "Learning from the Past: How History Education Shapes Support for Extreme Ideology," *CEPR Discussion Paper No. 19575*, 2024.

Campante, Filipe, Ruben Durante, and Francesco Sobbrio, "Politics 2.0: The Multifaceted Effect of Broadband Internet on Political Participation," *Journal of the European Economic Association*, 2018, *16* (4), 1094–1136.

Chou, Winston, Rafaela Dancygier, Naoki Egami, and Amaney A. Jamal, "Competing for Loyalists? How Party Positioning Affects Populist Radical Right Voting," *Comparative Political Studies*, 2021, *54* (12), 2226–2260.

Colantone, Italo and Piero Stanig, "Global Competition and Brexit," *American Political Science Review*, 2018, *112* (2), 201–218.

__ **and** __ , "The Trade Origins of Economic Nationalism: Import Competition and Voting Behavior in Western Europe," *American Journal of Political Science*, 2018, *62* (4), 936–953.

Costello, Rory, Jacques Thomassen, and Martin Rosema, "European Parliament Elections and Political Representation: Policy Congruence Between Voters and Parties," in "Assessing Political Representation in Europe," Routledge, 2020, pp. 10–32.

Dai, Yaoyao and Alexander Kustov, "The (In)Effectiveness of Populist Rhetoric: A Conjoint Experiment of Campaign Messaging," *Political Science Research and Methods*, 2024, *12* (4), 849–856.

Dalton, Russell J, "Party Representation Across Multiple Issue Dimensions," *Party Politics*, 2017, *23* (6), 609–622.

Danieli, Oren, Noam Gidron, Shinnosuke Kikuchi, and Ro'ee Levy, "Decomposing the Rise of the Populist Radical Right," *Unpublished Manuscript*, 2024.

Dechezleprêtre, Antoine, Adrien Fabre, Tobias Kruse, Bluebery Planterose, Ana Sanchez Chico, and Stefanie Stantcheva, "Fighting Climate Change: International Attitudes Toward Climate Policies," *American Economic Review*, 2025, *115* (4), 1258–1300.

Downs, Anthony, "An Economic Theory of Political Action in a Democracy," *Journal of Political Economy*, 1957, *65* (2), 135–150.

Falk, Armin, Anke Becker, Thomas Dohmen, Benjamin Enke, David Huffman, and Uwe Sunde, "Global Evidence on Economic Preferences," *The Quarterly Journal of Economics*, 2018, *133* (4), 1645–1692.

Fetzer, Thiemo, "Did Austerity Cause Brexit?," *American Economic Review*, 2019, *109* (11), 3849–3886.

Funke, Manuel, Moritz Schularick, and Christoph Trebesch, "Going to Extremes: Politics After Financial Crises, 1870–2014," *European Economic Review*, 2016, *88*, 227–260.

__ **,** __ **, and** __ , "Populist Leaders and the Economy," *American Economic Review*, 2023, *113* (12), 3249–3288.

Gallego, Maria and Norman Schofield, "Do Parties Converge to the Electoral Mean in All Political Systems?," *Journal of Theoretical Politics*, 2016, *28* (2), 288–330.

GLES, "GLES Tracking 2009–2023, Cumulation," GESIS, Cologne. ZA6832 Data file Version 2.0.0 2023.

Grewenig, Elisabeth, Philipp Lergetporer, Katharina Werner, and Ludger Woessmann, "Do Party Positions Affect the Public's Policy Preferences? Experimental Evidence on Support for Family Policies," *Journal of Economic Behavior & Organization*, 2020, *179*, 523–543.

Grigorieff, Alexis, Christopher Roth, and Diego Ubfal, "Does Information Change Attitudes Toward Immigrants?," *Demography*, 2023, *57* (3), 1117–1143.

Guenther, Laurenz, "Revisiting Immigration Attitudes: How Asylum Seekers Nuance Previous Findings," *SSRN Working Paper No. 5001788*, 2025.

__ , "Political Representation Gaps and Populism," *IAST Working Paper No. hal-05493259*, 2026.

Guriev, Sergei and Elias Papaioannou, "The Political Economy of Populism," *Journal of Economic Literature*, 2023, *60* (3), 753–832.

__ **, Nikita Melnikov, and Ekaterina Zhuravskaya**, "3G Internet and Confidence in Government," *The Quarterly Journal of Economics*, 2021, *136* (4), 2533–2613.

Haaland, Ingar, Christopher Roth, and Johannes Wohlfart, "Designing Information Provision Experiments," *Journal of Economic Literature*, 2023, *61* (1), 3–40.

Hager, Anselm, Lukas Hensel, Johannes Hermle, and Christopher Roth, "Group Size and Protest Mobilization Across Movements and Countermovements," *American Political Science Review*, 2022, *116* (3), 1051–1066.

Henkel, Luca and Christian Pugnaghi-Zimpelmann, "Proud to Not Own Stocks: How Identity Shapes Financial Decisions," *Review of Financial Studies*, 2026.

Heyna, Philipp, Violeta I. Haas, Hanno Hilbig, and Tim Wappenhans, "Electoral Responses to Local Cooperations with Radical-Right Parties," *Unpublished Manuscript*, 2026.

Hjorth, Frederik and Martin Vinæs Larsen, "When Does Accommodation Work? Electoral Effects of Mainstream Left Position Taking on Immigration," *British Journal of Political Science*, 2022, *52* (2), 949–957.

Hooghe, Liesbet and Gary Marks, "Cleavage Theory Meets Europe's Crises: Lipset, Rokkan, and the Transnational Cleavage," *Journal of European Public Policy*, 2018, *25* (1), 109–135.

Hoppmann, Eike, Simon Haas, and Olivia Meyer, "Germany's Federal Election Results Reveal Strong Voter Shifts," *Neue Zürcher Zeitung*, 2025.

Imbens, Guido W. and Joshua D. Angrist, "Identification and Estimation of Local Average Treatment Effects," *Econometrica*, 1994, *62* (2), 467–475.

Jäger, Simon, Christopher Roth, Nina Roussille, and Benjamin Schoefer, "Worker Beliefs about Outside Options," *The Quarterly Journal of Economics*, 2024, *139* (3), 1505–1556.

Keane, Michael P. and Timothy Neal, "A Practical Guide to Weak Instruments," *Annual Review of Economics*, 2024, *16*, 185–212.

Lee, David S, Enrico Moretti, and Matthew J Butler, "Do Voters Affect or Elect Policies? Evidence From the US House," *The Quarterly Journal of Economics*, 2004, *119* (3), 807–859.

Liberini, Federica, Michela Redoano, Antonio Russo, Angel Cuevas, and Ruben Cuevas, "Politics in the Facebook Era. Evidence from the 2016 US Presidential Elections," *European Journal of Political Economy*, 2025, *87*, 102641.

Link, Sebastian, Andreas Peichl, Christopher Roth, and Johannes Wohlfart, "Information Frictions Among Firms and Households," *Journal of Monetary Economics*, 2023, *135*, 99–115.

Mudde, Cas and Cristobal Rovira Kaltwasser, *Populism: A Very Short Introduction*, Oxford University Press, 2017.

Nyhan, Brendan, Ethan Porter, and Thomas J Wood, "Time and Skeptical Opinion Content Erode the Effects of

Science Coverage on Climate Beliefs and Attitudes," *Proceedings of the National Academy of Sciences*, 2022, *119* (26), e2122069119.

Olbrich, Elisabeth and Sven Banisch, "The Rise of Populism and the Reconfiguration of the German Political Space," *Frontiers in Big Data*, 2021, *4*.

Otjes, Simon and Christoffer Green-Pedersen, "Why Is Immigration Important to You? A Revisit to Public Issue Salience and Elite Cues," *European Journal of Political Research*, 2024.

Persson, Torsten and Guido Tabellini, *Political Economics: Explaining Economic Policy*, MIT Press, 2002.

Roth, Christopher and Johannes Wohlfart, "How Do Expectations About the Macroeconomy Affect Personal Expectations and Behavior?," *Review of Economics and Statistics*, 2020, *102* (4), 731–748.

__ **, Sonja Settele, and Johannes Wohlfart**, "Risk Exposure and Acquisition of Macroeconomic Information," *American Economic Review: Insights*, 2022, *4* (1), 34–53.

Schaub, Max and Davide Morisi, "The Enduring Effect of Immigration Attitudes on Vote Choice in Germany," *German Politics and Society*, 2022, *40* (4), 62–87.

Schofield, Norman, "A Valence Model of Political Competition in Britain: 1992–1997," *Electoral Studies*, 2005, *24* (3), 347–370.

__ **and Alexei Zakharov**, "A Stochastic Model of the 2007 Russian Duma Election," *Public Choice*, 2010, *142* (1–2), 177–194.

Silva, Bruno Castanho and Christopher Wratil, "Do Parties' Representation Failures Affect Populist Attitudes? Evidence from a Multinational Survey Experiment," *Political Science Research and Methods*, 2023, *11* (2), 347–362.

Stantcheva, Stefanie, "How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible," *Annual Review of Economics*, 2023, *15* (1), 205–234.

Tella, Rafael Di, Randy Kotti, Caroline Le Pennec, and Vincent Pons, "Keep Your Enemies Closer: Strategic Platform Adjustments During US and French Elections," *American Economic Review*, 2025, *115* (8), 2488–2528.

Turnbull-Dugarte, Stuart J., Jack Bailey, Daniel Devine, Zachary Dickson, Sara B. Hobolt, Will Jennings, Robert Johns, and Katharina Lawall, "The Electoral Risks of Accommodating the Radical Right: Evidence from the British Labour Government," *SocArXiv Preprint*, 2026.

Vasilopoulou, Sofia and Katjana Gattermann, "Matching Policy Preferences: The Linkage Between Voters and MEPs," *Journal of European Public Policy*, 2013, *20* (4), 606–625.

Walczak, Agnieszka and Wouter Van der Brug, "Representation in the European Parliament: Factors Affecting the Attitude Congruence of Voters and Candidates in the EP Elections," *European Union Politics*, 2013, *14* (1), 3–22.

Zhuravskaya, Ekaterina, Maria Petrova, and Ruben Enikolopov, "Political Effects of the Internet and Social Media," *Annual Review of Economics*, 2020, *12* (1), 415–438.

Appendix

Contents

A. Manipulation Checks

B. Party Competition in the German Policy Space

C. Regression Tables

D. Heterogeneity

E. Descriptive Binscatters

F. CAPE Analysis: Individual Outcomes

G. Learning Rate Correlates

H. Pre-Registered Specification: Perceived Representation Gap

I. Alternative Outcome Indexes

J. Supplementary Tables and Figures

K. Questionnaire (Translated from German)

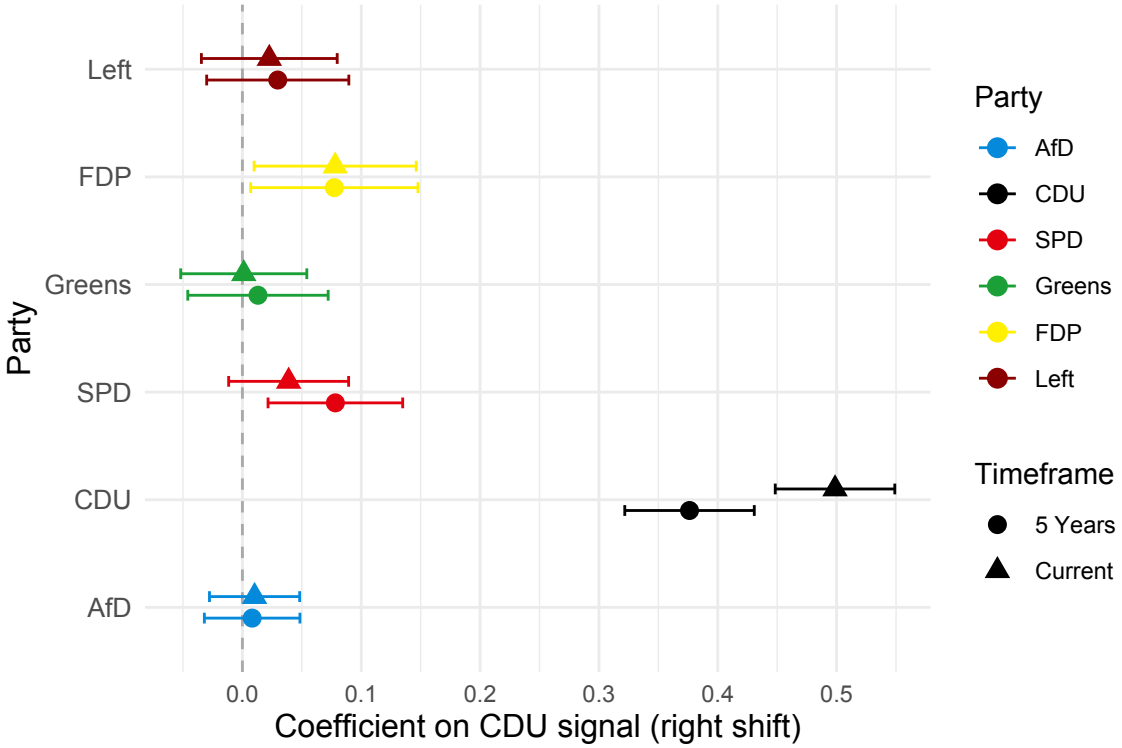


Figure A1: Treatment Effects on Beliefs about Parties' Immigration Stances

A Manipulation Checks

A.1 Posterior Beliefs

Table A1: First Stage: Effect of Signal on CDU Posterior Belief

	CDU Posterior Belief
Signal (CDU position)	0.497***
	(0.026)
F-statistic	375.3
Observations	5,040
Controls	Yes

Note: OLS regression of the CDU posterior belief on the randomly assigned signal, both coded so that higher values mean a more *liberal* immigration stance (the uniform sign convention used throughout the paper). Controls: age, gender, immigration background, East Germany indicator, education. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

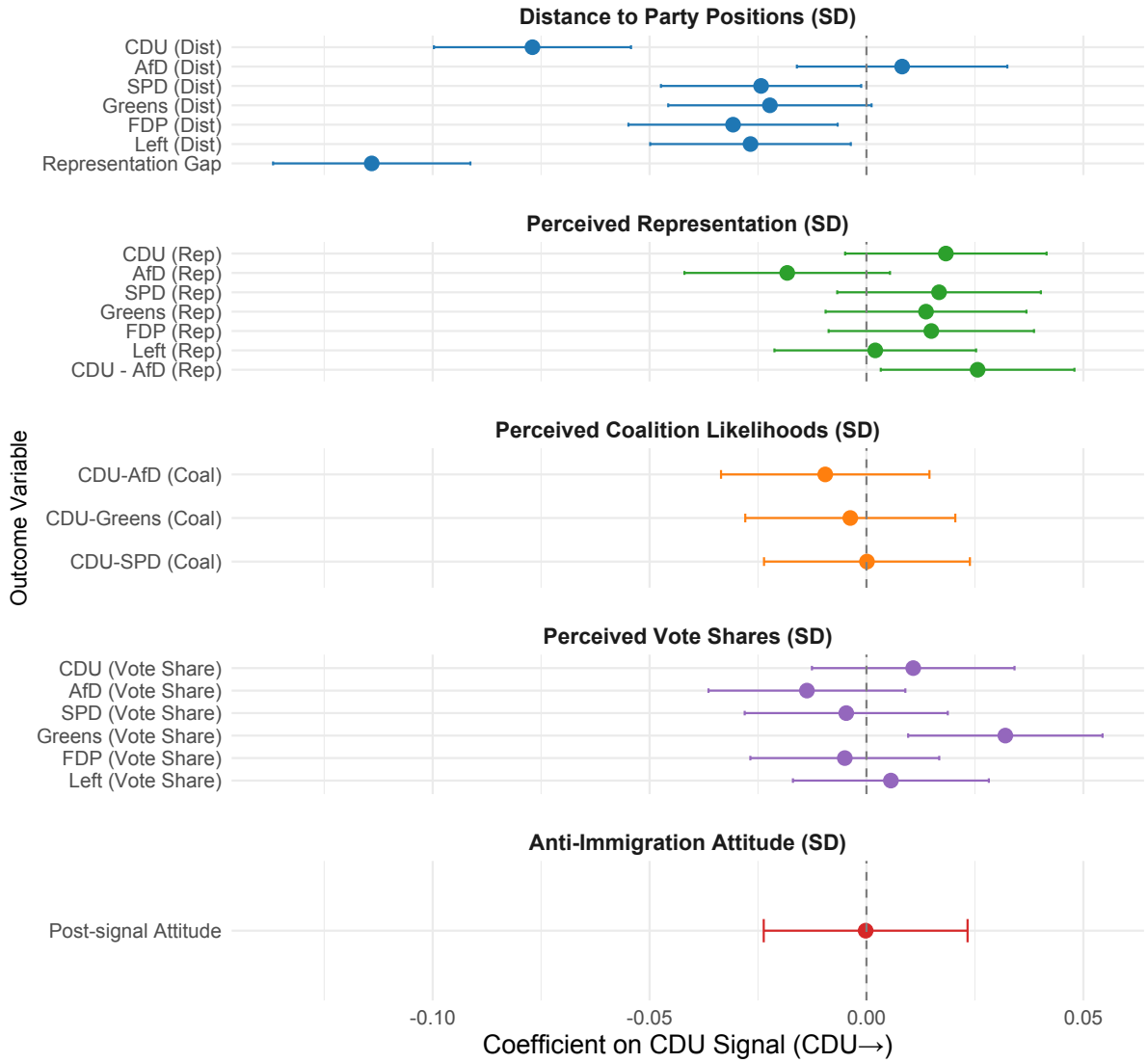


Figure A2: Effects of Treatment (RF) on Channel Variables
Note: Bars indicate 95% confidence intervals. Signal takes values 4, 5, 6, 7. Based on equation (4). N = 5,040.

A.2 Cross-Learning

A.3 Instrument Relevance and First Stage Heterogeneity

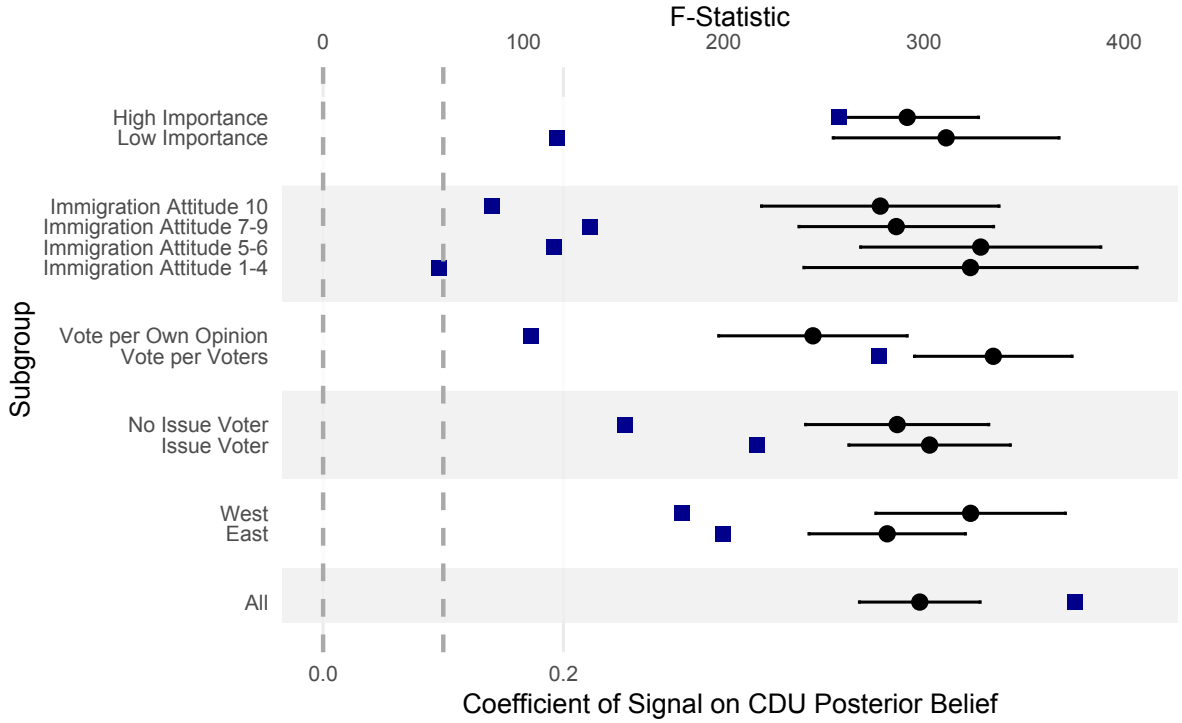


Figure A3: Effect of the Signal on the CDU Posterior Belief and F-Statistics by Subgroup
Note: Black dots: OLS coefficients of the signal on the CDU posterior belief (0–10 scale), with 95% confidence intervals. Blue squares: F-statistics (right axis). All regressions include demographic controls (age, gender, immigration background, East Germany, education).

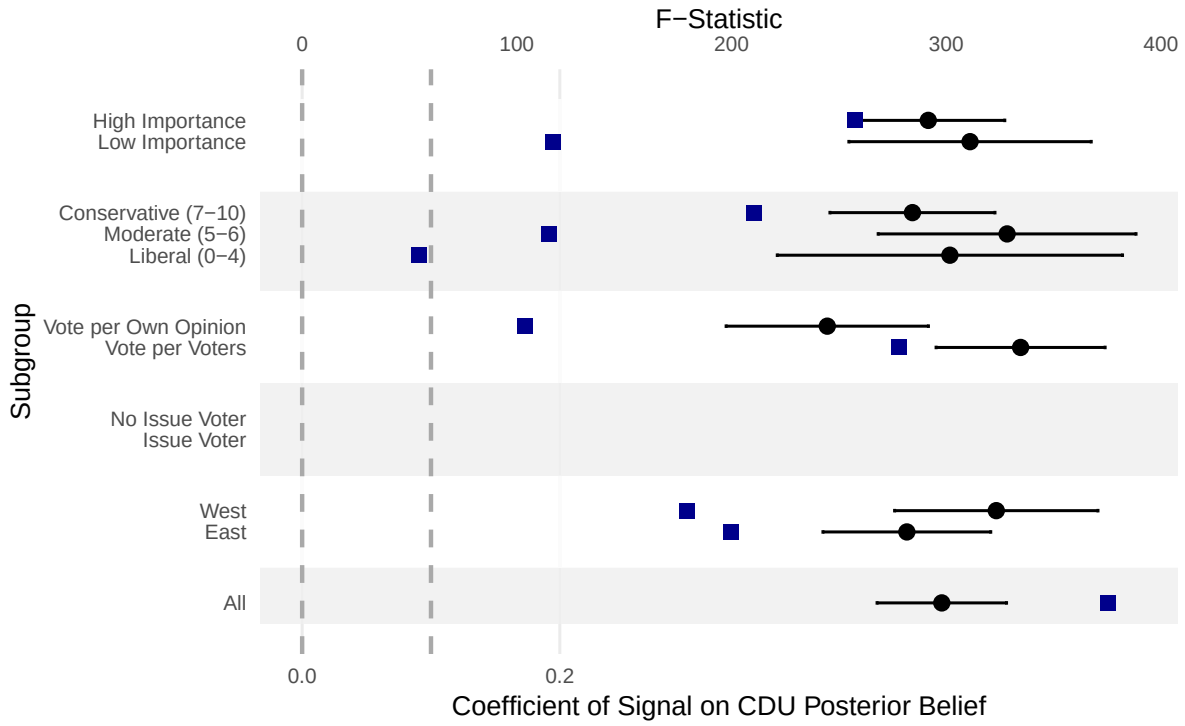


Figure A4: Effect of the Signal on the CDU Posterior Belief and F-Statistics by Subgroup (3-Bin Immigration Attitudes)

Note: Same as the previous figure, but with immigration attitudes grouped into three bins (Liberal: 0–4, Moderate: 5–6, Conservative: 7–10) matching the categorization used in the main text heterogeneity analysis.

B Party Competition in the German Policy Space

This appendix applies the spatial perspective of Section 2 descriptively to the German policy space. Using data from the German Longitudinal Election Study (GLES, 2023)—which contains responses from 52,341 German citizens across 48 representative surveys between 2009 and 2023—we track the two-dimensional policy space over four political periods. The exercise has three descriptive objectives: to examine whether the AfD occupies a distinct region of the policy space, whether its separation from the mainstream parties appears primarily along the immigration dimension, and whether the CDU borders the part of the policy space occupied by AfD voters. These patterns are qualitatively consistent with the framework of Section 2, though the exercise is descriptive and cannot establish causality—that is the task of our experiment.

Data and Measurement. Prior research has found that the policy space in Germany and many other European countries is well-characterized by a two-dimensional structure with one economic and one non-economic "cultural" dimension (Bakker et al., 2012; Bonomi et al., 2021; Danieli et al., 2024; Guenther, 2026; Hooghe and Marks, 2018; Olbrich and Banisch, 2021). Regarding the cultural dimension, the most important issue is immigration, which voters in Germany usually consider to be the most important issue in general (Guenther, 2026; Otjes and Green-Pedersen, 2024; Schaub and Morisi, 2022). We use one survey item to measure the position on each dimension. For the economic dimension, we use an item that asks subjects whether they prefer lower taxes and social security contributions, even if this means fewer welfare benefits, or vice versa. We label this dimension "redistribution" for short. For the cultural dimension, we use an item that asks participants whether immigration opportunities should be made easier or restricted. For both items, subjects rated their own opinion on a balanced 11-point scale. They also rated all major parties on the same scale, which enables comparisons between where voters are and where they **think** parties are. We also use an item that asks participants how important they find each of these issues, elicited on a five-point scale, ranging from "Very important" to "Completely unimportant." In addition, we use "scalometer" items that ask subjects on an 11-point scale how positive or negative their views of each party are in general. Finally, we use subjects' responses on their intention to vote or abstain, and which party they would vote for.

Not every GLES wave fields the full battery of items this exercise requires. We therefore restrict the sample to respondents with valid answers on both issue-importance items and on their own position and the perceived position of every included party on both dimensions, with a valid CDU/CSU placement, non-missing survey weights, and either a reported vote for one of the included parties or a declared abstention. These item-availability and complete-case restrictions reduce the sample from 52,341 respondents to the 12,391 observations underlying Figure A5 and Figure A6; the figure notes report observation counts by period.

Nearest-Party Assignment. We reconstruct four political periods: (a) the period before the AfD was founded (2009–2013), (b) the years after its foundation but before the refugee crisis (2013–2015), (c) the years of the refugee crisis (2015–2016), and (d) the time after the refugee crisis (2017–2023). We ask how the geography of party competition changed as the AfD entered this space. For transparency, we use a deliberately stripped-down assignment rule: each respondent is assigned to the party closest to them in weighted policy space, where the weights reflect the relative importance the individual places on the two issues. This deterministic, two-dimensional rule is not the model of Section 2—it abstracts from valence and idiosyncratic preference shocks—and we use it descriptively, to characterize which party is best positioned to attract each voter, not to predict individual votes.

Results. Figure A5 shows the German 2D policy space and visualizes the attitudes of citizens, the perceived positions of parties, and the party assignments implied by the nearest-party rule. Figure A6 shows the same policy space with actual voting behavior. Three descriptive patterns emerge. First, once the AfD enters, the nearest-party rule assigns it a large and distinct region of the policy space—concentrated among immigration-conservative voters—and actual AfD voters cluster in the same region. Second, the visual separation between the AfD and the mainstream parties occurs primarily along the immigration dimension rather than the economic dimension. Third, the CDU borders the AfD's region: it is the spatially closest mainstream party for most voters near the AfD. These patterns are qualitatively consistent with the spatial framework, but they establish neither predictive accuracy at the individual level nor causality—the experiment in the main text addresses the latter.

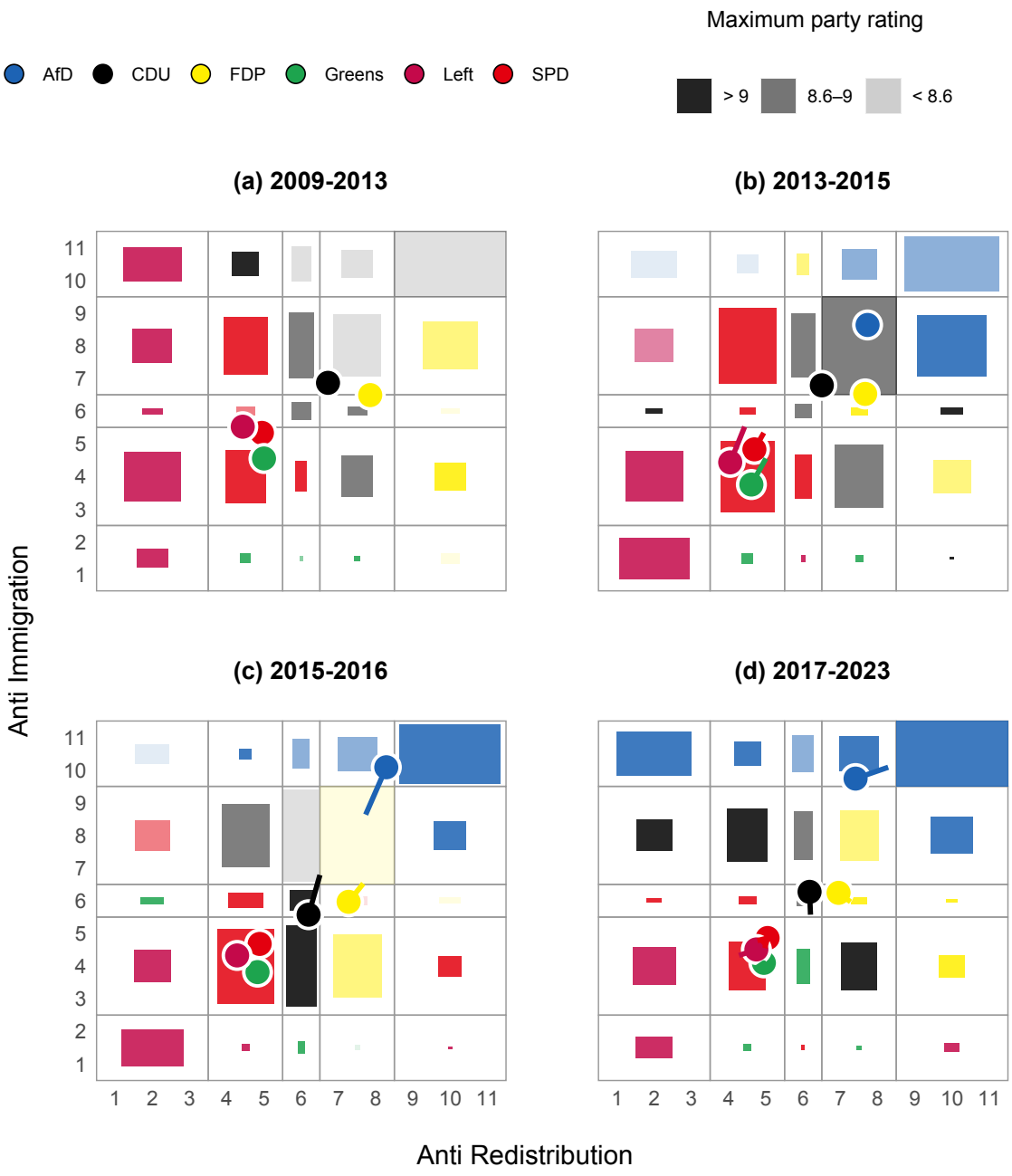


Figure A5: German Policy Space over Time: Nearest-Party Assignment
Note: Distribution of citizens and perceived party positions in two-dimensional policy space. Colors indicate the party selected by the nearest-party assignment rule described in the text; they are not predictions from the model in Section 2. Points show average perceived party positions. Rectangle area is proportional to voter share. The numbers of observations in panels a, b, c, and d are 2,410, 680, 415, and 8,886, respectively.



Figure A6: German Policy Space over Time: Actual Voting and Turnout
Note: Same as Figure A5 but colors indicate actual voting behavior rather than the nearest-party assignment.

C Regression Tables

Table A2: Effects of the Signal on AfD Outcome Variables (RF)

	Dependent variable:								
	Main Study						Follow-Up Study		
	Index	Voting	Thermometer	Campaign	Donation	Newsletter	Index	Voting	Thermometer
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
-Signal (4–7)	0.013	0.011**	0.099**	−0.005	−0.003	0.002	−0.004	0.001	−0.009
	(0.011)	(0.005)	(0.047)	(0.011)	(0.012)	(0.012)	(0.018)	(0.008)	(0.018)
Age	−0.003***	−0.001**	−0.011***	−0.002**	−0.006***	0.006***	−0.002	−0.001*	−0.002
	(0.001)	(0.0004)	(0.004)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Male	0.178***	0.063***	0.423***	0.134***	0.216***	0.105***	0.115***	0.048***	0.112***
	(0.026)	(0.012)	(0.106)	(0.026)	(0.028)	(0.028)	(0.040)	(0.018)	(0.040)
Immigrant	−0.113***	−0.082***	−0.325**	−0.111***	−0.007	0.009	−0.163***	−0.085***	−0.122**
	(0.038)	(0.018)	(0.159)	(0.040)	(0.043)	(0.038)	(0.054)	(0.026)	(0.056)
East	0.300***	0.144***	1.226***	0.243***	0.156***	0.006	0.262***	0.115***	0.243***
	(0.026)	(0.012)	(0.108)	(0.026)	(0.028)	(0.029)	(0.040)	(0.018)	(0.041)
Education	0.171***	0.077***	0.700***	0.161***	0.023	0.095***	0.195***	0.082***	0.189***
	(0.028)	(0.013)	(0.119)	(0.028)	(0.031)	(0.029)	(0.043)	(0.020)	(0.045)
Constant	0.683***	0.625***	1.603***	0.667***	0.478***	−0.133	0.551***	0.587***	0.430***
	(0.093)	(0.043)	(0.379)	(0.101)	(0.097)	(0.100)	(0.147)	(0.067)	(0.150)
Prior beliefs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	5,040	5,040	5,040	5,040	5,040	5,040	2,182	2,182	2,182
R^2	0.180	0.175	0.141	0.159	0.069	0.022	0.171	0.169	0.136

Note: For the Signal, we estimate the effect of a unit decrease, i.e., displaying the CDU as more liberal on immigration. Index is the PCA Index of all AfD variables. Age is measured in years. Male, University Degree, and East (Germany) are indicators. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table A3: Effects of the Signal on CDU Outcome Variables (RF)

	Dependent variable:								
	Main Study						Follow-Up Study		
	Index	Voting	Thermometer	Campaign	Donation	Newsletter	Index	Voting	Thermometer
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
-Signal (4–7)	−0.015	−0.006	−0.018	−0.009	−0.020	0.002	−0.009	−0.011	0.009
	(0.012)	(0.005)	(0.038)	(0.012)	(0.012)	(0.012)	(0.019)	(0.008)	(0.018)
Age	0.003***	0.001	0.003	0.006***	−0.005***	0.006***	0.005***	0.002***	0.003**
	(0.001)	(0.0004)	(0.003)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)
Male	0.043	0.025**	0.002	0.018	0.068**	−0.001	0.124***	0.045***	0.113***
	(0.027)	(0.011)	(0.085)	(0.027)	(0.028)	(0.029)	(0.042)	(0.017)	(0.042)
Immigrant	0.109**	0.024	0.369***	0.071	0.047	0.089*	0.101	0.044	0.074
	(0.042)	(0.018)	(0.133)	(0.045)	(0.043)	(0.048)	(0.065)	(0.028)	(0.062)
East	−0.186***	−0.068***	−0.483***	−0.088***	−0.162***	−0.055*	−0.166***	−0.078***	−0.108**
	(0.029)	(0.012)	(0.088)	(0.029)	(0.029)	(0.030)	(0.043)	(0.018)	(0.042)
Education	−0.0001	0.026*	−0.047	0.040	−0.117***	−0.022	−0.016	−0.011	−0.003
	(0.032)	(0.013)	(0.097)	(0.032)	(0.031)	(0.035)	(0.049)	(0.021)	(0.048)
Constant	−0.827***	0.0003	−3.526***	−0.805***	−0.119	−0.370***	−0.847***	−0.064	−0.827***
	(0.094)	(0.037)	(0.298)	(0.098)	(0.099)	(0.093)	(0.142)	(0.058)	(0.149)
Prior beliefs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	5,040	5,040	5,040	5,040	5,040	5,040	2,182	2,182	2,182
R^2	0.099	0.049	0.127	0.052	0.048	0.018	0.081	0.053	0.086

Note: For the Signal, we estimate the effect of a unit decrease, i.e., displaying the CDU as more liberal on immigration. Index is the PCA Index of all CDU variables. Age is measured in years. Male, University Degree, and East (Germany) are indicators. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table A4: Effects on AfD Outcomes: Reduced Form and Standard 2SLS

	Dependent variable:					
	Index	Voting	Thermometer	Campaign	Donation	Newsletter
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Reduced Form (OLS)						
Signal (CDU ←)	0.013	0.011**	0.025**	−0.005	−0.003	0.002
	(0.011)	(0.005)	(0.012)	(0.011)	(0.012)	(0.012)
Panel B: Standard 2SLS						
CDU Posterior (→)	0.026	0.021**	0.049**	−0.010	−0.006	0.005
	(0.023)	(0.011)	(0.024)	(0.023)	(0.025)	(0.024)
Controls	✓	✓	✓	✓	✓	✓
Prior beliefs FE	✓	✓	✓	✓	✓	✓
First-stage F			375.3			
Observations			5,040			

Note: Panel A reports the reduced-form effect of the CDU signal (more liberal direction). Panel B reports 2SLS estimates where the endogenous variable is the posterior belief about CDU immigration stance (more conservative direction), instrumented by the randomized signal. Controls include age, gender, immigration background, East Germany indicator, and education. Prior beliefs are absorbed via fixed effects. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table A5: Effects on CDU Outcomes: Reduced Form and Standard 2SLS

	Dependent variable:					
	Index	Voting	Thermometer	Campaign	Donation	Newsletter
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Reduced Form (OLS)						
Signal (CDU ←)	−0.015	−0.006	−0.006	−0.009	−0.020	0.002
	(0.012)	(0.005)	(0.012)	(0.012)	(0.012)	(0.012)
Panel B: Standard 2SLS						
CDU Posterior (→)	−0.029	−0.013	−0.011	−0.018	−0.040	0.004
	(0.024)	(0.010)	(0.024)	(0.025)	(0.025)	(0.025)
Controls	✓	✓	✓	✓	✓	✓
Prior beliefs FE	✓	✓	✓	✓	✓	✓
First-stage F				375.3		
Observations				5,040		

Note: Panel A reports the reduced-form effect of the CDU signal (more liberal direction). Panel B reports 2SLS estimates where the endogenous variable is the posterior belief about CDU immigration stance (more conservative direction), instrumented by the randomized signal. Controls include age, gender, immigration background, East Germany indicator, and education. Prior beliefs are absorbed via fixed effects. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

D Heterogeneity

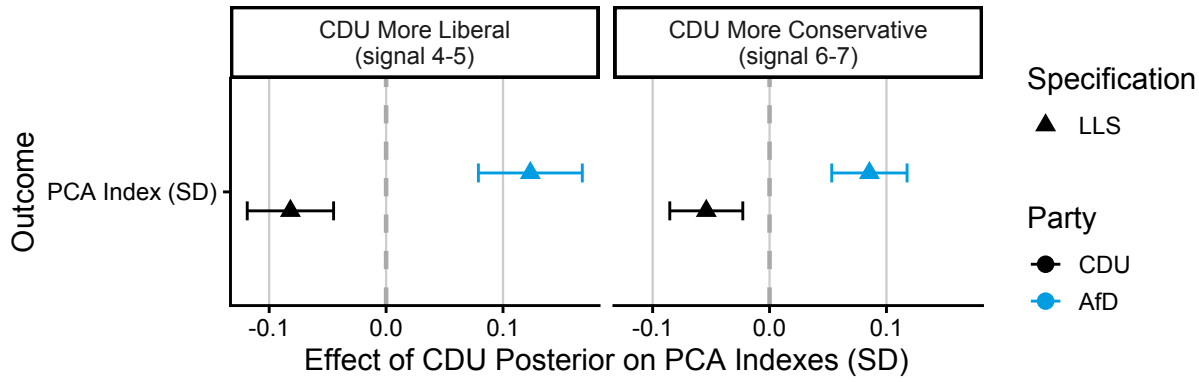


Figure A7: Testing for Asymmetric Effects
Note: LLS effects of CDU posterior belief on PCA indexes. The left panel uses only signals that portray the CDU as more liberal (4–5); the right panel uses signals that portray the CDU as more conservative (6–7). Bars indicate 95% confidence intervals.

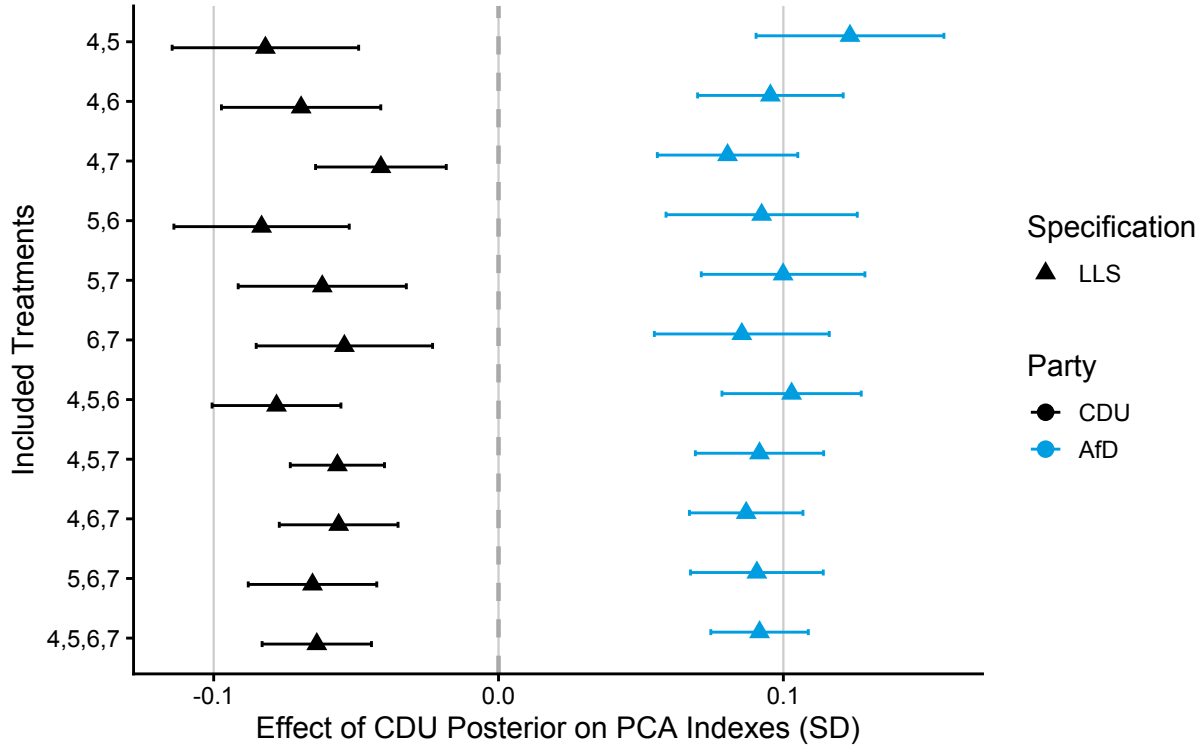


Figure A8: Effects on PCA Indexes Using Different Treatment Subsets
Note: Each estimate uses a different subset of treatment arms. Bars indicate 95% confidence intervals.

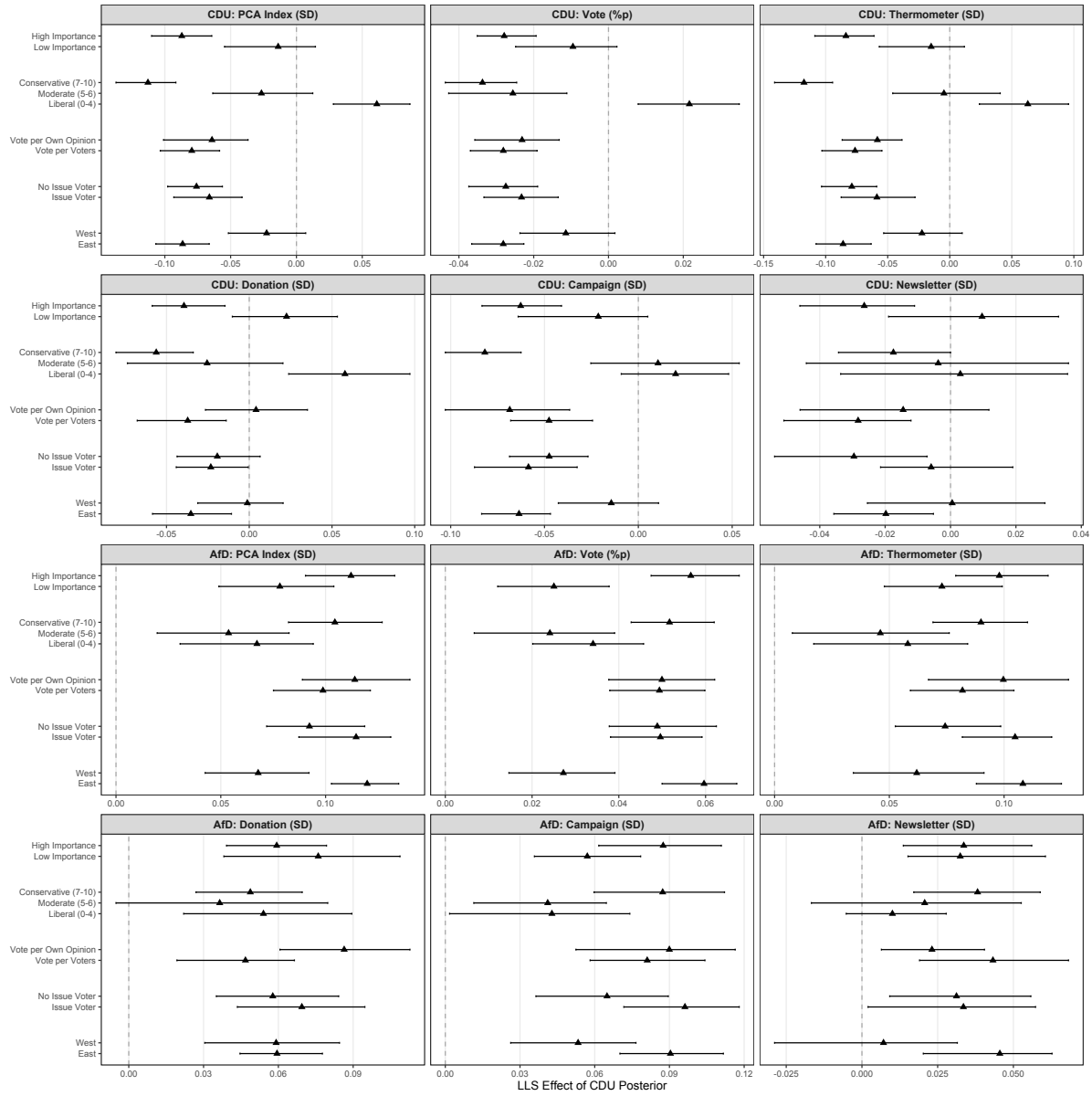


Figure A9: Heterogeneous LLS Effects by Subgroup: All Individual Outcomes
Note: LLS estimates of the effect of perceived CDU positioning on each individual outcome, separately by subgroup. Bars indicate 95% bootstrap confidence intervals.

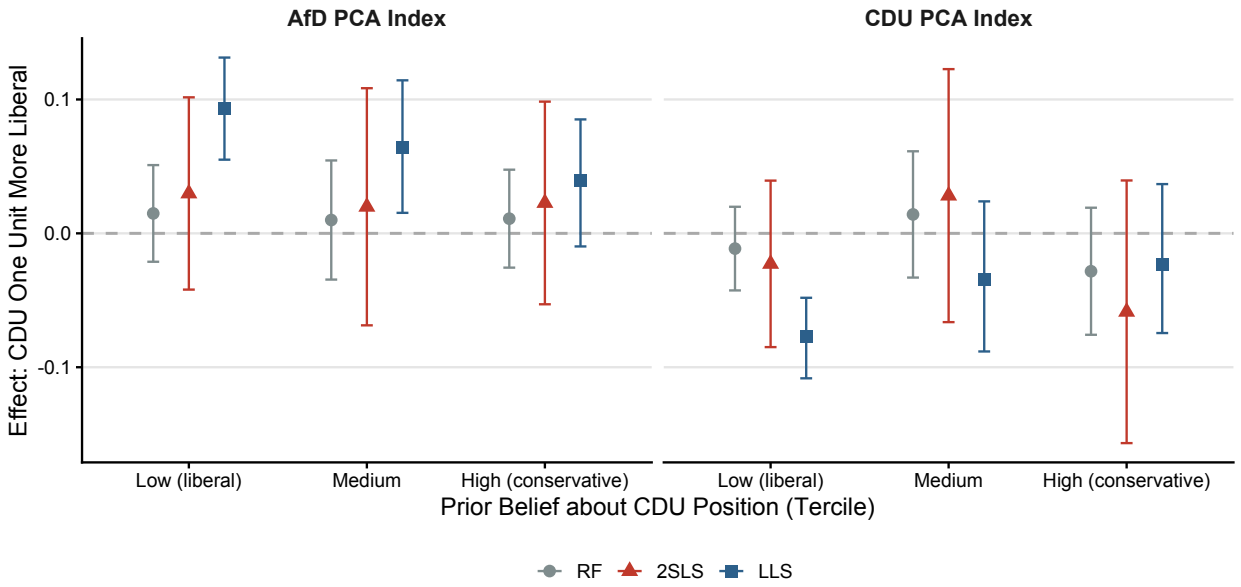
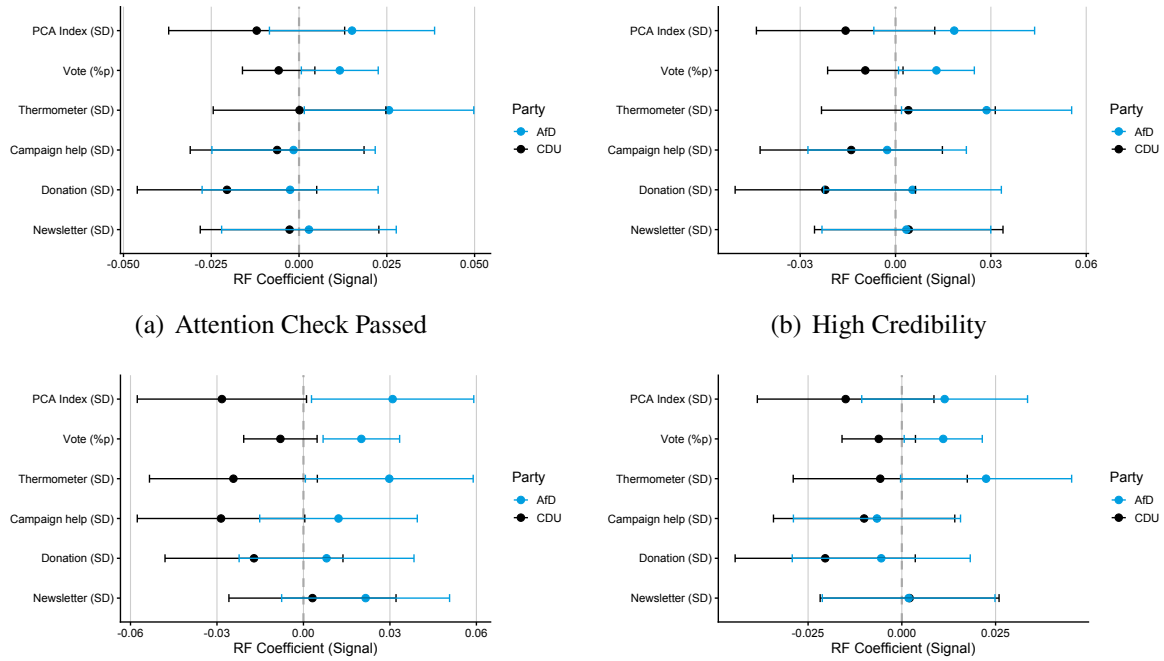


Figure A10: Heterogeneous Effects by Prior Beliefs about the CDU's Immigration Stance
Note: The sample is split into terciles of prior beliefs about the CDU's immigration position. "Low (liberal)" tercile: participants who initially perceived the CDU as most liberal on immigration. RF, 2SLS, and LLS estimates for the AfD and CDU PCA indexes within each subgroup. RF and 2SLS: 95% robust confidence intervals; LLS: 95% bootstrap confidence intervals.



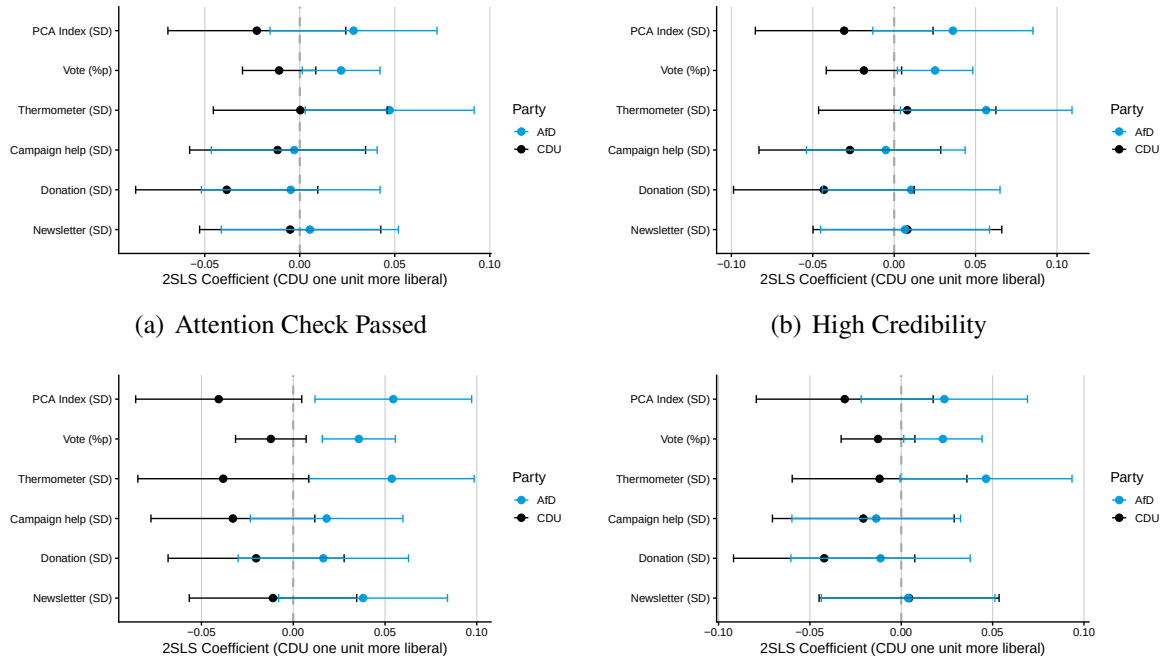
(a) Attention Check Passed

(b) High Credibility

(c) Update Toward Signal

(d) Including Speeders

Figure A11: Robustness Checks: Reduced-Form Estimates
Note: Each panel shows OLS reduced-form estimates (effect of signal) for all AfD and CDU outcomes on a restricted sample. Bars indicate 95% confidence intervals.



(a) Attention Check Passed

(b) High Credibility

(c) Update Toward Signal

(d) Including Speeders

Figure A12: Robustness Checks: 2SLS Estimates
Note: Each panel shows 2SLS estimates (CDU posterior instrumented by signal) for all AfD and CDU outcomes on a restricted sample. Bars indicate 95% confidence intervals.

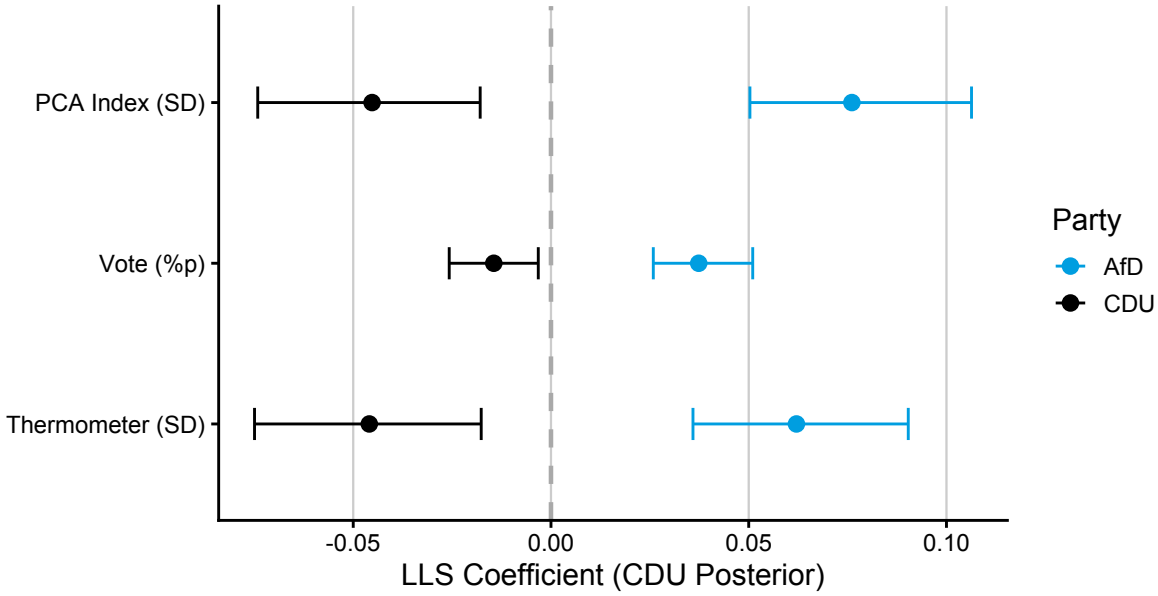


Figure A13: Follow-Up Survey: LLS Estimates
Note: LLS estimates from the obfuscated follow-up survey, conducted approximately one week after the main study. Each coefficient shows the effect of perceiving the CDU as one unit more liberal on immigration. Bars indicate 95% bootstrap confidence intervals. N = 2,182. Figure A14 reports the corresponding RF and 2SLS estimates.

Table A6: Follow-Up Survey: Effect of Perceived CDU Positioning on Political Outcomes

	CDU Outcomes			AfD Outcomes		
	RF	2SLS	LLS	RF	2SLS	LLS
PCA Index (SD)	-0.009	-0.017	-0.045***	-0.004	-0.007	0.076***
	(0.019)	(0.034)	[-0.074, -0.018]	(0.018)	(0.033)	[0.050, 0.106]
Vote (%p)	-0.011	-0.020	-0.014**	0.001	0.001	0.037***
	(0.008)	(0.014)	[-0.026, -0.003]	(0.008)	(0.015)	[0.026, 0.051]
Thermometer (SD)	0.009	0.017	-0.046***	-0.009	-0.016	0.062***
	(0.018)	(0.034)	[-0.075, -0.018]	(0.018)	(0.034)	[0.036, 0.090]
Observations	2,182	2,182	2,182	2,182	2,182	2,182
First-stage F	—	194.6	—	—	194.6	—
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: RF: reduced-form OLS of outcome on signal. 2SLS: CDU posterior instrumented by signal. LLS: Local Least Squares estimator of Balla-Elliott (2026). All coefficients show the effect of perceiving the CDU as one unit more liberal on immigration. RF and 2SLS report heteroskedasticity-robust standard errors in parentheses; LLS reports 95% Bayesian-bootstrap percentile confidence intervals in brackets, with significance stars based on normal-approximation z-tests using trimmed bootstrap standard errors. Controls: age, gender, immigration background, East Germany indicator, education. N = 2,182. *p< 0.1, **p< 0.05, ***p< 0.01.

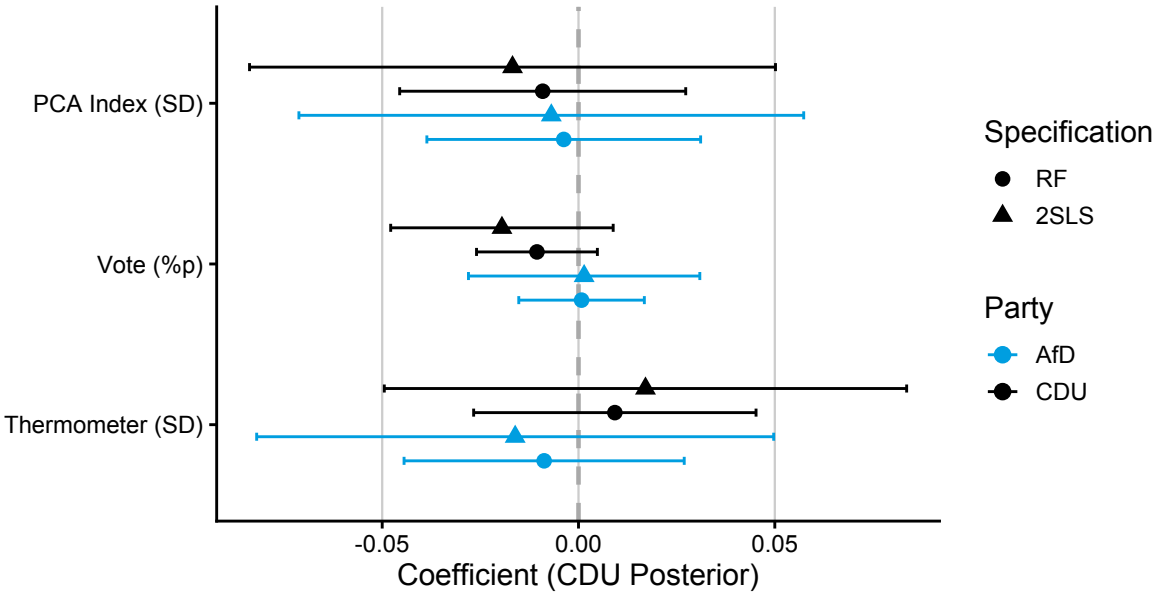
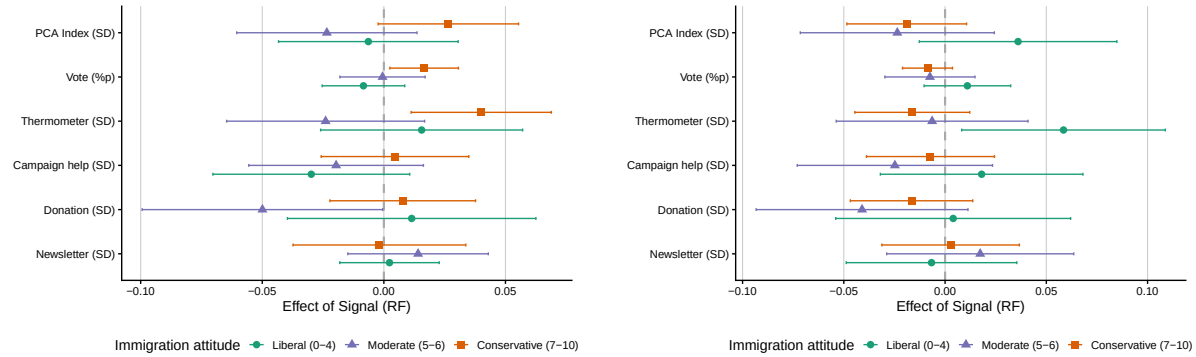


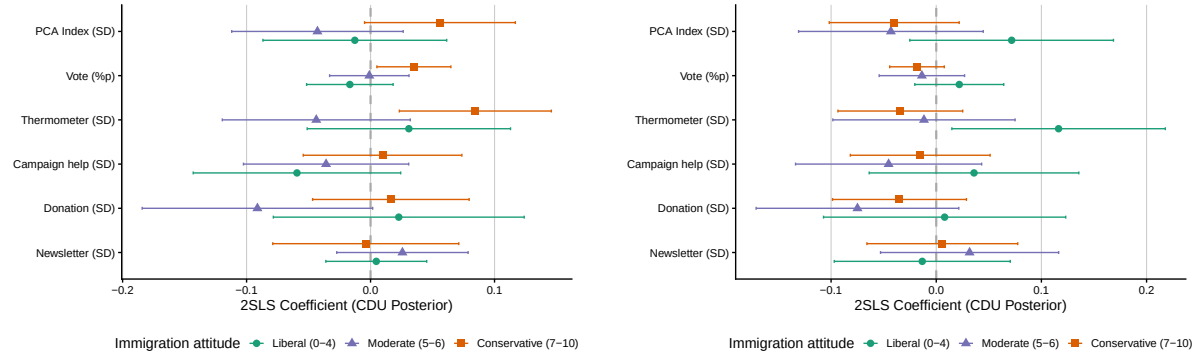
Figure A14: Follow-Up Survey: Reduced-Form and 2SLS Estimates
Note: RF (circles) and 2SLS (triangles) estimates from the obfuscated follow-up survey. RF coefficients show the effect of the signal; 2SLS coefficients show the effect of the CDU posterior instrumented by the signal. Bars indicate 95% confidence intervals. N = 2,182.



(a) AfD outcomes (b) CDU outcomes

Figure A15: Reduced-Form Effects of the Signal by Pre-Treatment Immigration Attitudes

Note: OLS estimates, shown separately for AfD outcomes (left) and CDU outcomes (right). For each dependent variable, the figure reports three estimates, one per pre-treatment immigration-attitude subsample (Liberal: 0–4, Moderate: 5–6, Conservative: 7–10 on the 0–10 scale), matching the subgroups used in Figure 9. Bars indicate 95% confidence intervals. Controls: age, gender, immigration background, East Germany indicator, education, prior-belief fixed effects.



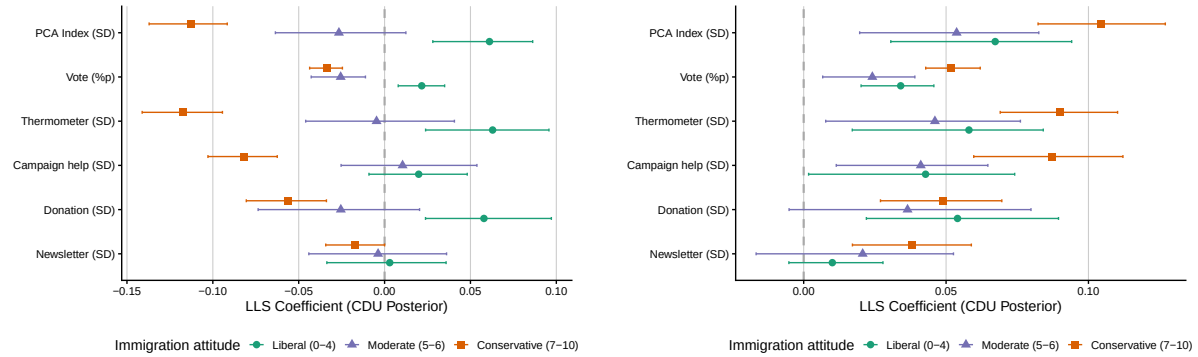
(a) AfD outcomes (b) CDU outcomes

Figure A16: Standard 2SLS Estimates by Pre-Treatment Immigration Attitudes

Note: 2SLS estimates instrumenting the CDU posterior belief with the randomized signal, shown separately for AfD outcomes (left) and CDU outcomes (right). For each dependent variable, the figure reports three estimates, one per pre-treatment immigration-attitude subsample (Liberal: 0–4, Moderate: 5–6, Conservative: 7–10 on the 0–10 scale), matching the subgroups used in Figure 9. Bars indicate 95% confidence intervals. Controls: age, gender, immigration background, East Germany indicator, education, prior-belief fixed effects.

D.1 Individual Outcomes by Immigration Attitude, Across Estimators

Figure A15, Figure A16, and Figure A17 report, for each individual outcome, the reduced-form, 2SLS, and LLS estimates separately by pre-treatment immigration-attitude subsample (Liberal 0–4, Moderate 5–6, Conservative 7–10). They complement the PCA-index results in Subsection 5.4.



(a) CDU outcomes

(b) AfD outcomes

Figure A17: LLS Estimates by Pre-Treatment Immigration Attitude

Note: LLS estimates of the effect of perceiving the CDU as one unit more liberal on immigration, separately for the three pre-treatment immigration-attitude subsamples: Liberal (0–4), Moderate (5–6), and Conservative (7–10). Bars indicate 95% bootstrap confidence intervals.

E Descriptive Binscatters

This appendix presents descriptive, nonparametric evidence on the relationship between perceived CDU positioning, respondents' own immigration attitudes, and political outcomes. Unlike the experimental estimates in the main text, these binscatters exploit both experimental and non-experimental variation in beliefs; we present them as complementary descriptive evidence.

Figure A18 and Figure A19 present conditional binscatters of political outcomes against the CDU posterior belief, controlling for demographic characteristics. For AfD outcomes, all five measures decline monotonically as the perceived CDU position moves rightward. For CDU outcomes, the mirror image holds: all measures increase as the perceived CDU position moves rightward. These monotonic relationships are consistent with a proximity model where voters respond to the distance between their preferences and the CDU's perceived stance, and inconsistent with protest voting (which predicts no relationship) or legitimation (which predicts the opposite pattern for AfD outcomes).

Figure A20 and Figure A21 present analogous binscatters where the x-axis is the respondent's own immigration attitude (pre-treatment). AfD support increases monotonically with anti-immigration attitudes, while CDU support peaks among moderate conservatives—consistent with the spatial voting framework where voters sort to their closest party.

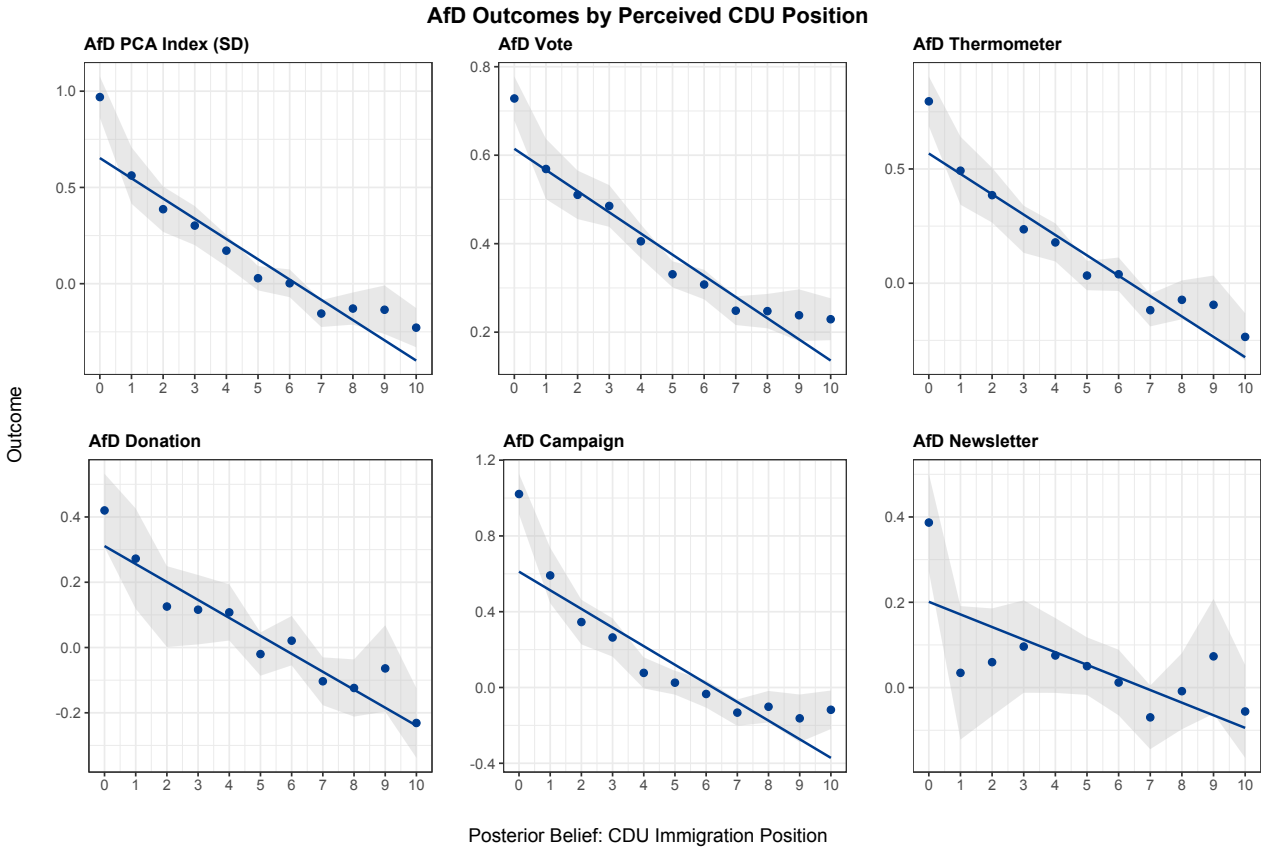


Figure A18: AfD Outcomes by Perceived CDU Immigration Position: Conditional Binscatter
Note: Binscatter of AfD outcomes against the posterior belief about the CDU's immigration position, conditional on demographic controls (age, gender, immigration background, East Germany, education). Higher values on the x-axis indicate a more conservative perceived CDU stance. Markers show covariate-adjusted mean outcomes at each value of the 0–10 scale (demographic controls partialled out); shaded bands are 95% confidence intervals and the solid line is a covariate-adjusted linear fit. N = 5,040.

F CAPE Analysis: Individual Outcomes

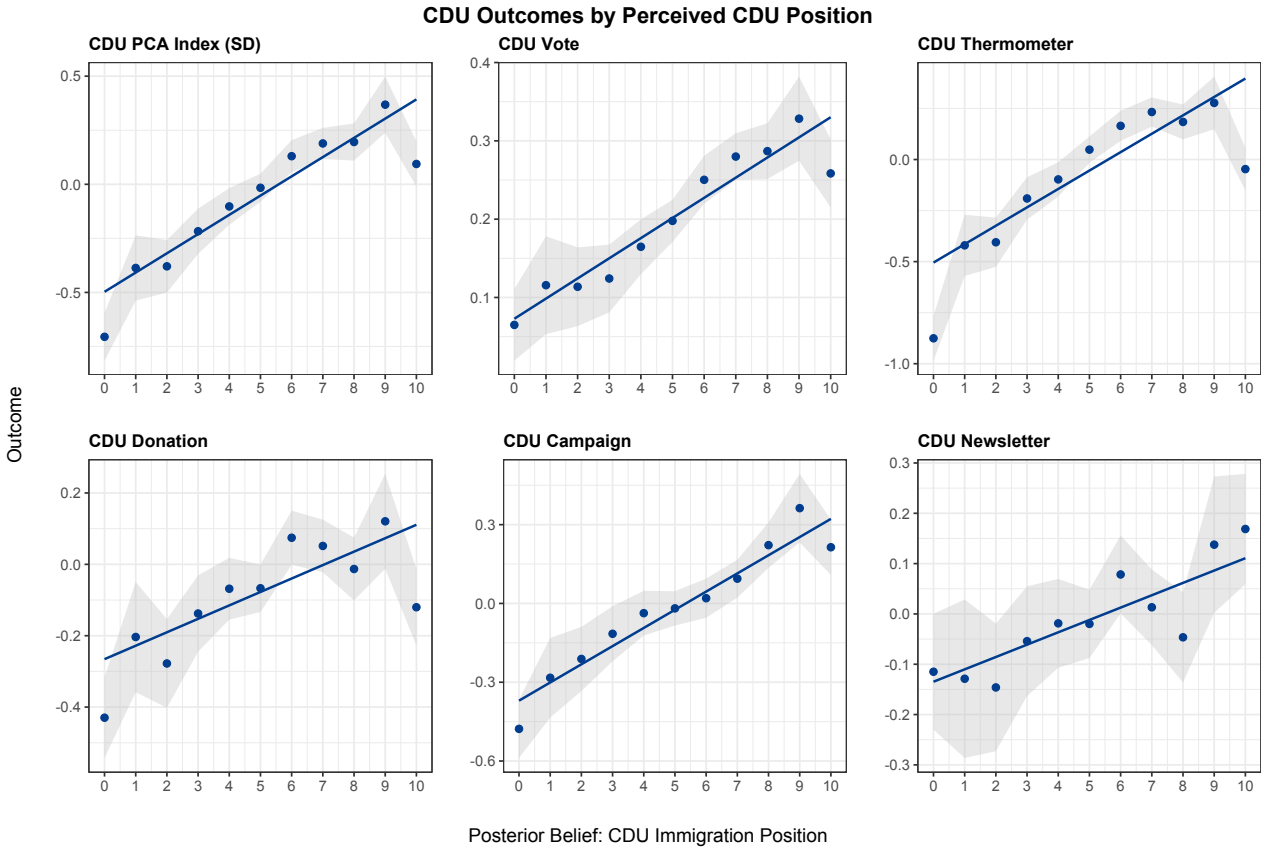


Figure A19: CDU Outcomes by Perceived CDU Immigration Position: Conditional Binscatter

Note: Binscatter of CDU outcomes against the posterior belief about the CDU's immigration position, conditional on demographic controls. See Figure A18 for details. N = 5,040.

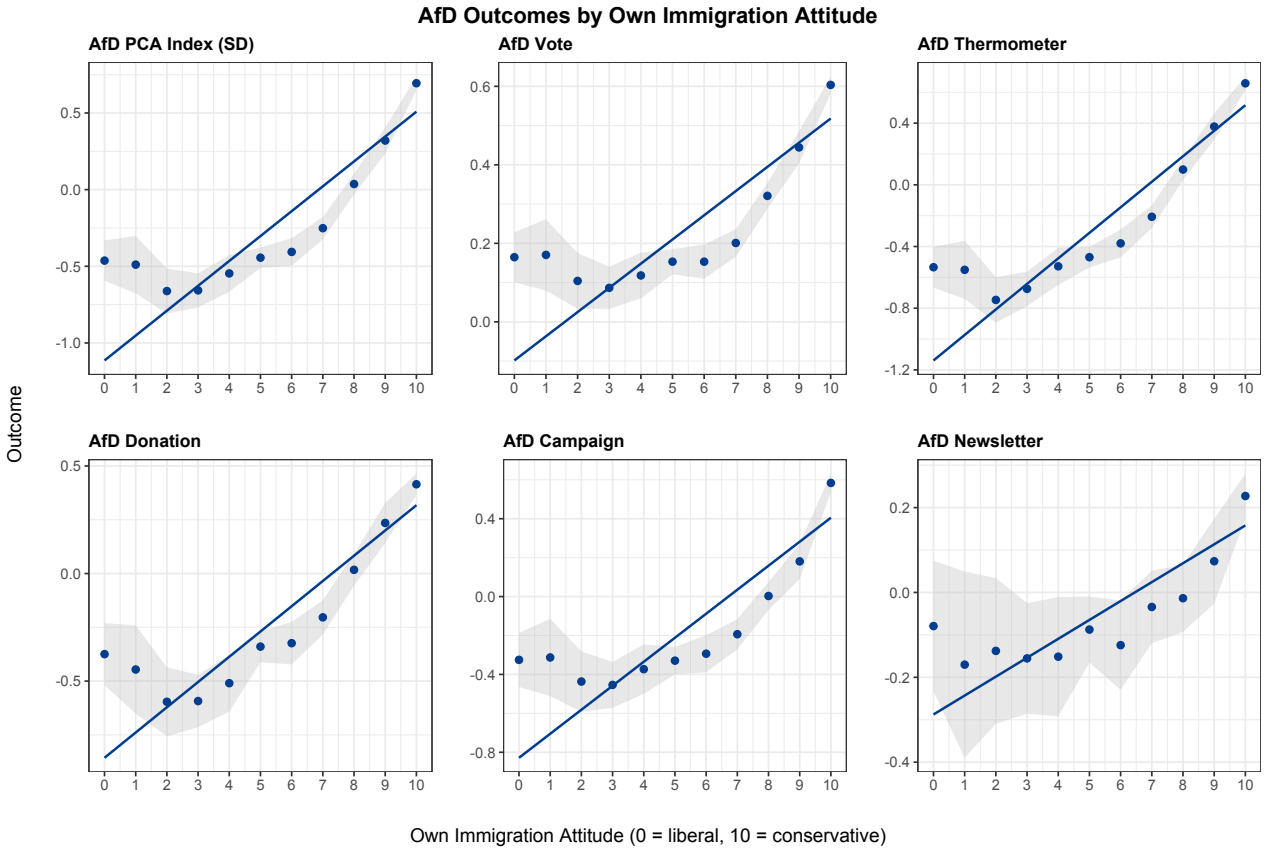


Figure A20: AfD Outcomes by Own Immigration Attitude: Conditional Binscatter

Note: Binscatter of AfD outcomes against the respondent's own pre-treatment immigration attitude (0 = liberal, 10 = conservative), conditional on demographic controls (age, gender, immigration background, East Germany, education). Markers show covariate-adjusted mean outcomes at each value of the 0–10 scale (demographic controls partialled out); shaded bands are 95% confidence intervals and the solid line is a covariate-adjusted linear fit. N = 5,040.

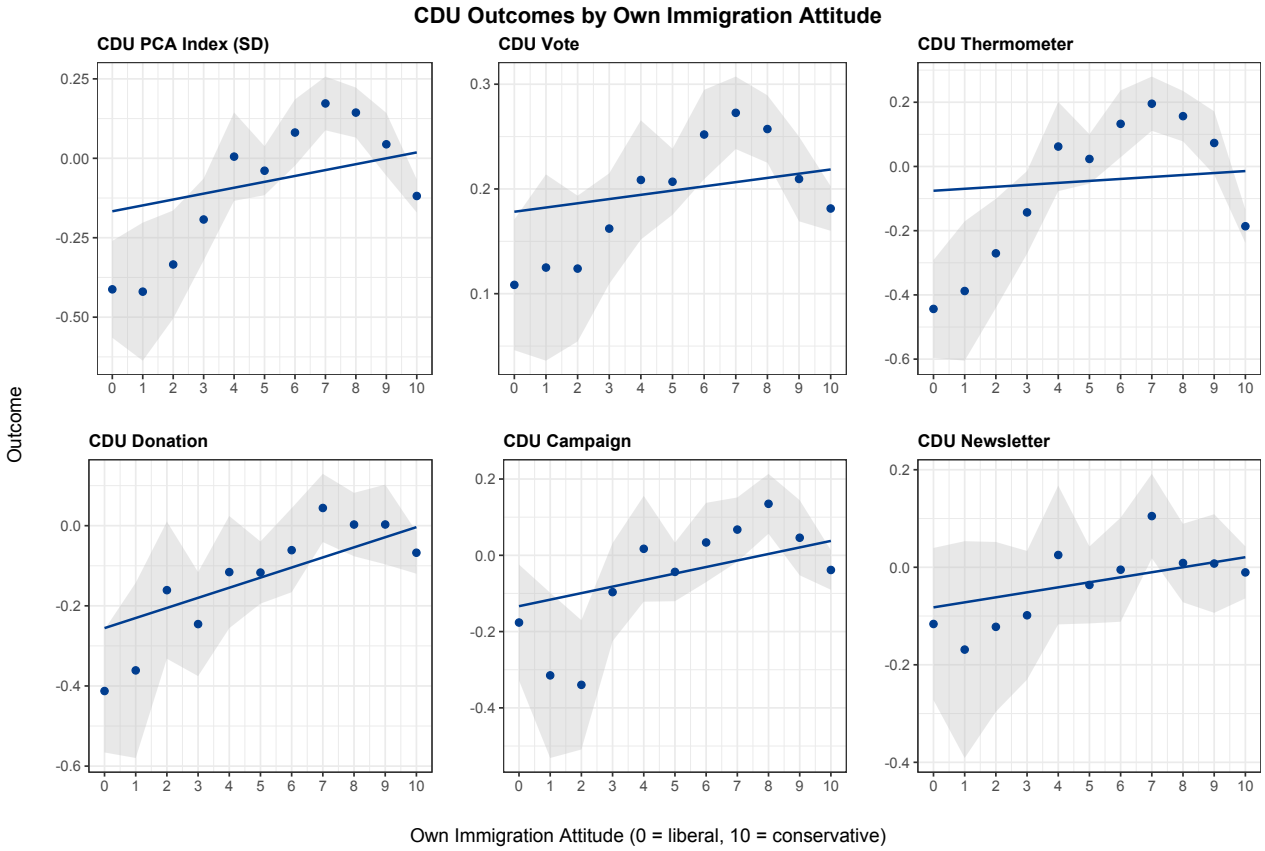


Figure A21: CDU Outcomes by Own Immigration Attitude: Conditional Binscatter

Note: Binscatter of CDU outcomes against the respondent's own pre-treatment immigration attitude. See Figure A20 for details. N = 5,040.

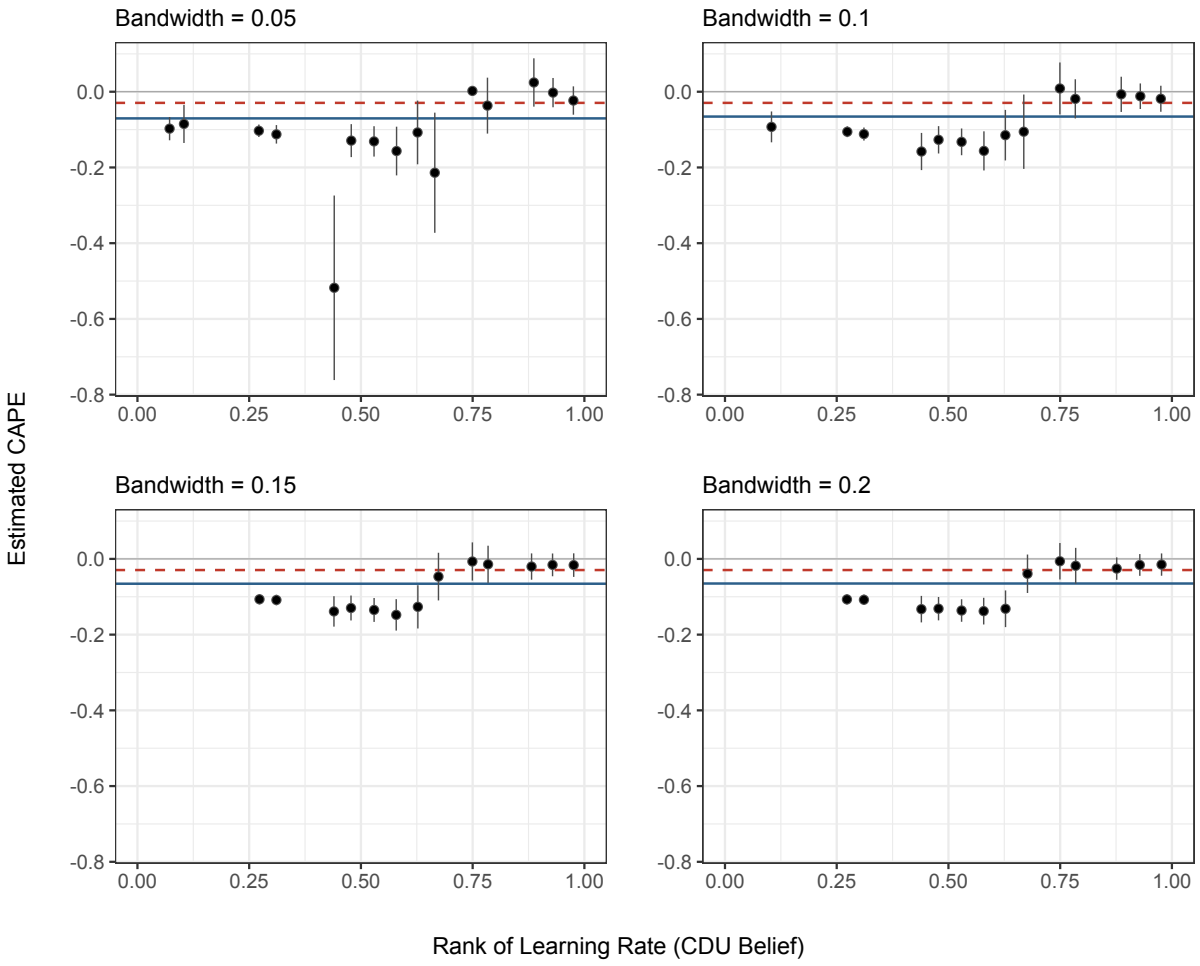


Figure A22: CAPE Estimates for CDU PCA Index

Note: Conditional Average Partial Effects of the CDU posterior belief on the CDU PCA index across four bandwidth choices. See Figure 7 for details.

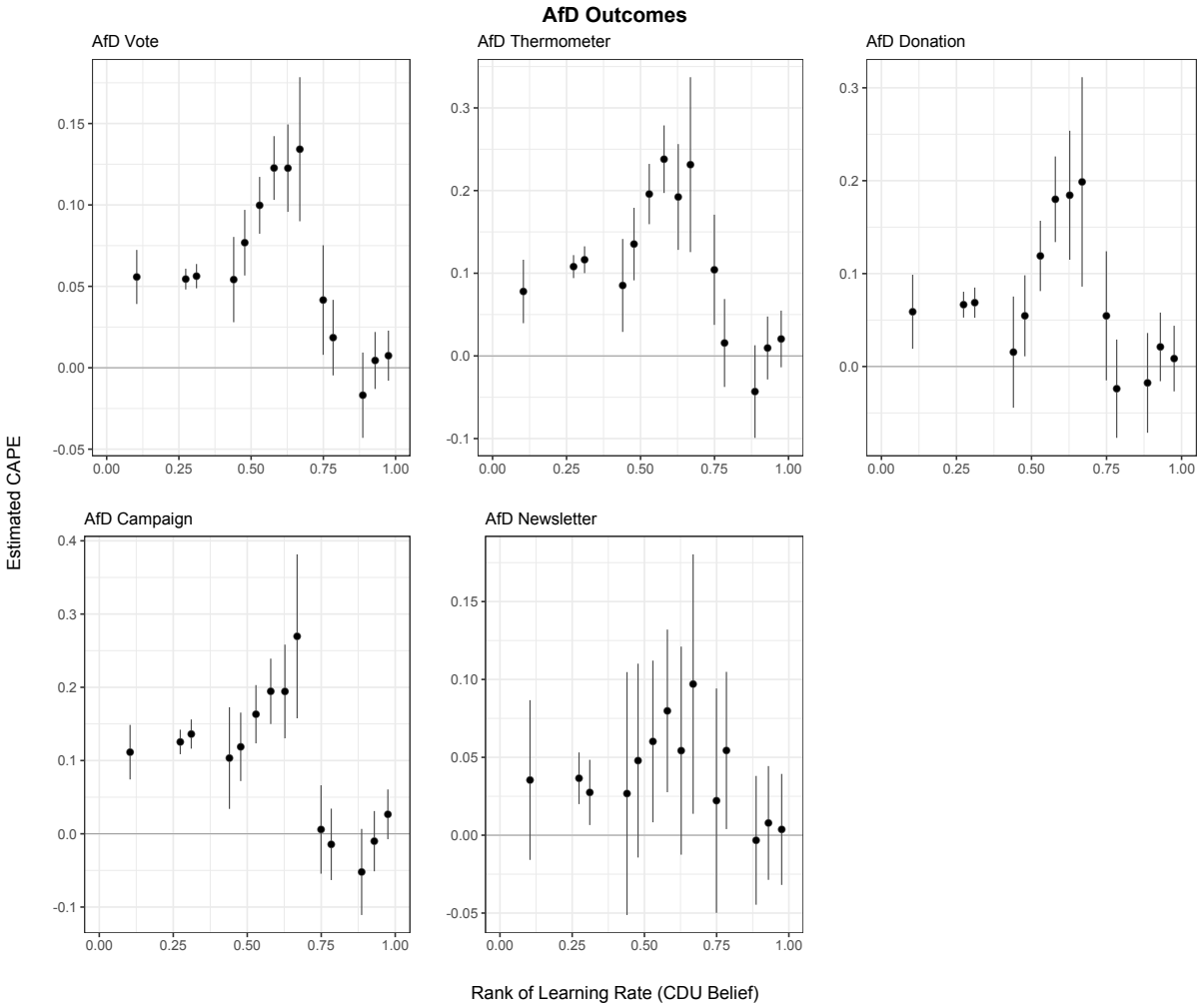


Figure A23: CAPE for Individual AfD Outcomes

Note: CAPE at bandwidth 0.1 for each individual AfD outcome. Points are binned estimates with pointwise 95% bootstrap confidence intervals; see Figure 7 for details. Non-learners included at the lowest learning-rate rank (N = 5,040).

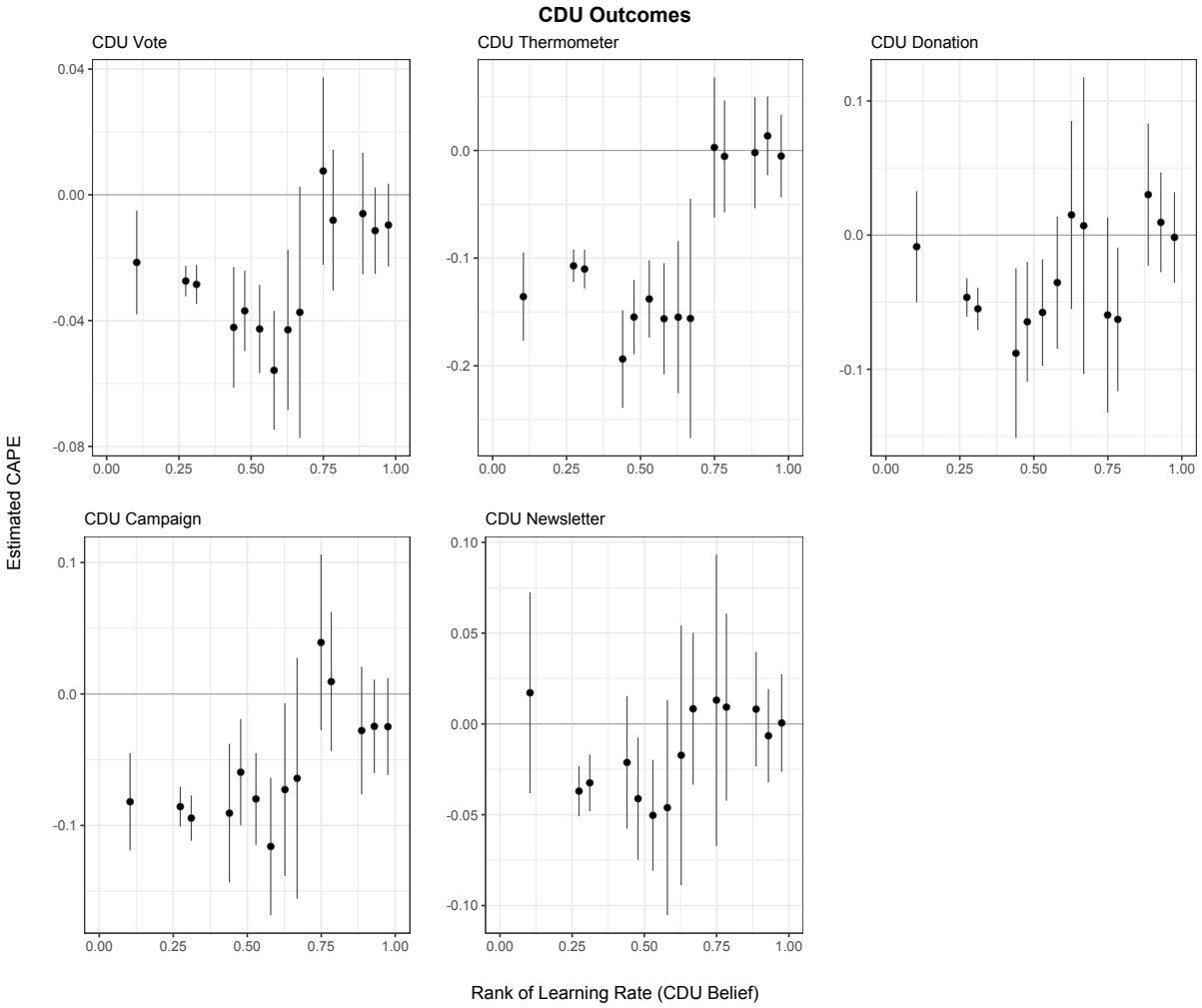


Figure A24: CAPE for Individual CDU Outcomes

Note: Same as Figure A23 but for CDU outcomes.

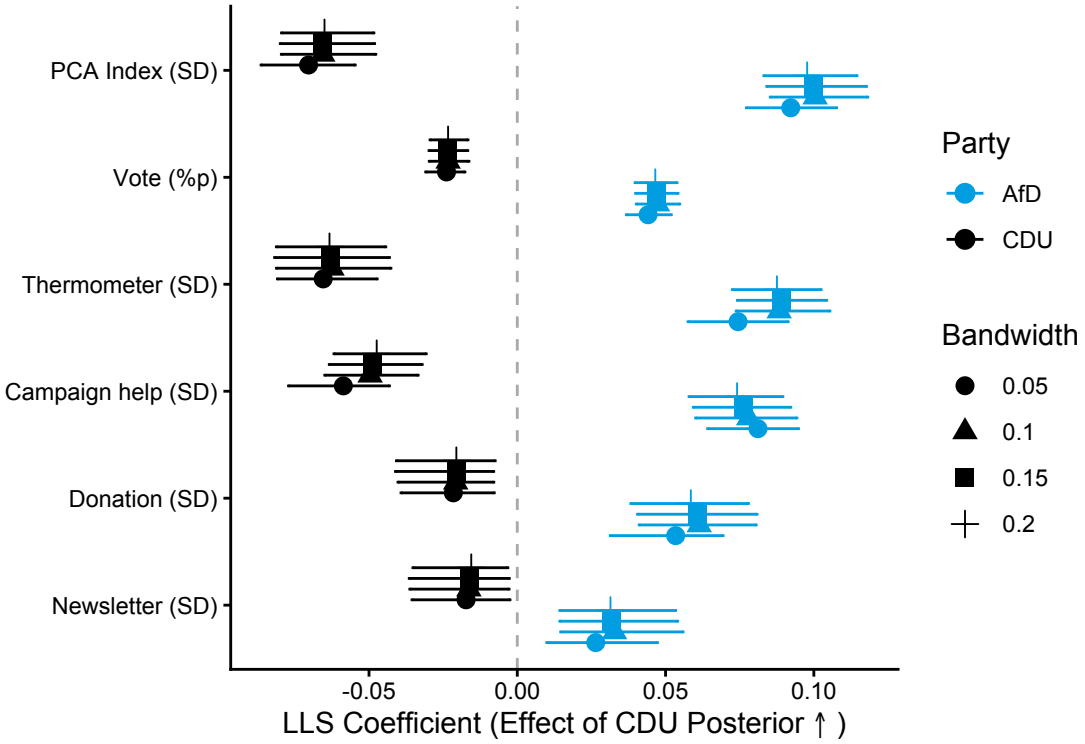


Figure A25: Bandwidth Sensitivity of LLS Estimates
Note: LLS estimates of the effect of perceived CDU positioning across four bandwidth choices. Bars indicate 95% bootstrap confidence intervals. Non-learners included at the lowest learning-rate rank. N = 5,040.

G Learning Rate Correlates

The LLS estimator relies on variation in the Bayesian learning rate $\alpha_i = (\text{Posterior}_i - \text{Prior}_i)/(\text{Signal}_i - \text{Prior}_i)$, which captures how much each participant updates their CDU belief in response to the signal. To validate the estimator and understand what drives heterogeneous updating, we regress the learning rate on individual characteristics (Table A7 and Figure A26). Non-learners (participants for whom the signal equals the prior, so α is undefined) are excluded ($N = 572$).

The strongest predictors of lower learning rates are residing in East Germany and being male—groups with arguably stronger pre-existing political views. Participants with more conservative prior beliefs about the CDU update more, consistent with the Bayesian interpretation: those whose priors are further from the signal have more to learn. Participants who rate immigration as more important and those with more conservative immigration attitudes update less, consistent with motivated reasoning or more entrenched priors. Self-reported credibility of the information, perceived survey bias, attention check performance, and completion time are not significant predictors, suggesting that the variation in learning rates reflects genuine heterogeneity in belief updating rather than differences in engagement or trust.

Table A7: Correlates of the Learning Rate (α)
Note: OLS regressions with the learning rate as dependent variable. Bivariate: separate regressions for each correlate. Multivariate: all correlates included jointly. Continuous variables are standardized (mean zero, unit SD) in the multivariate regression. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

	Bivariate		Multivariate	
	Coefficient	(SE)	Coefficient	(SE)
Age	-0.0002	(0.0010)	0.0049	(0.0165)
Male	-0.0733**	(0.0286)	-0.0816***	(0.0287)
University Degree	-0.0172	(0.0292)	-0.0456	(0.0304)
Residing in East Germany	-0.0988***	(0.0291)	-0.0858***	(0.0295)
Immigration Importance	-0.0474***	(0.0144)	-0.0280*	(0.0160)
Own Immigration Attitude	-0.0203***	(0.0052)	-0.0422**	(0.0171)
Prior CDU Belief	0.0171***	(0.0033)	0.0481***	(0.0113)
Prior Inaccuracy (|Prior - Signal|)	-0.0182**	(0.0089)	-0.0201	(0.0151)
Prior AfD Belief	-0.0091	(0.0080)	–	
Prior SPD Belief	0.0142**	(0.0057)	–	
Prior Greens Belief	0.0175***	(0.0062)	–	
Prior FDP Belief	0.0082*	(0.0045)	–	
Prior Left Belief	0.0082*	(0.0048)	–	
Found Information Trustworthy	0.0088	(0.0142)	0.0046	(0.0148)
Found Survey Biased	0.0045	(0.0341)	0.0063	(0.0354)
Completion Time (Minutes)	0.0003	(0.0004)	0.0112	(0.0134)
Failed 2nd Attention Check	-0.0121	(0.0636)	-0.0341	(0.0659)
Observations	4,468		4,468	
R^2			0.0120	

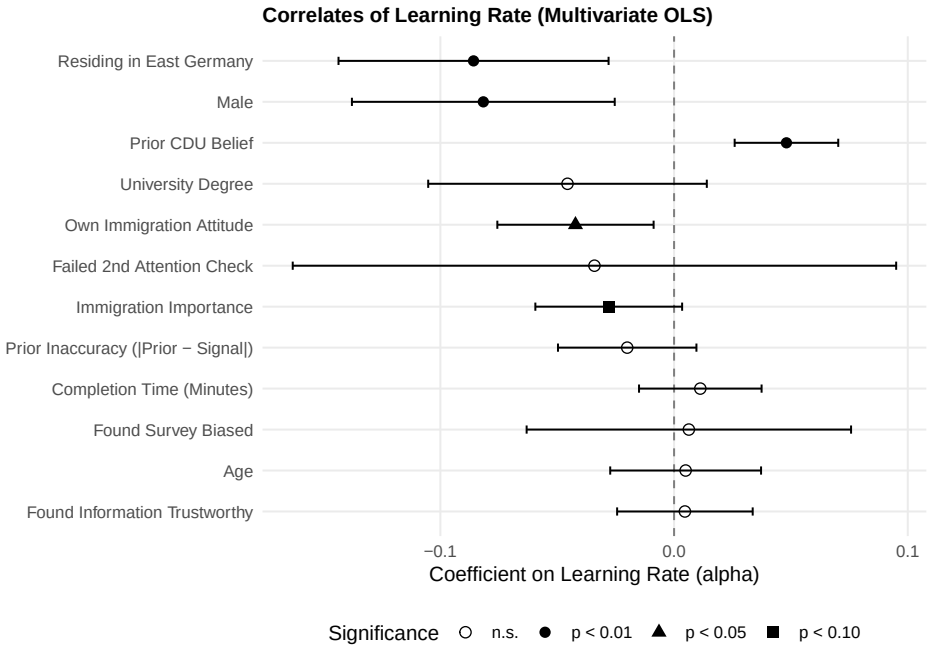


Figure A26: Correlates of the Learning Rate: Multivariate OLS
Note: Coefficients from a multivariate OLS regression of the learning rate (α) on individual characteristics. Continuous variables are standardized. Bars indicate 95% confidence intervals. N = 4,468.

H Pre-Registered Specification: Perceived Representation Gap

As pre-registered, our original specification used the perceived representation gap—the distance between a participant's own immigration attitude and the perceived stance of the closest mainstream party—as the endogenous variable. The representation gap is defined as:

$$\text{Perceived Gap}_i = \min_{p \in \mathcal{P} \setminus \{\text{AfD}\}} \left| \text{Own Attitude}_i - \text{Perceived Party Position}_{ip} \right|, \tag{7}$$

where $\mathcal{P}$ denotes the set of parties represented in parliament, so that the minimum is taken over all *mainstream* (non-AfD) parties. We present these results here for transparency and to demonstrate that our conclusions are robust to the choice of endogenous variable.

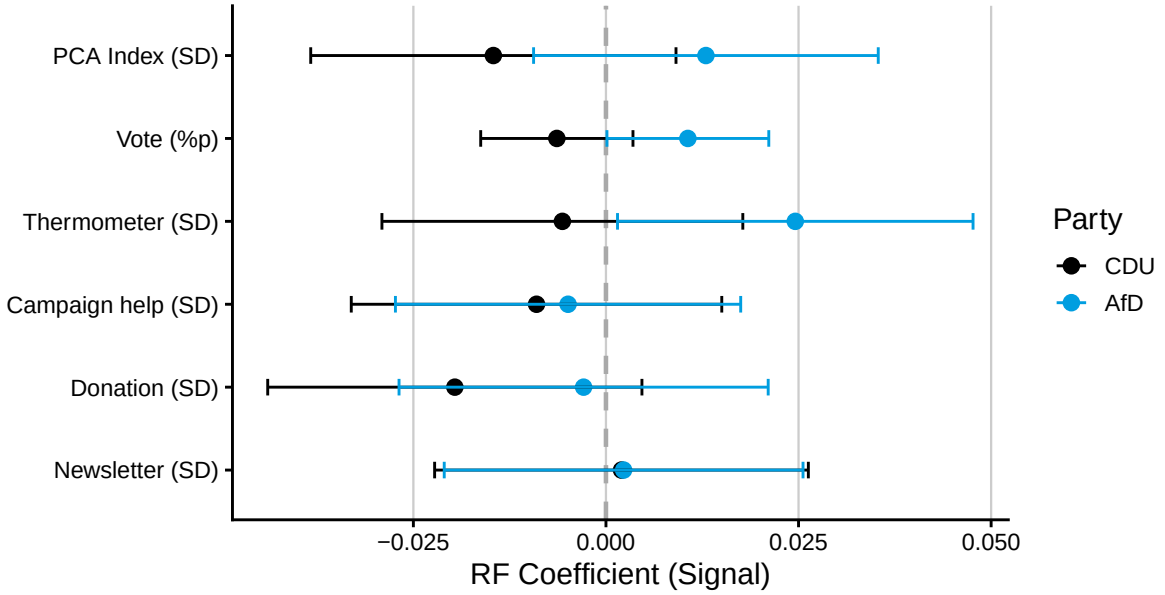


Figure A27: Pre-Registered Specification: Reduced-Form Estimates
Note: OLS reduced-form estimates (effect of signal on outcomes). The PCA index includes all outcome variables. Bars indicate 95% confidence intervals. N = 5,040.

The results are qualitatively identical to the main specification. Under the LLS estimates, a one-unit increase in the perceived representation gap increases AfD voting by approximately 6.5 percentage points and decreases CDU voting by approximately 1.7 percentage points. The larger AfD magnitude relative to the CDU posterior specification reflects the weaker first stage ($F \approx 97$), which mechanically scales up the IV coefficient. The first-stage F-statistic, while well

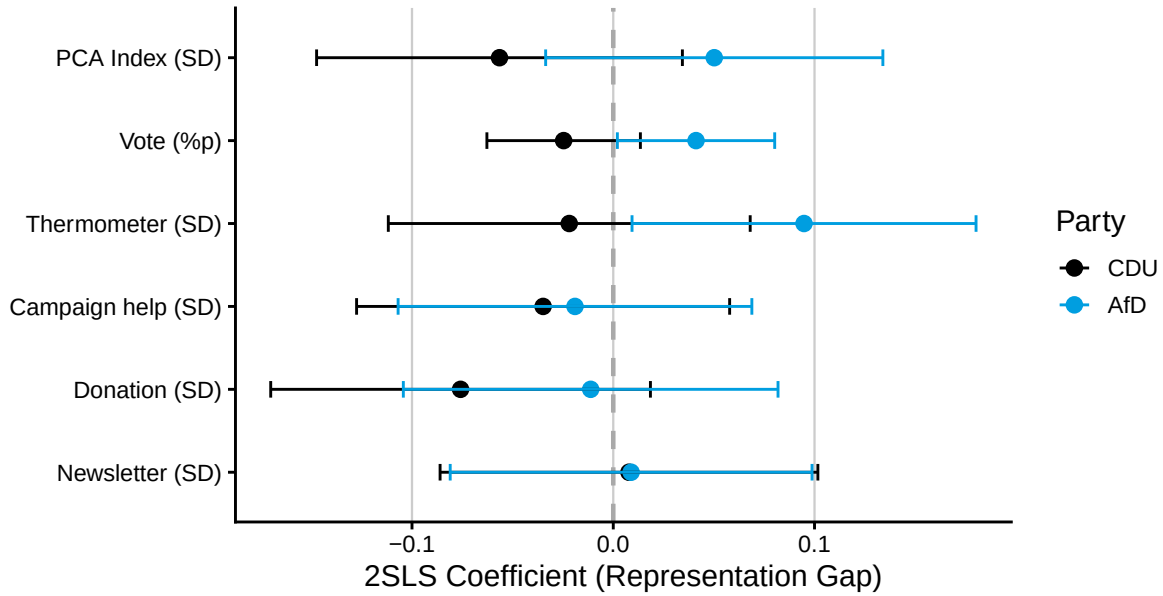


Figure A28: Pre-Registered Specification: 2SLS Estimates
Note: 2SLS estimates using the perceived representation gap as endogenous variable, instrumented by the randomized signal. Bars indicate 95% confidence intervals. N = 5,040.

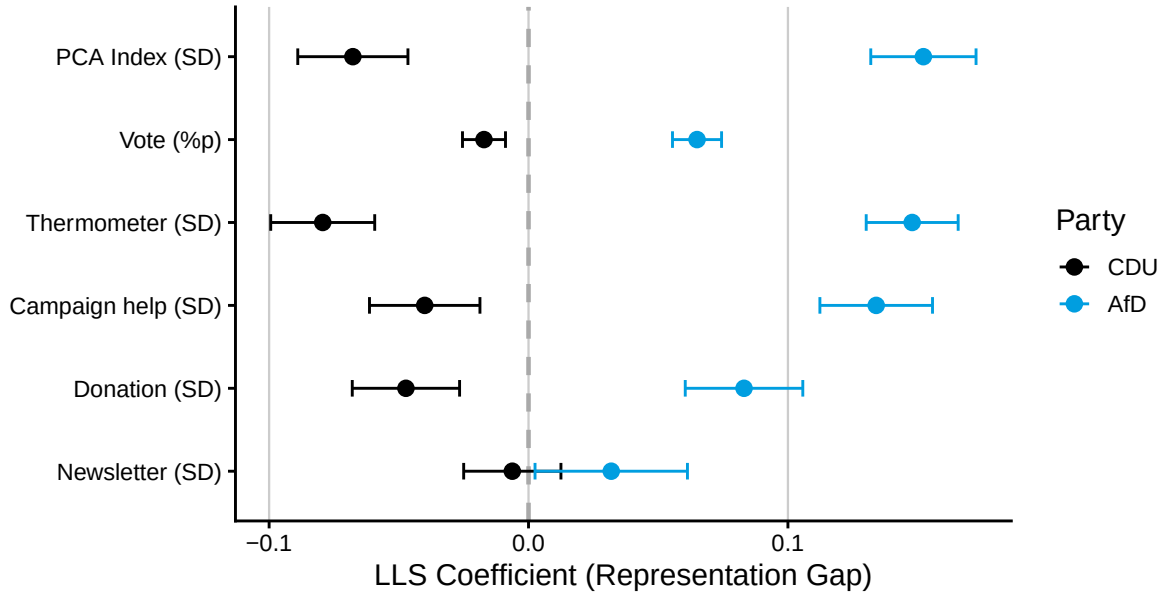


Figure A29: Pre-Registered Specification: LLS Estimates
Note: LLS estimates using the perceived representation gap as endogenous variable. Bars indicate 95% bootstrap confidence intervals. N = 5,040.

above 10, is substantially weaker than the $F = 375$ obtained with the CDU posterior, supporting our decision to use the CDU posterior as the main specification.

Table A8: Effects of the Perceived Representation Gap on AfD Outcomes (Pre-Registered IV)

	Dependent variable:					
	Main Study					
	Index	Voting	Thermometer	Campaign	Donation	Newsletter
	(1)	(2)	(3)	(4)	(5)	(6)
Perceived RG (SD)	0.152***	0.065***	0.148***	0.134***	0.083***	0.032**
	(0.010)	(0.004)	(0.011)	(0.011)	(0.010)	(0.015)
Constant	✓	✓	✓	✓	✓	✓
Dem. Controls	✓	✓	✓	✓	✓	✓
Prior beliefs	✓	✓	✓	✓	✓	✓
Observations	5,040	5,040	5,040	5,040	5,040	5,040

Note: LLS estimates using the perceived representation gap as endogenous variable, instrumented by the r
in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table A9: Effects of the Perceived Representation Gap on CDU Outcomes (Pre-Registered IV)

	Dependent variable:								
	Main Study						Follow-Up Study		
	Index	Voting	Thermometer	Campaign	Donation	Newsletter	Index	Voting	Thermometer
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Perceived RG (SD)	−0.068***	−0.017***	−0.079***	−0.040***	−0.047***	−0.006	−0.034**	−0.009	−0.039**
	(0.011)	(0.004)	(0.011)	(0.010)	(0.011)	(0.009)	(0.016)	(0.007)	(0.015)
Constant	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dem. Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Prior beliefs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	5,040	5,040	5,040	5,040	5,040	5,040	2,182	2,182	2,182

Note: LLS estimates using the perceived representation gap as endogenous variable, instrumented by the randomized signal. Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

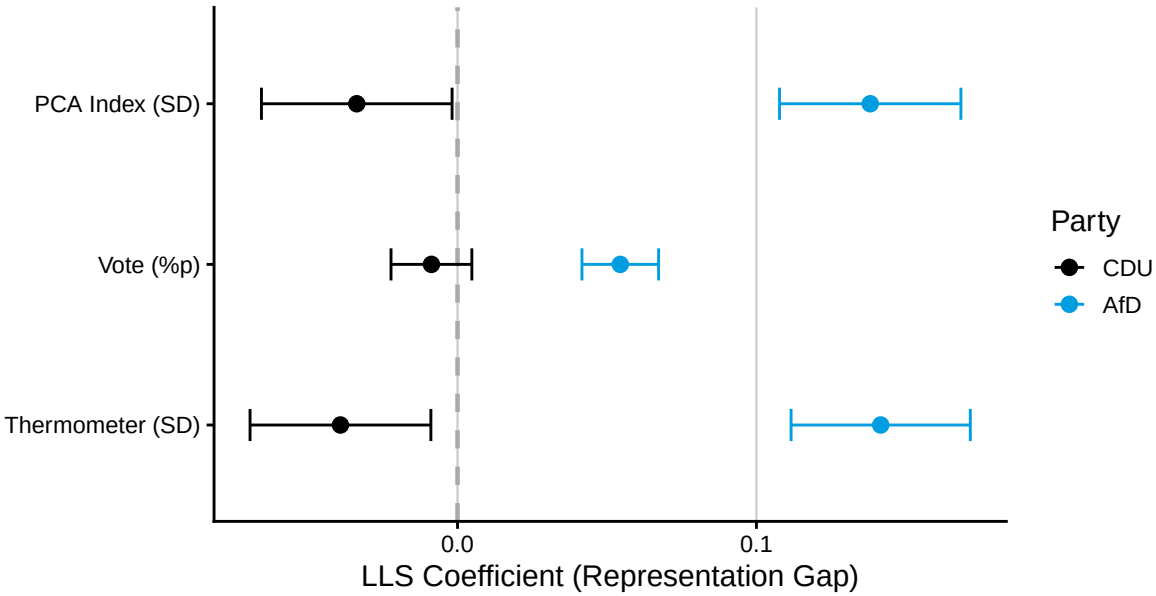


Figure A30: Follow-Up Pre-Registered Specification: LLS Estimates
Note: LLS estimates using the perceived representation gap as endogenous variable in the follow-up survey. Bars indicate 95% bootstrap confidence intervals.

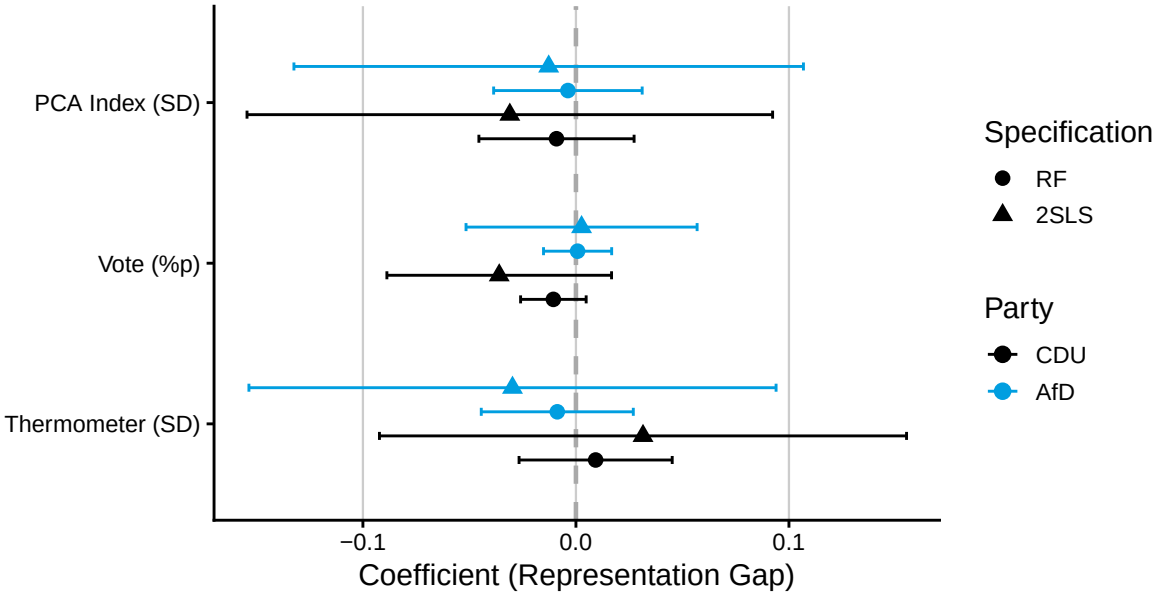


Figure A31: Follow-Up Pre-Registered Specification: RF and 2SLS Estimates
Note: RF (circles) and 2SLS (triangles) estimates from the follow-up survey using the perceived representation gap as endogenous variable. Bars indicate 95% confidence intervals.

Table A10: Effect of Signal and Perceived Representation Gap on Turnout (Pre-Registered IV)

	Dependent variable: Vote Abstention (SD)			
	Main Study		Follow-Up Study	
	RF	IV	RF	IV
	(1)	(2)	(3)	(4)
Treatment (CDU Signal)	0.001		0.009	
	(0.012)		(0.018)	
Perceived Representation Gap		0.017		0.003
		(0.012)		(0.017)
Demographic controls	✓	✓	✓	✓
Observations	5,040	5,040	2,182	2,182

Note: Dependent variable: standardized self-reported likelihood to vote. Pre-registered specification using the perceived representation gap as endogenous variable. Robust standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

I Alternative Outcome Indexes

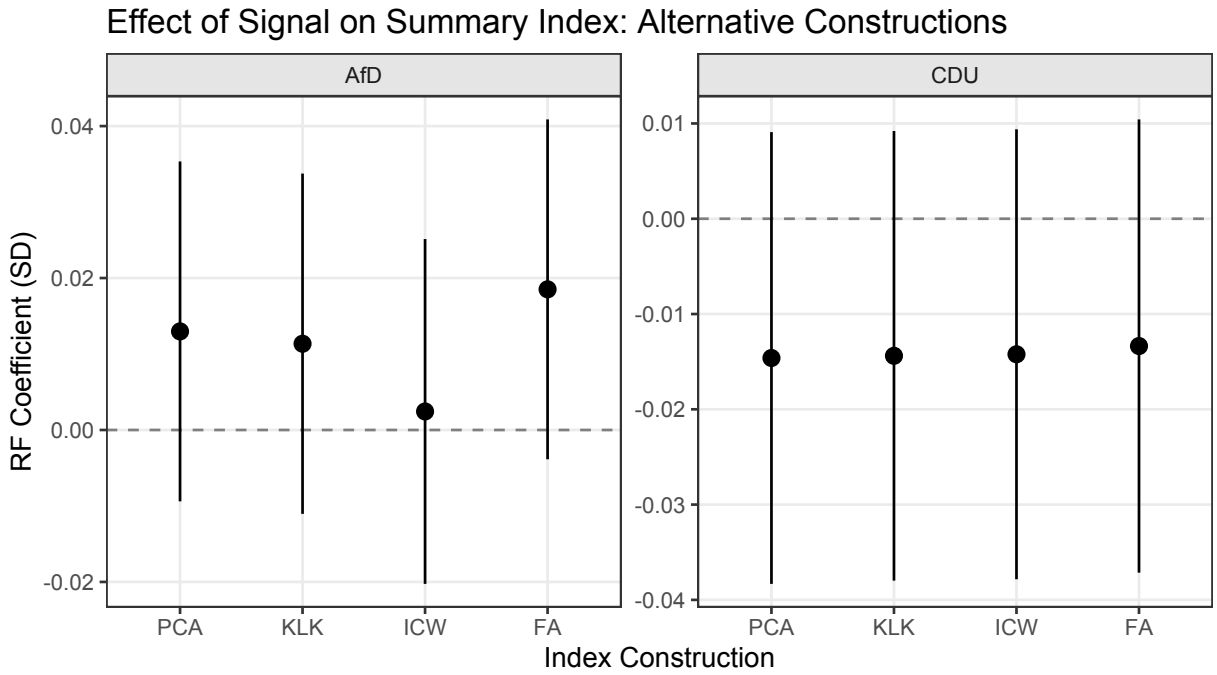


Figure A32: Reduced-Form Effects Using Alternative Index Constructions
Note: KLK mean-effects index, Anderson ICW index, and factor analysis index alongside PCA. Bars indicate 95% confidence intervals.

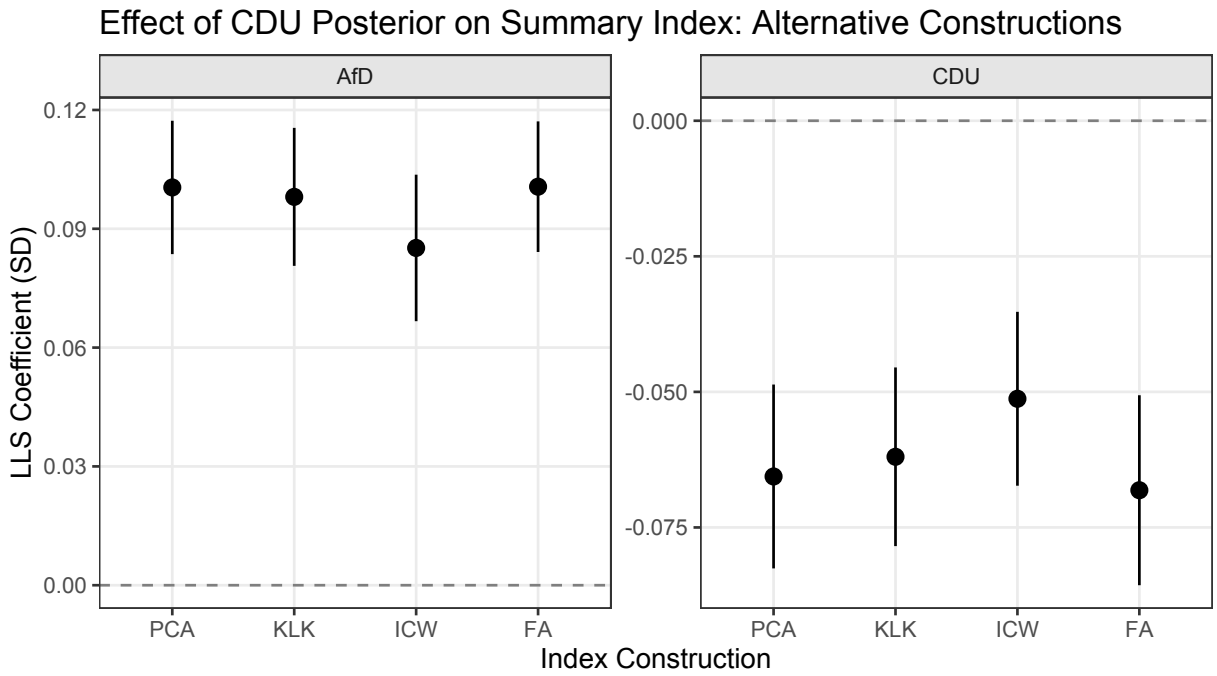


Figure A33: LLS Effects Using Alternative Index Constructions
Note: Same as Figure A32 but for LLS estimates. The endogenous variable is the CDU posterior belief.

J Supplementary Tables and Figures

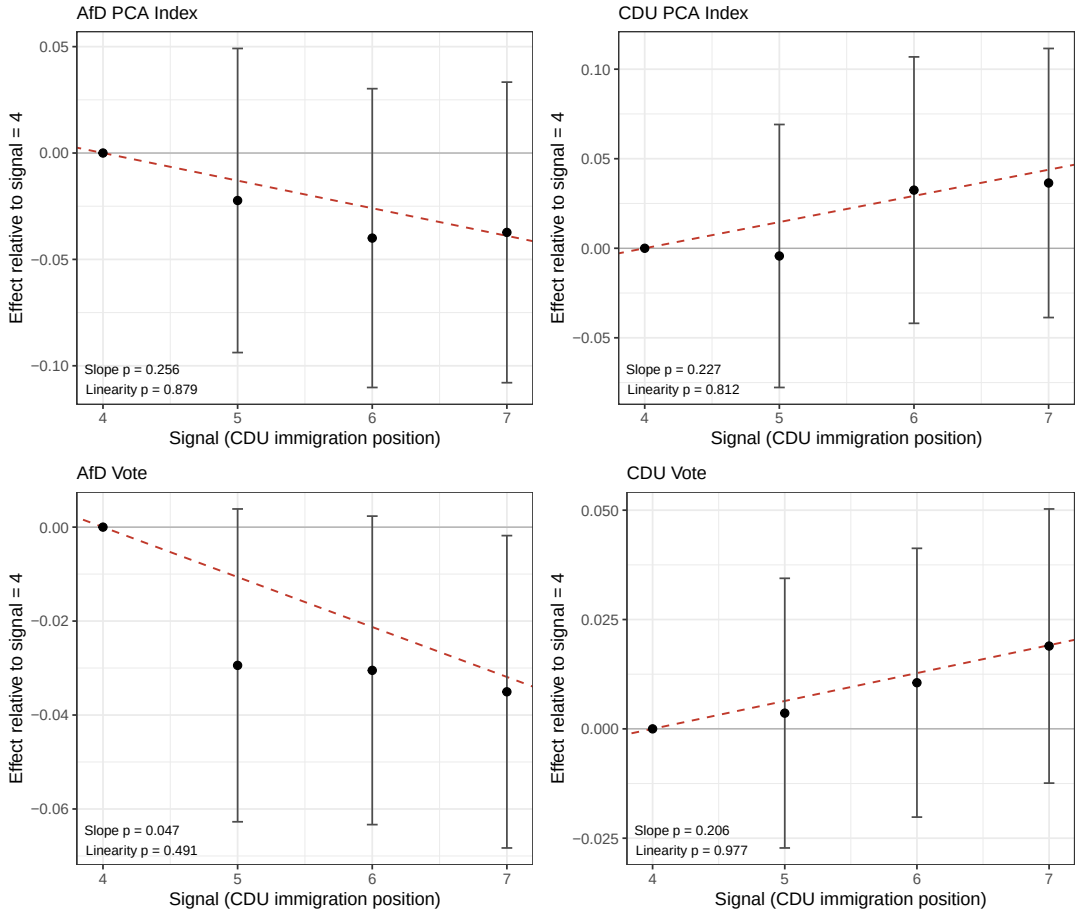


Figure A34: Saturated Reduced Form by Signal Arm

Note: Adjusted effect of each signal arm (5, 6, 7) relative to the signal = 4 arm, from OLS regressions of each outcome on signal-arm indicators. Controls: age, gender, immigration background, East Germany indicator, education, prior-belief fixed effects. Bars are 95% robust confidence intervals. The dashed line is the fitted linear reduced form. "Linearity" reports the p-value of a Wald test that the arm effects are linear in the signal (two restrictions). N = 5,040.

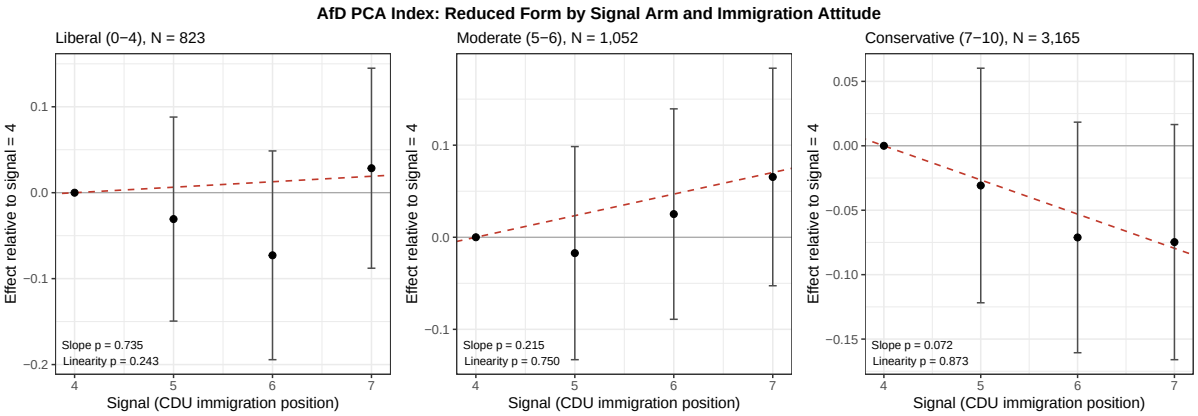


Figure A35: Saturated Reduced Form by Signal Arm and Immigration Attitude
Note: Same specification as Figure A34, for the AfD PCA index, estimated separately within pre-treatment immigration-attitude groups (Liberal 0–4, Moderate 5–6, Conservative 7–10).

Table A11: Follow-Up Results: Attrition Diagnostics and Inverse-Probability Weighting

	Reduced Form		2SLS		LLS	
Outcome	Unweighted	IPW	Unweighted	IPW	Unweighted	IPW
AfD PCA Index	-0.004	-0.005	-0.007	-0.009	0.076***	0.078***
	(0.018)	(0.018)	(0.033)	(0.034)	(0.013)	(0.013)
AfD Vote	0.001	-0.000	0.001	-0.000	0.037***	0.038***
	(0.008)	(0.008)	(0.015)	(0.015)	(0.006)	(0.006)
AfD Thermometer	-0.009	-0.009	-0.016	-0.017	0.062***	0.065***
	(0.018)	(0.019)	(0.034)	(0.034)	(0.013)	(0.013)
CDU PCA Index	-0.009	-0.006	-0.017	-0.011	-0.045***	-0.046***
	(0.019)	(0.019)	(0.034)	(0.034)	(0.014)	(0.014)
CDU Vote	-0.011	-0.009	-0.020	-0.017	-0.014**	-0.014**
	(0.008)	(0.008)	(0.014)	(0.014)	(0.006)	(0.005)
CDU Thermometer	0.009	0.011	0.017	0.021	-0.046***	-0.049***
	(0.018)	(0.019)	(0.034)	(0.035)	(0.015)	(0.015)

Note: Follow-up estimates of the effect of perceiving the CDU as one unit more liberal on immigration, unweighted and weighted by the inverse of the estimated probability of follow-up participation. Participation probabilities are from a logit on age, gender, immigration background, East Germany residence, education, pre-treatment immigration attitudes, and prior beliefs about the CDU's position; weights are stabilized and capped at the 99th percentile. Attrition is unrelated to treatment: a joint test of the signal-arm indicators in a participation regression yields $p = 0.68$. RF and 2SLS: robust standard errors in parentheses; LLS: bootstrap standard errors (500 draws), bandwidth = 0.2 as in the main follow-up estimates. N = 2,182. *p< 0.1, **p< 0.05, ***p< 0.01.

Table A12: Descriptives and Representativeness

	Our Sample (%)		German Population (%)
	Main Study	Follow-Up Study	
Male	53.5	55.2	49.5
Immigration Background	13	14	30
University Degree	29.8	29.5	33.3
Residing in East Germany	60.3	53	14.9
Age Groups			
18–39	33.5	36.7	31.2
40–59	40.2	40.9	32.2
60+	26.3	22.4	36.6
Observations	5,040	2,182	

Note: Data on age for the German adult (18+) population refer to 2023 and are from the 2022 census available here. Percentages for age brackets are calculated using as denominator the adult (18+) population.

Table A13: Balance Tests

	Treatment				
	Signal = 4	**Signal = 5**	**Signal = 6**	**Signal = 7**	**p-value**
Age	47.685	46.828	48.173	47.387	0.158
Male	0.529	0.550	0.526	0.536	0.622
University Degree	0.311	0.312	0.273	0.297	0.114
Residing in East Germany	0.602	0.604	0.612	0.593	0.807
Pre-Treatment Own Immigration Stance	7.090	7.118	7.223	6.976	0.169
Pre-Treatment Belief about Parties					
CDU Immigration Stance	5.905	5.676	5.808	5.871	0.220
AfD Immigration Stance	8.823	8.880	8.855	8.794	0.809
SPD Immigration Stance	3.600	3.536	3.600	3.593	0.891
Greens Immigration Stance	2.595	2.543	2.586	2.582	0.959
FDP Immigration Stance	5.336	5.209	5.442	5.365	0.229
Left Immigration Stance	4.810	4.711	4.793	4.891	0.412
Found Information Trustworthy	2.990	2.937	2.974	2.979	0.589
Observations	1,249	1,232	1,295	1,264	

Note: The last column reports the p-value for the test of no difference in means across the four treatment groups. For continuous variables, from an ANOVA F-test; for binary variables, from a chi-squared test. Male, University Degree, Residing in East Germany are indicators. Immigration stances vary between 0 and 10. Immigration Importance varies between 1 and 5. Found Information Trustworthy varies between 0 and 5.

Table A14: Participants Who Update Beliefs Consistently vs. Inconsistently

	Consistent Updating	**Inconsistent Updating**	**p-value**
Age	48.084	44.823	0.000
Male	0.540	0.513	0.149
University Degree	0.302	0.278	0.156
Residing in East Germany	0.604	0.597	0.736
Immigration Importance	4.006	4.021	0.687
Pre-Treatment Own Immigration Stance	7.099	7.121	0.833
Pre-Treatment Belief about Parties			
CDU Immigration Stance	5.889	5.459	0.000
AfD Immigration Stance	8.927	8.403	0.000
SPD Immigration Stance	3.526	3.860	0.000
Greens Immigration Stance	2.507	2.913	0.000
FDP Immigration Stance	5.368	5.201	0.095
Left Immigration Stance	4.820	4.711	0.223
Most Conservative ≠ AfD	0.084	0.153	0.000
AfD Liberal	0.059	0.104	0.000
Post-Treatment Belief about Parties			
Most Conservative ≠ AfD	0.076	0.157	0.000
AfD Liberal	0.048	0.094	0.000
Found Information Trustworthy	2.961	3.015	0.148
Found Survey Biased	0.224	0.237	0.417
Completion Time (Minutes)	15.261	24.920	0.238
Failed 2nd Attention Check	0.073	0.110	0.001
Observations	4,178	862	

Note: Consistent Updating: participants whose post-treatment CDU beliefs are weakly closer to the signal than their pre-treatment beliefs. Inconsistent Updating: participants whose post-treatment beliefs are strictly further from the signal.

Table A15: Effect of Signal and Perceived CDU Positioning on Turnout

	Main Study			*Follow-Up Study*		
	RF	2SLS	LLS	RF	2SLS	LLS
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment (CDU Signal)	0.001			0.009		
	(0.012)			(0.018)		
CDU Posterior Belief		0.003	-0.007		0.016	0.006
		(0.024)	[-0.026, 0.010]		(0.034)	[-0.020, 0.031]
Demographic controls	✓	✓	✓	✓	✓	✓
Observations	5,040	5,040	5,040	2,182	2,182	2,182

Note: Dependent variable: standardized self-reported likelihood to vote. Columns (1) and (4): OLS reduced form (effect of signal). Columns (2) and (5): 2SLS using CDU posterior belief as endogenous variable. Columns (3) and (6): LLS estimates. Robust standard errors in parentheses for RF and 2SLS; 95% bootstrap confidence intervals in brackets for LLS. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

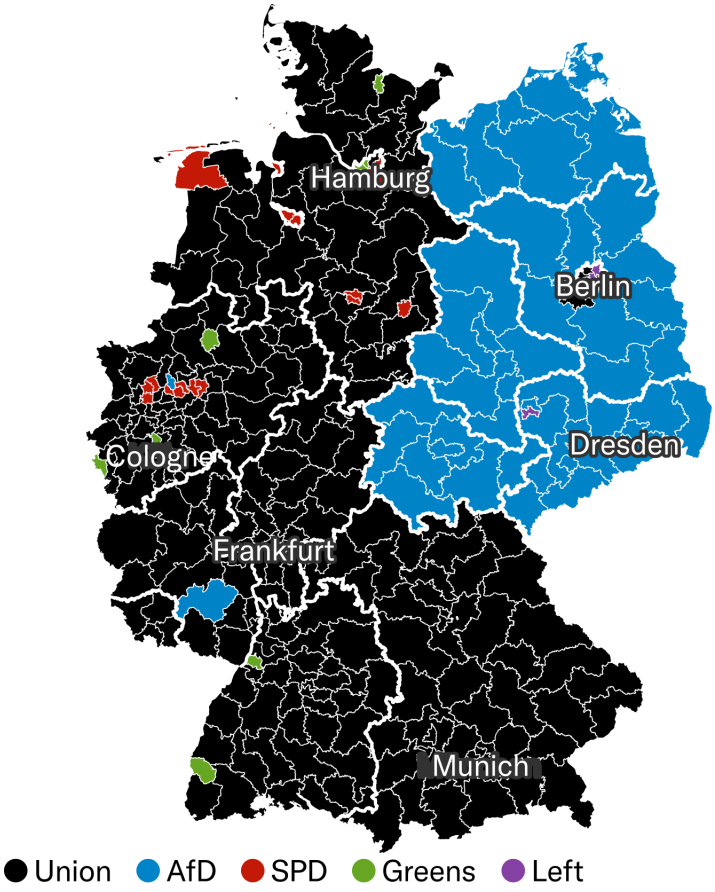


Figure A36: Party with Greatest Vote Share by Region in 2025
Note: Source is Hoppmann et al. (2025). "Union" = CDU/CSU.

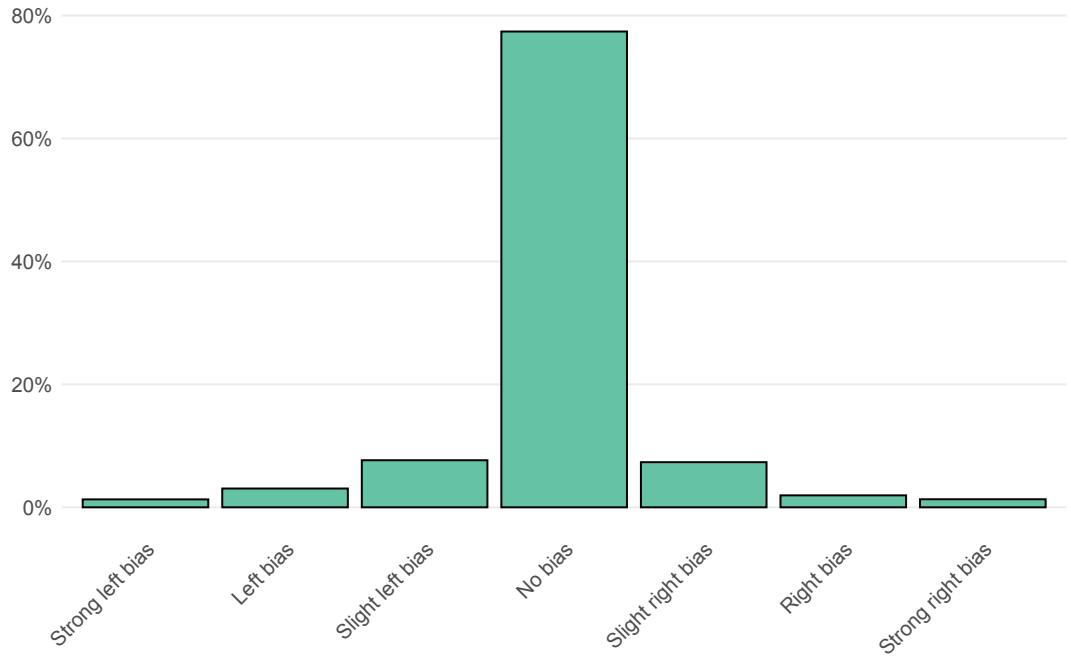


Figure A37: Distribution of Survey's Perceived Political Bias

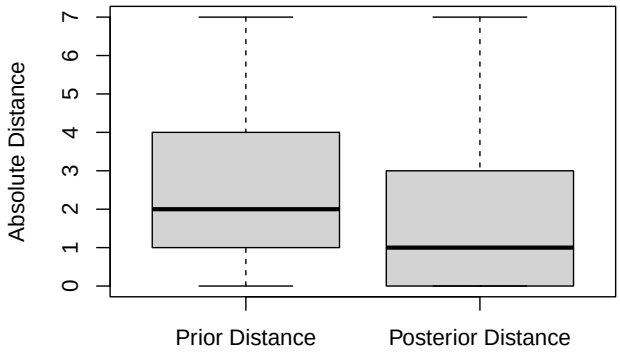


Figure A38: Difference (in Absolute Amount) Between CDU Signal and (a) Pre-Treatment Belief about CDU Immigration Stance (Left) or (b) Post-Treatment Belief about CDU Immigration Stance (Right)

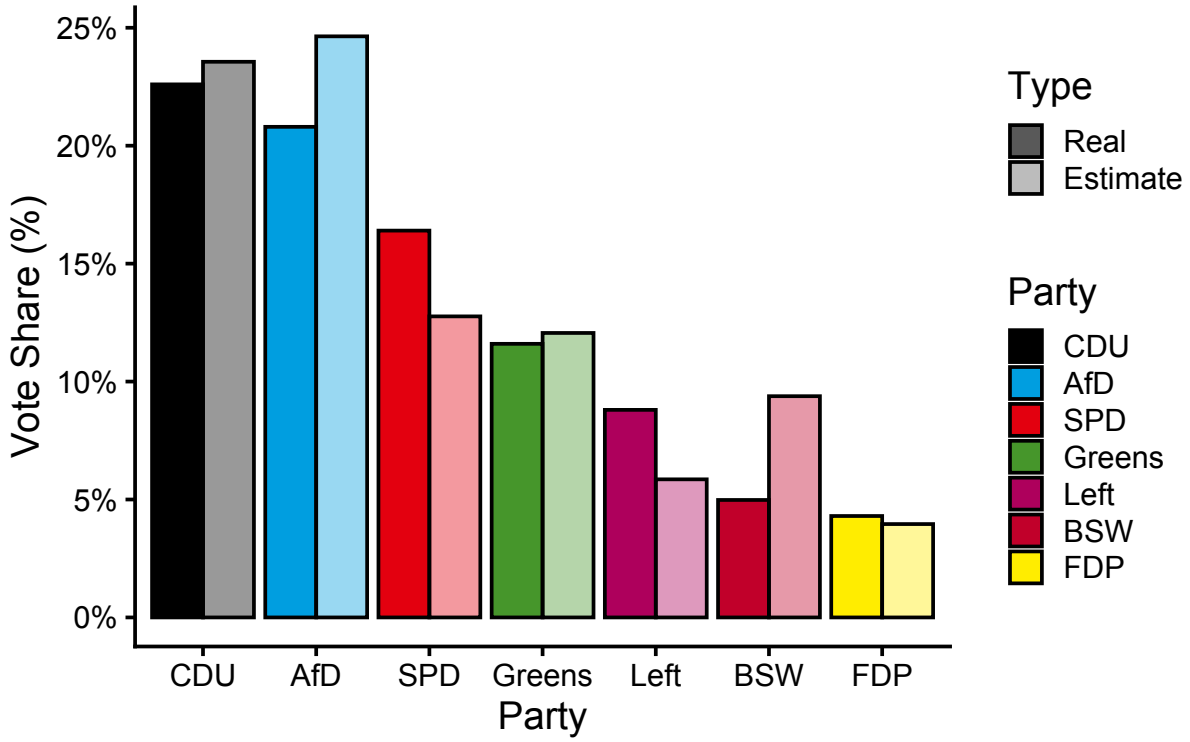


Figure A39: Reweighted Sample Vote Shares vs. Actual 2025 Election Results
Note: The figure compares vote intentions from our reweighted experimental sample to the official results of the 2025 German federal election. Reweighting accounts for oversampling of East Germany.

K Questionnaire (Translated from German)

K.1 Pre-Treatment Block (Fixed Order & Constant Across Treatments)

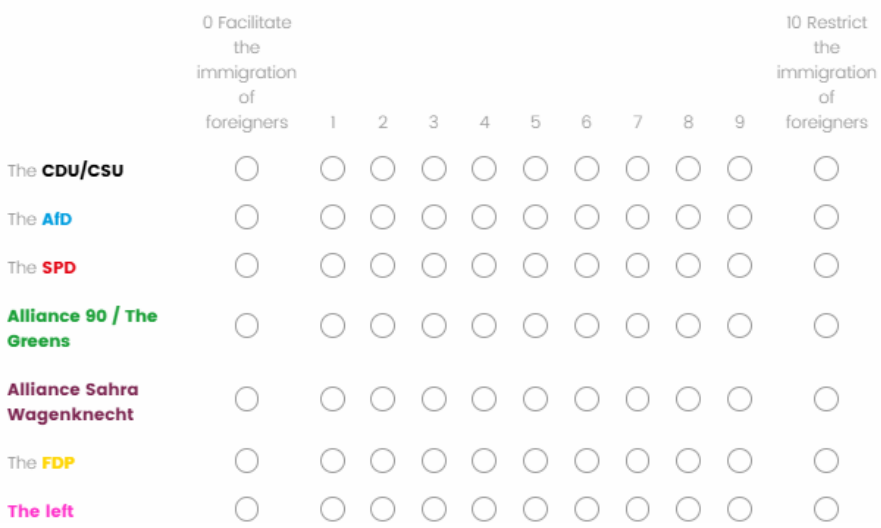
The survey started with a screen obtaining participants' informed consent and a CAPTCHA. It continued with an attention check and demographic questions on gender, age, educational attainment, and German federal state of residency. In the fourth screen, we measured participants' immigration attitudes and beliefs about parties' stances on immigration as shown below.

Some want to facilitate immigration opportunities for foreigners, while others want to restrict them.

What is your personal opinion *on the topic of immigration opportunities? Please use the following scale from 0 to 10. 0 on the scale means that, in your opinion, immigration opportunities should be strongly facilitated. A value of 10 means that, in your opinion, immigration opportunities should be strongly restricted. You can use the values between 0 and 10 to grade your opinion.*



What do you think, ***which policy represents****...?*



The fifth screen introduced the information provision as follows.

Scientific studies show that most Germans misjudge the positions of political parties. This is partly due to the fact that politicians often do not reveal their positions in public. Thus, scientists have long been researching how to measure the positions of political parties. As recent studies have shown, one valid method is to evaluate the responses of parties' candidates in an anonymous survey.

We used this method to measure the immigration stance of German parties. For this purpose, we analyzed the answers of some Bundestag candidates to the same question we just asked you. The currently available data is for candidates to the 2021 Bundestag election, so we have no data for Sarah Wagenknecht's Alliance yet. We will provide you with information on the responses of 503 candidates. These are broken down by party as follows CDU/CSU: 11, AfD: 58, SPD: 65, Greens: 131, FDP: 117, Left: 121.

The next page shows the results.

K.2 Treatment Block (Fixed Order & Randomized Across Treatments)

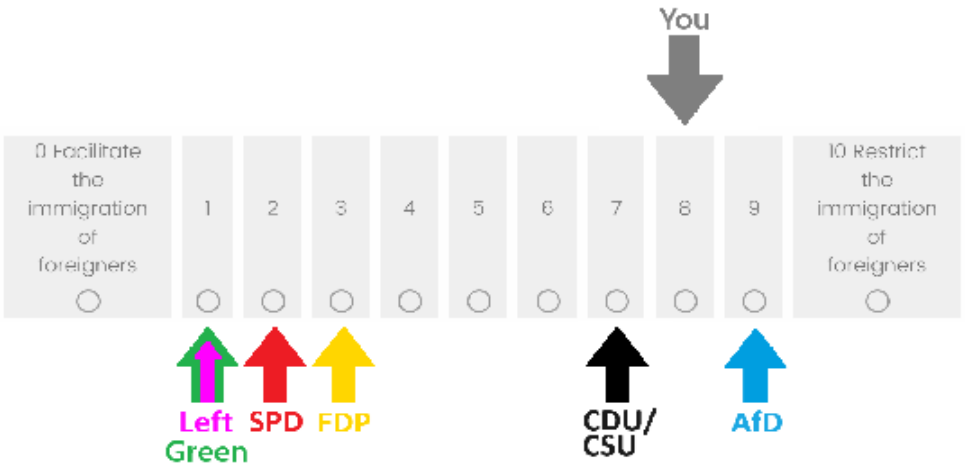
The sixth screen provided the randomized information. The following text was displayed at the top of the screen for all treatments:

The following figure illustrates the positions of German parties. The colored arrows show the middle position (median) of some Bundestag candidates of the respective party. The gray arrow shows your own position, which you indicated earlier.

Click here for details. [The middle position (also called median) is defined as follows: if you rank all positions in order of size, it lies exactly in the middle. Example: 5 CDU candidates have positions 5, 6, 7, 8, 8. Then, the middle position is 7, because two positions are smaller and two are larger than 7.]

The remainder of the screen was treatment-specific. Figure A40 shows the second half of the screen for participants assigned to the treatment where the CDU position is 7. Treatments differed by (i) the position of the "CDU/CSU" arrow and (ii) the message below the figure.

Some want to facilitate immigration opportunities. Others want to restrict immigration opportunities. What is your opinion?



As can be seen, the middle Bundestag candidate of the ...

- CDU/CSU wants to restrict immigration opportunities
- AfD wants to severely restrict immigration opportunities
- SPD wants to facilitate immigration opportunities
- Greens wants to strongly facilitate immigration opportunities
- FDP wants to facilitate immigration opportunities
- Left Party wants to strongly facilitate immigration opportunities

Figure A40: Information Provision, Treatment CDU = 7

K.3 Post-Treatment Block (Fixed Order & Constant Across Treatments)

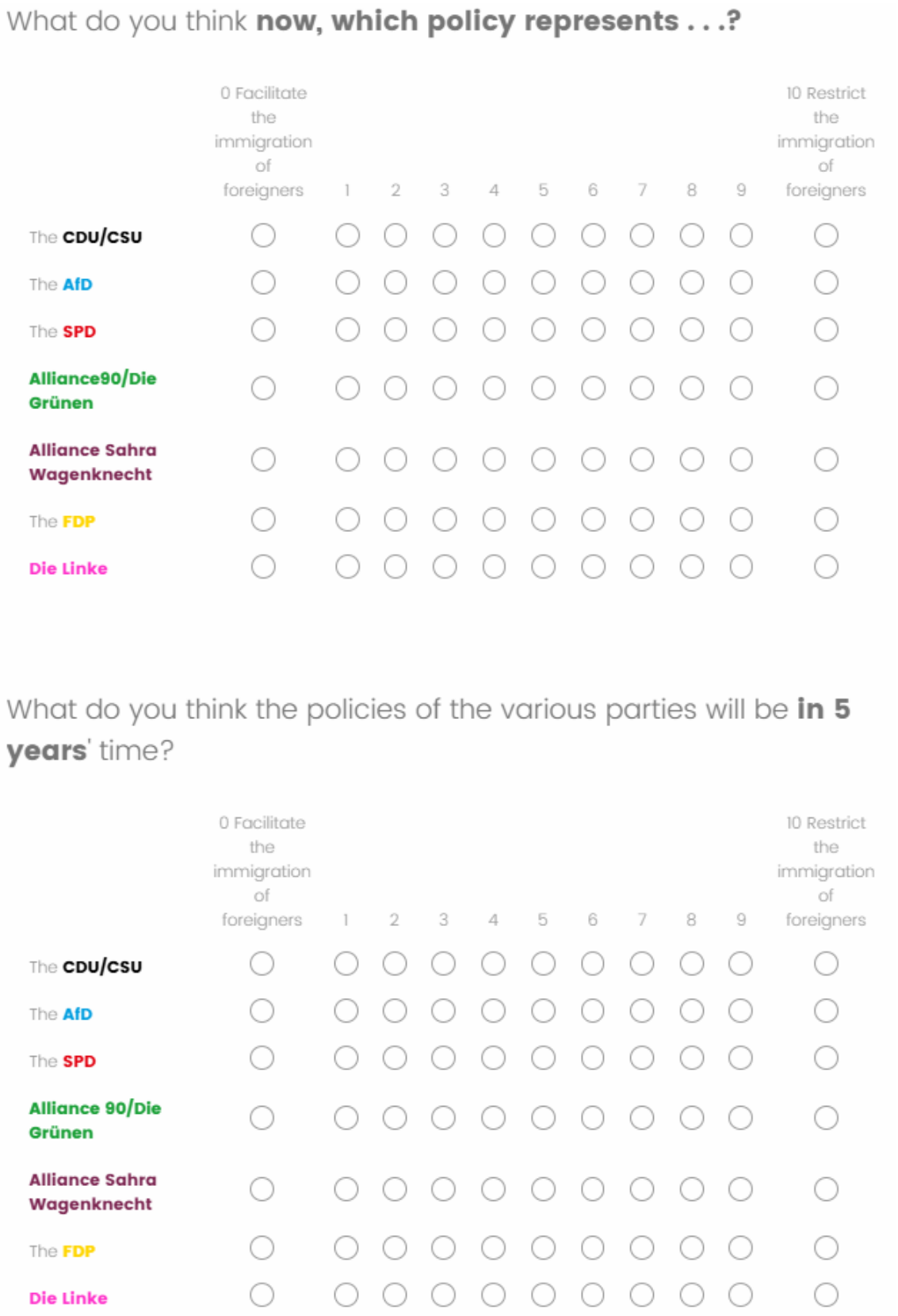


Figure A41: Post-Treatment Beliefs about Parties' Immigration Stance

K.4 Main Outcomes (Randomized Order & Constant Across Treatments)

The order of the following four blocks (Voting, Campaign Support, Newsletter Subscription, Feeling Thermometer) was randomized. The order within each of these blocks was held constant. The final main outcome (Donation) was always shown after these four main outcomes.

K.4.1 Voting

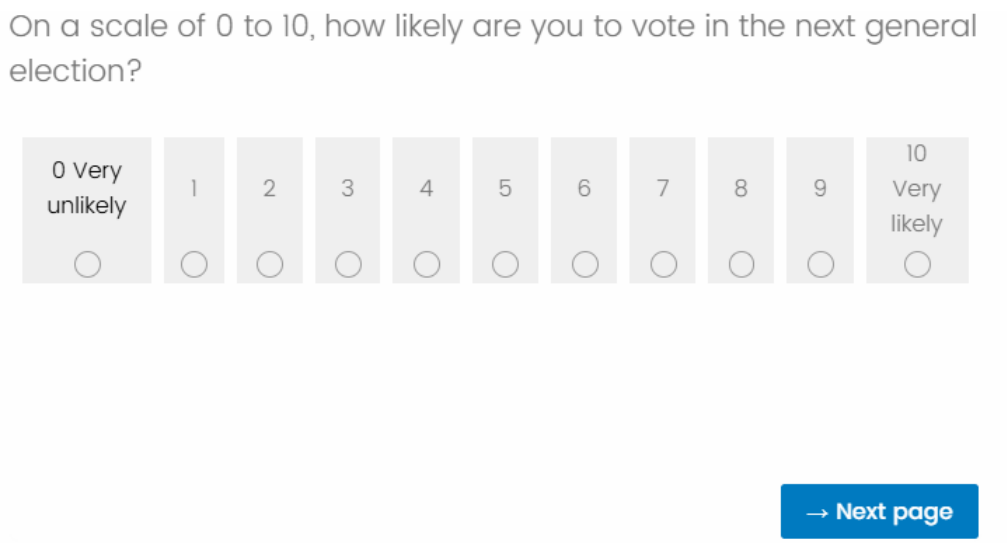


Figure A42: Turnout

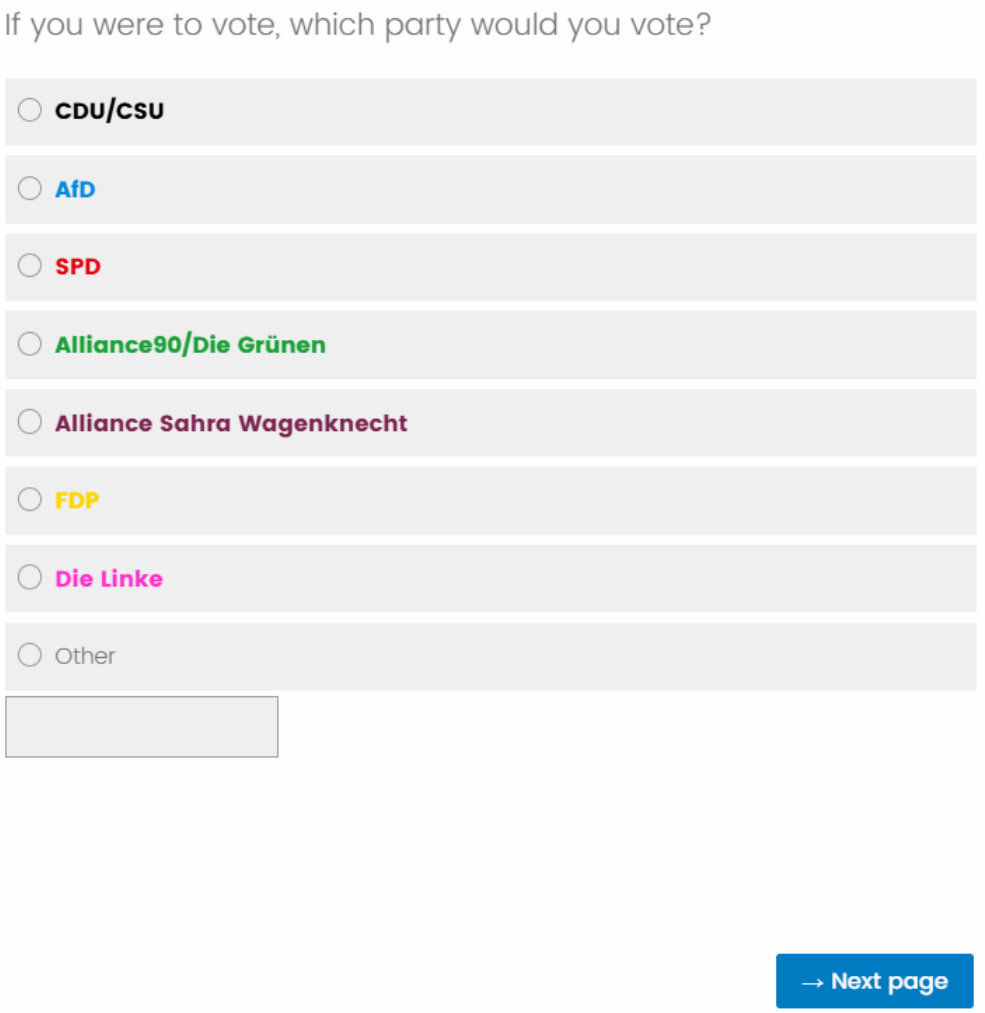


Figure A43: Voting Intention

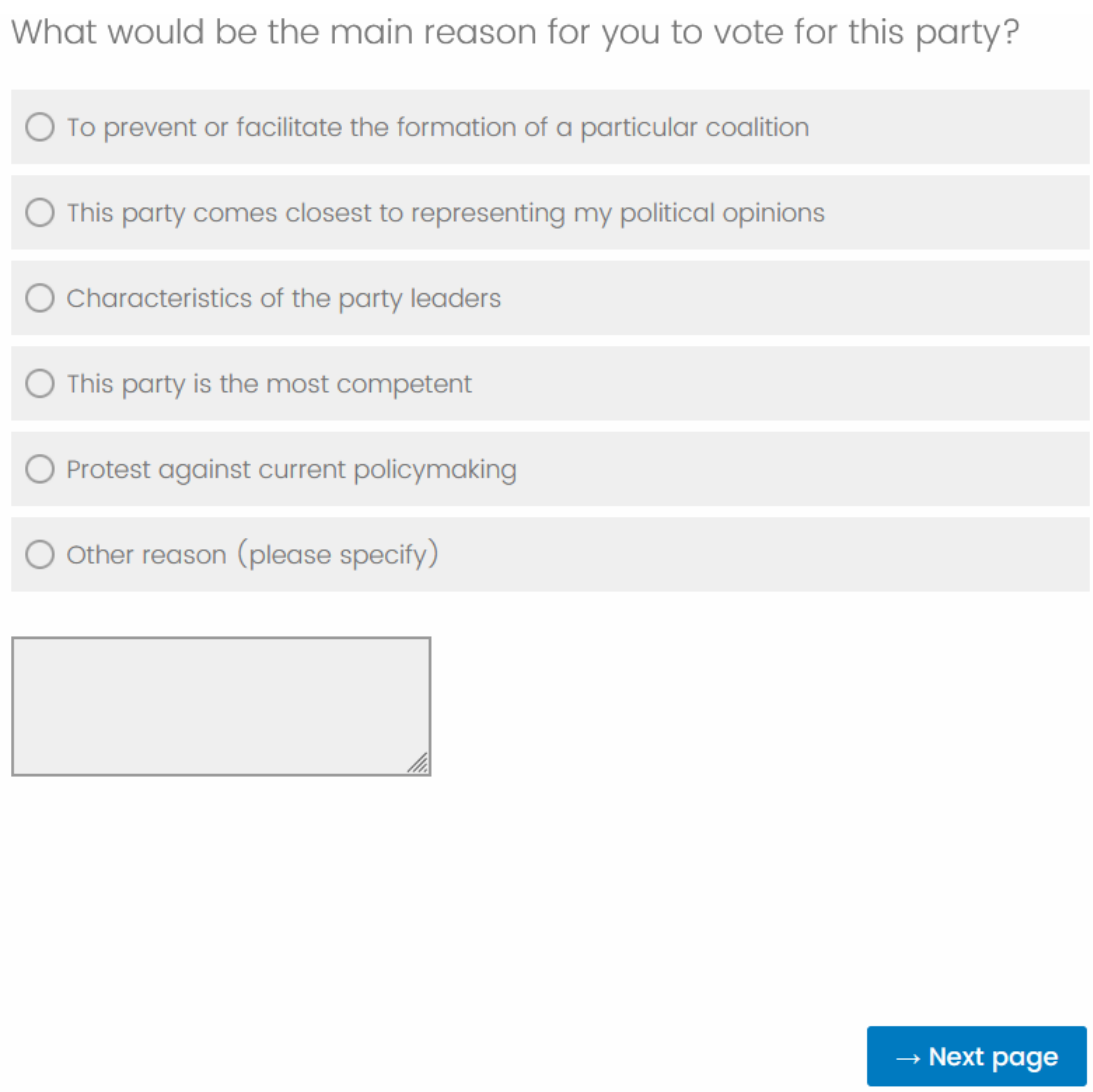


Figure A44: Voting Reason

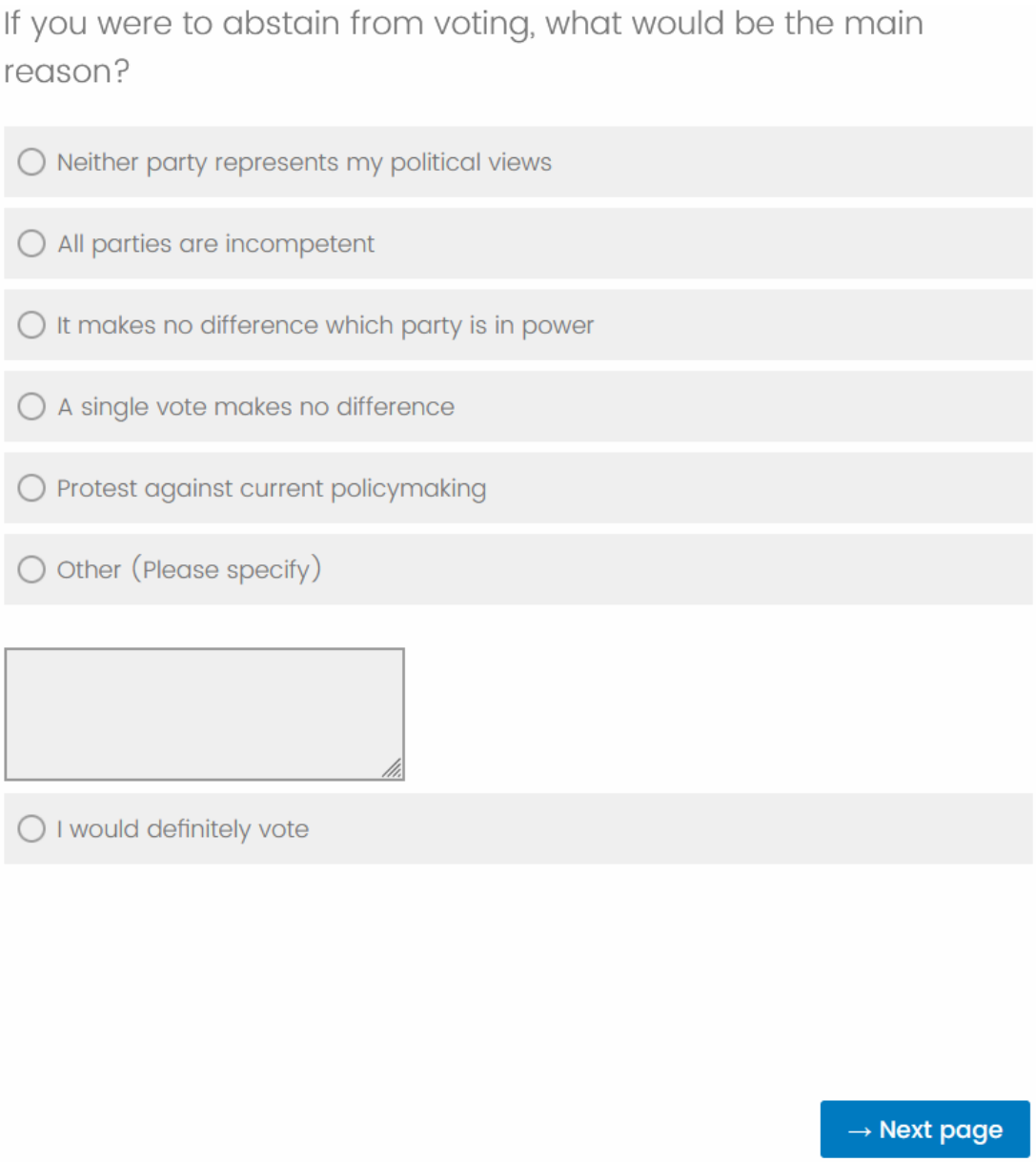


Figure A45: Turnout Reason

K.4.2 Campaign Support

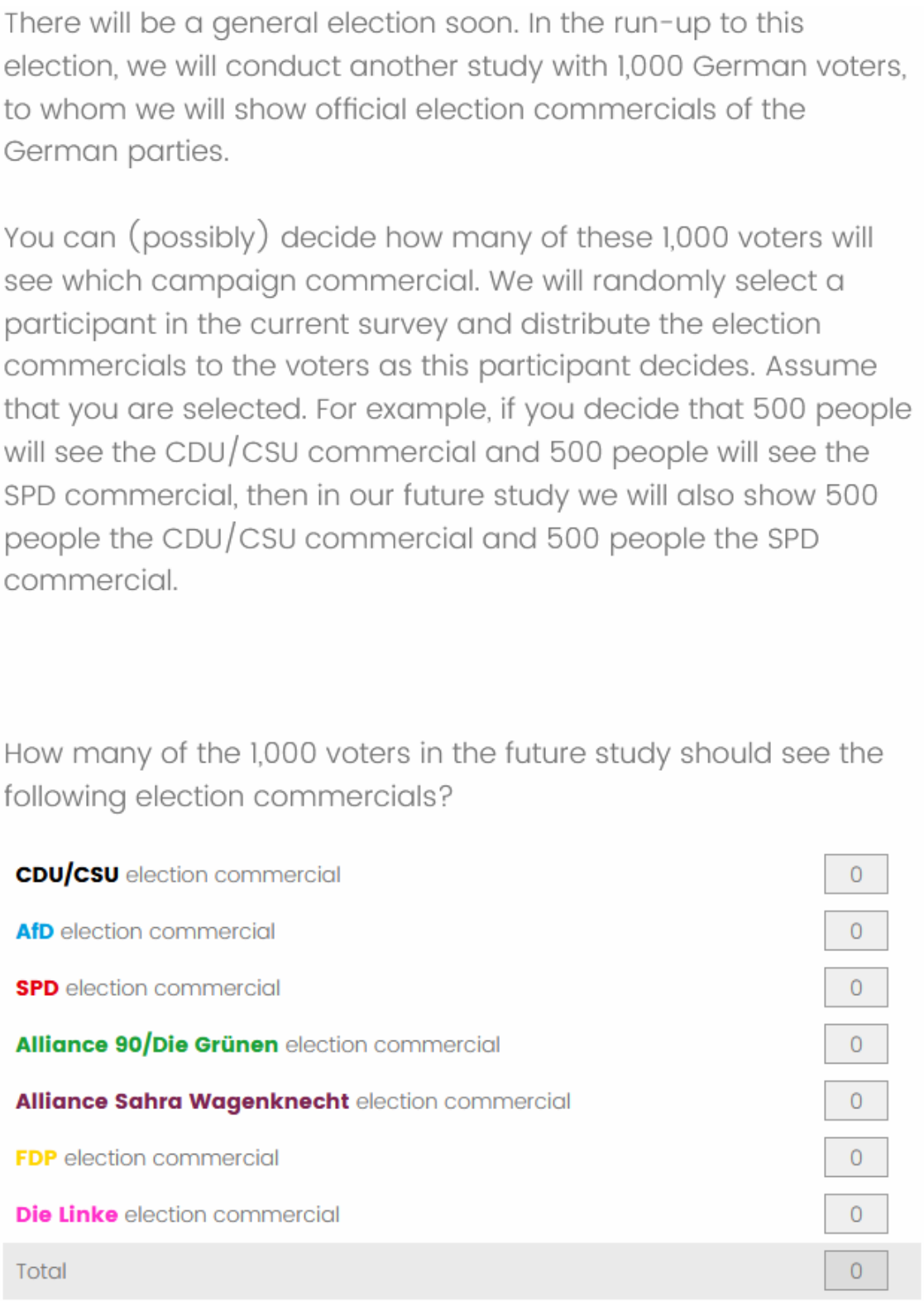


Figure A46: Campaign Support

K.4.3 Newsletter Subscription

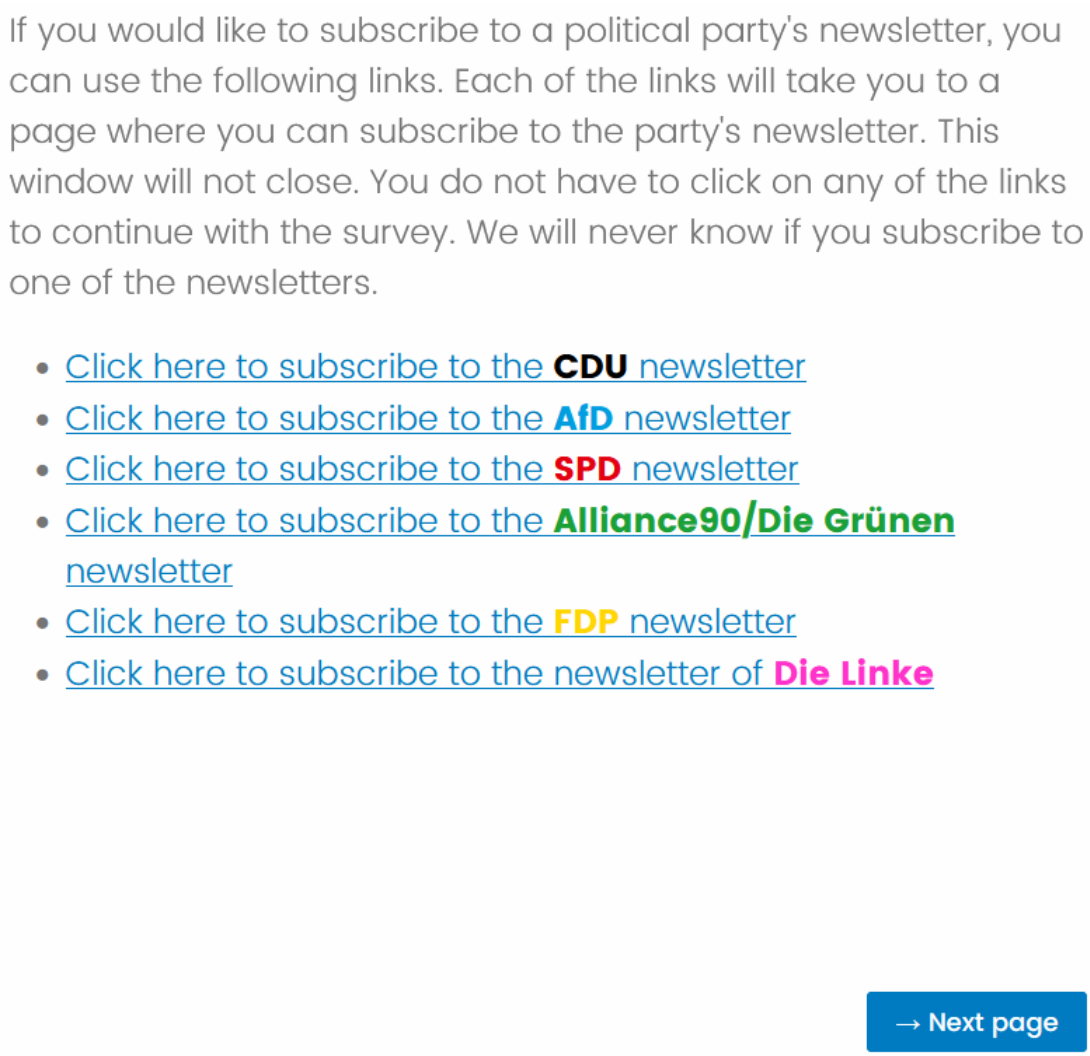


Figure A47: Newsletter Subscription

K.4.4 Feeling Thermometer

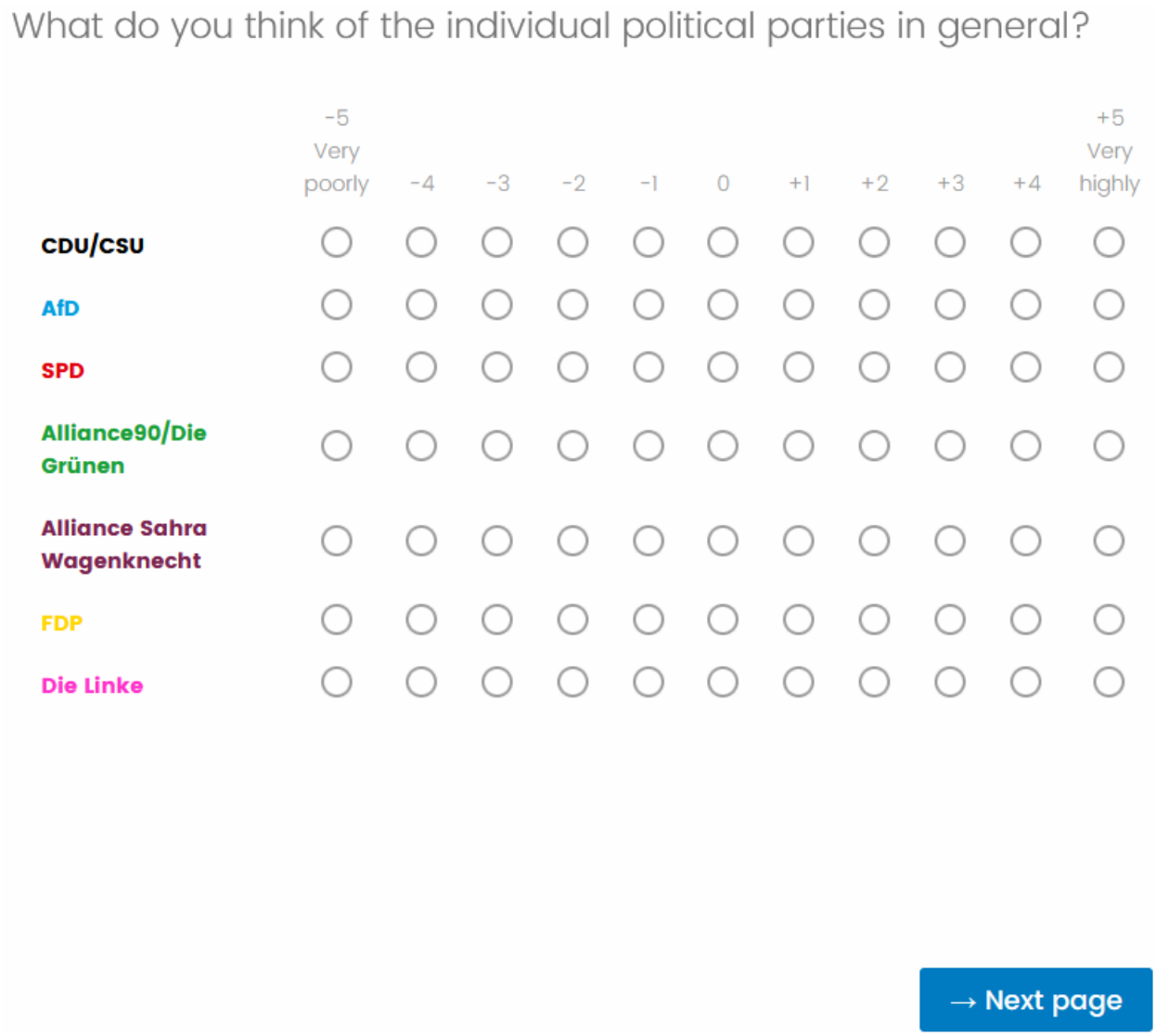


Figure A48: Attitude Thermometer

K.5 Donations (Randomized Order & Constant Across Treatments)

Donation items were organized as a sequence of four to five questions, using the staircase method (Falk et al., 2018). In each decision, participants chose between receiving a payment for themselves combined with a €5 donation to the party's donation account, and receiving nothing (with no donation made). Participants were truthfully informed that we would randomly select 10 participants and implement one randomly chosen decision for each of them. Figure A49 shows the first question participants were asked regarding the CDU. If they indicated that they were willing to allow the donation for a payment of €50 to them, we asked them whether they would also be willing to allow the donation for a lower amount. Similarly, if they did not allow the donation when offered €50, we asked whether they would be willing to do so for a higher amount. Hence, depending on their answer to the first question, the payments offered to them in the second question of the sequence were either €20 (if they allowed the donation) or €80 (if they did not allow the donation). Following the same logic, and depending on their response in this second question, amounts in the third question were either 5, 35, 65, or 95. Similarly, the fourth question included the values 0, 10, 30, 40, 60, 70, 90, and 105. Finally, we asked a fifth question offering a payment of €200 only to participants who were not willing to allow the donation for a payment of €105. This adaptive sequence was identical for the CDU and AfD and we randomized whether participants first answered questions about a donation to the AfD or to the CDU.

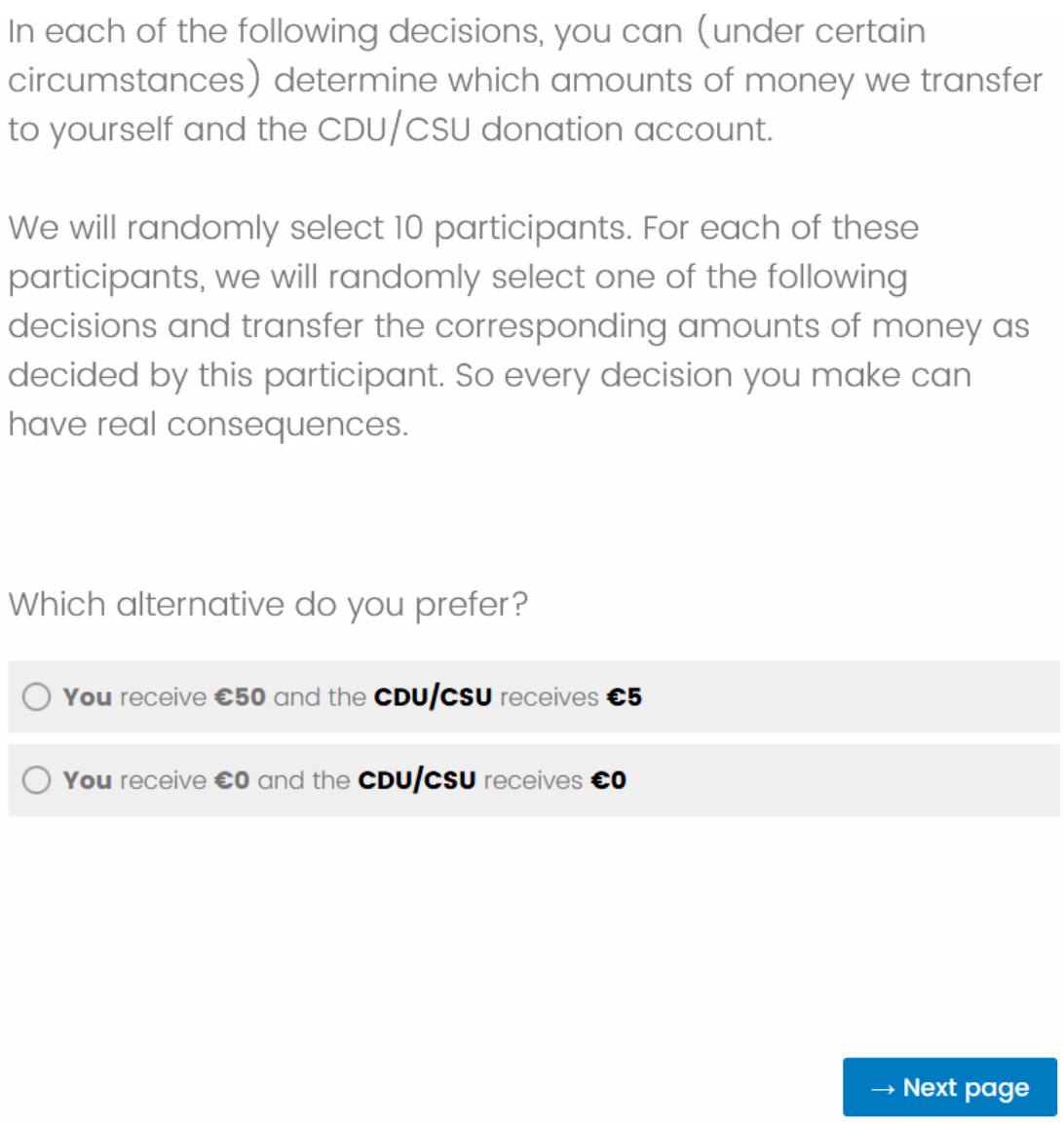


Figure A49: The First CDU Donation Item

K.6 Other Outcomes (Randomized Order & Constant Across Treatments)

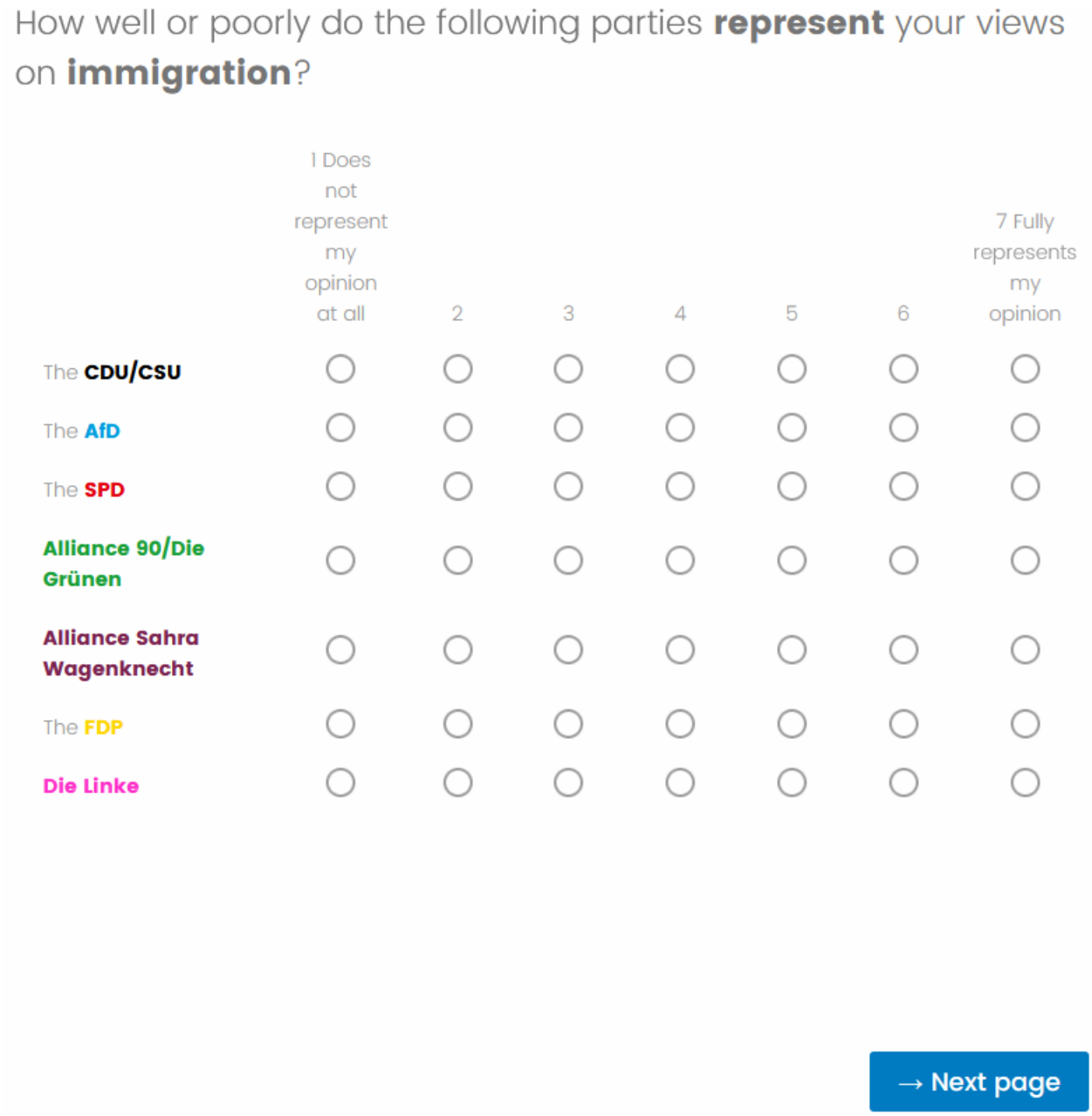


Figure A50: Perceived Representation

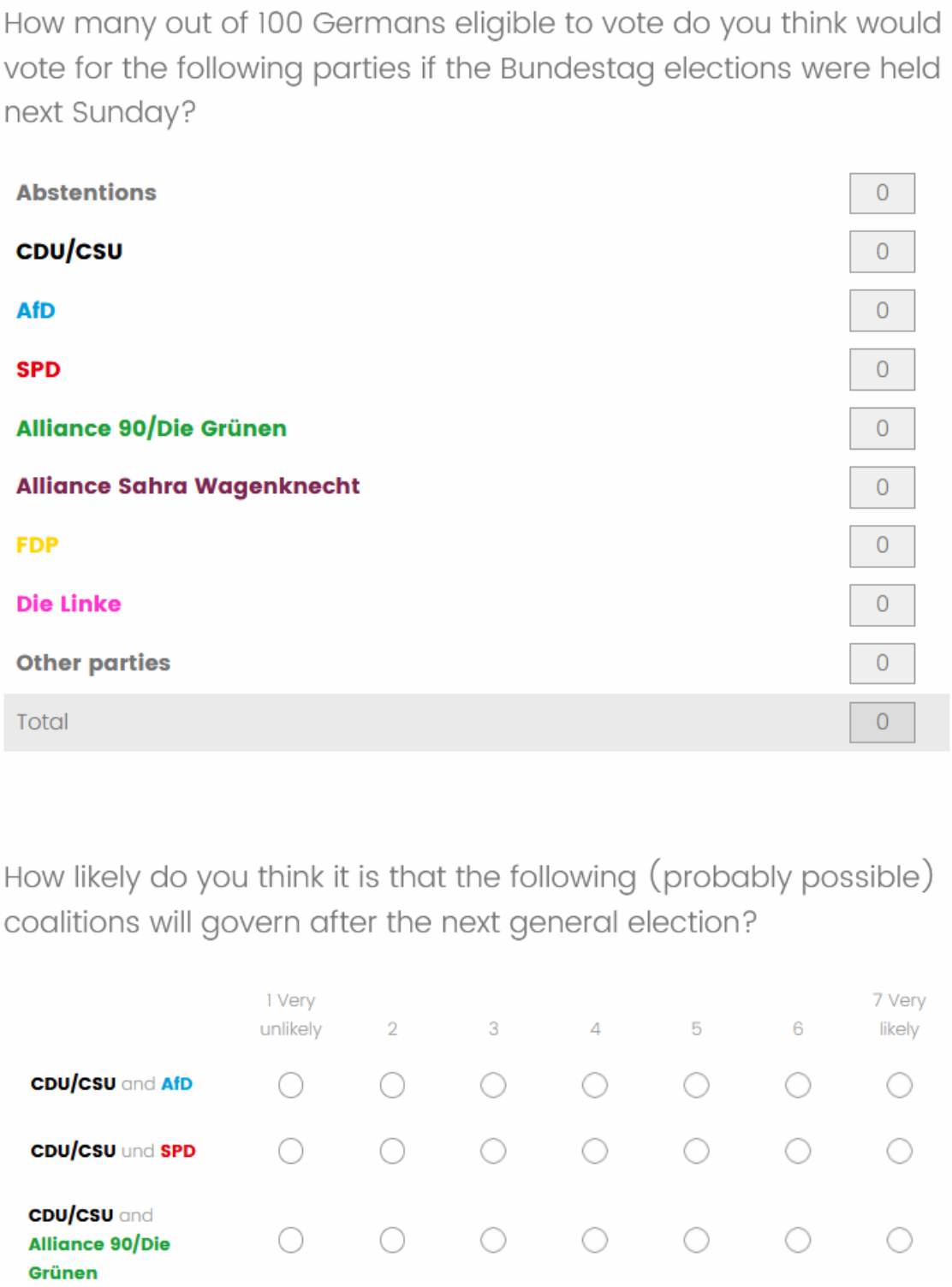


Figure A51: Expected Vote Shares

K.7 Final Block (Fixed Order & Constant Across Treatments)



Figure A52: Final Block, Screen 1

Did you find this survey politically biased or politically unbiased?

- ○ Strong left bias
- ○ Left bias
- ○ Slight left bias
- ○ No or nearly no bias
- ○ Slight right bias
- ○ Right bias
- ○ Strong right bias

You have received information about the opinions of members of the Bundestag as part of this study. Do you find this information credible or untrustworthy?

- ○ Very credible
- ○ Credible
- ○ Rather credible
- ○ Rather implausible
- ○ Implausible
- ○ Very implausible

Were you or at least one of your parents born abroad?

- ○ Yes
- ○ No

Figure A53: Final Block, Screen 2